



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Programme: RDS (Group: 4)

Assignment

BMDS DATA VISUALIZATION

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FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Plagiarism Statement and Guideline for Late Submission of Coursework

Read, complete, and sign this statement to be submitted with the written report.

We confirm that the submitted works are all our own work and are in our own words.

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1.0 Company Background

1.1 Nike

Based on responses from the questionnaire we distributed, Nike emerged as the brand most people are likely to purchase their footwear.

1.1.0 Background History

Nike was founded on January 25, 1964 by University of Oregon track athlete Phil Knight and his coach, Bill Bowerman, originally under the name Blue Ribbon Sports (BRS). Initially based in Eugene, Oregon, the company started by distributing shoes from Japanese manufacturer Onitsuka Tiger, selling them at track meets from



Phil Knight's car. During its first year, BRS sold 1,300 pairs of shoes, generating a profit of \$8,000.

In 1966, the company opened its first retail store in Santa Monica, California, and expanded to the East Coast a year later. By May 30, 1971, BRS rebranded as Nike, which is the name of the Greek goddess of victory. The company's first major advertising campaign, "There is no finish line," aired in 1977, and it famously did not feature a Nike product. Nike's iconic slogan, "Just Do It," was introduced in 1988, cementing the brand's identity globally.

Nike which is an American multinational corporation, is the world's largest manufacturer and supplier of athletic shoes apparel, and sports equipment. Nike's business operations cover the design, development, production, marketing, and sales of footwear, apparel, equipment, and accessories. Its product range includes shoes, cleats, jerseys, baselayers, shorts, and accessories that are designed for sports such as basketball, tennis, and golf, which cater to men, women, and children. The "Just Do It" slogan became an iconic slogan and the recognizable Swoosh logo has become synonymous with the brand.

Other than sports, Nike also evolved into a cultural symbol, mainly in the hip-hop fashion in time. This crossover into mainstream fashion originated from its strong association with sporting success. Since the 1980s, Nike apparel, such as the Air Jordan, Air Force 1, and

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Air Max sneakers, has become iconic especially in American youth culture. The brand's exclusive limited-edition releases and regional product launches have helped in fueling its popularity among sneaker enthusiasts, which has helped build a large loyal and passionate fan base.

In the late 20th century, American and European teenagers started to adopt Nike's products such as sneakers and sportswear and ingrained them as a part of their everyday outfits, pairing them with other clothings such as jeans, skirts, and sweaters. As Nike continued to meet the needs of athletes and broadened its appeal to casual wearers, it still prioritized design, innovation, and trendiness while also maintaining a focus on functionality. This approach has made Nike a key player in both athletic and casual fashion markets.

1.2 Adidas

The second brand that most people will buy according to the response is Adidas.

1.2.0 Background History

Adidas is a German athletic apparel and footwear corporation and it was founded by Adolf “Adi” Dassler on August 18, 1949, in Herzogenaurach, Germany.

Adidas is the largest sportswear manufacturing in Europe and also the second largest in the world, after



Nike. There are three stripes logos that are usually visible and can be seen at the Adidas product. The three stripes logo was acquired from the Finnish athletic footwear brand Krhu Sports, for two bottles of whiskey. Originally a family business, Dassler worked with his brother, Rudolf in 1924 by assisting in the development of spiked running shoes for multiple athletic events. In the 1936 Summer Olympics, Dassler Brothers defied all odds to provide shoe spikes to Jesse Owens. As a result, Jesse Owens won four Olympic gold medals in a row, and the Dassler Brothers Shoe Factory became an instant success. In 1949, following a breakdown in relationship between the brothers, Adolf created Adidas and Rudolf established Puma which became Adidas’ business rival. In time, Adidas quickly gained fame for its innovative sports footwear, particularly after the German national football team won the 1954 FIFA World Cup while wearing Adidas’ groundbreaking screw-in-stud football boots.

Adidas’ reputation continued to grow throughout the 1960s and 1970s. Adidas had been known worldwide for its high-quality sports equipment, apparel and footwear. Today, Adidas has become one of the largest sportswear companies globally and it is recognized for its commitment to both performance and style across different sports.

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In the 1980s, Adidas began its journey from sportswear giant to fashion trendsetter, particularly through its strong association with hip-hop culture. Adidas had marked a turning point by collaborating with Run-D.M.C which is an American hip-hop rap group. The group embraced Adidas' sneakers and tracksuits by showcasing the brand in their song "My Adidas". With the help of this partnership, Adidas is able to make its products part of everyday street fashion.

Adidas continued to blend performance with fashion, particularly through high-profile collaborations in the following years. By collaborating and partnering up with Kanye West and the launch of the Yeezy line, and well-known designers such as Stella McCartney and Pharrell Williams, Adidas had significantly boosted its presence in the lifestyle market. Adidas also put out limited-edition releases such as the Yeezy Boost which sparked major interest in the sneakerhead community while also reinforcing Adidas' status as a major trend in streetwear.

Today, Adidas has become a fashion icon with its three stripes recognized globally not just for athletic performance but for making a bold statement in the world of casual and street fashion. Adidas has solidified its place at the intersection of sport and style through its ability to stay ahead of trends while maintaining its sports heritage.

1.3 Skechers

The third brand that most people will buy according to the response is Skechers.

1.3.0 Background History

Skechers U.S.A. Inc. was founded in 1992 by Robert Greenberg, shortly after he left L.A. Gear. Initially, Skechers focused on the men's market, launching stylish street shoes. In 1993, with the launch of the "Chrome Dome" boot, Skechers quickly became popular and began to be sold in retailers such as Foley and Nordstrom.



In 1995, Skechers signed an agreement with Genova and Signal/American to begin producing casual clothing for boys and men, and expanded to Southeast Asia and Eastern Europe in 1997. By 2001, Skechers entered the British market and opened its first concept store.

In 2002, the first Canadian retail store opened, and Skechers launched the Skechers Work series for blue-collar workers and the Skechers Active series for women's casual shoes. In 2003, Skechers signed licensing agreements with multiple brands to further expand its market share.

In 2011, Skechers filed a lawsuit against Sears, alleging infringement of its product line. In 2012, Skechers Japan was established and signed marketing agreements with several celebrities. In 2013, Skechers official footwear and apparel sponsor of the Chevron Houston Marathon

In 2014, Skechers announced its interest in the LA Clippers against Reedbok and Fila, alleging copyright infringement. At the same time, Skechers expanded its European distribution center in Belgium and opened its 1000 th store. In 2015, Skechers Performance continued to serve as the official footwear and apparel sponsor of the Chevron Houston Marathon, marking the company's continued growth and expansion.

2.0 Products Details

No	Brands	Products
1	Adidas	● Casual Shoes ● Sport Shoes ● Sandals ● Formal Shoes
2	Puma	● Casual Shoes ● Sport Shoes ● Sandals ● Formal Shoes
3	Nike	● Casual Shoes ● Sport Shoes ● Sandals ● Sneakers
4	Vans	● Casual Shoes ● Formal Shoes ● Boots
5	Skechers	● Casual Shoes ● Sport Shoes ● Sandals ● Formal Shoes
6	Bata	● Casual shoes
7	Lining	● Sport Shoes

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8	New Balance	• Casual Shoes • Sport Shoes
9	Asics	• Sport Shoes
10	361°	• Casual Shoes
11	Cotton on	• Sandals
12	Line Seven	• Sport Shoes
13	B.U.M	• Casual Shoes
14	Vincci	• Casual Shoes
15	Under Armour	• Sport Shoes
16	The Ballerina	• Formal Shoes
17	Converse	• Sneakers

3.0 Data Preparation

In this process, we will be discussing the preparation of the raw data in order to ensure that the data is clean, accurate and in the right format before moving on to the next step for further analysis.

3.1 Data Collection

The initial step in the data preparation process is data collection. The data that we are going to collect is about customers' most demanded footwear and shoe products and the most popular competitor's brands. In this stage, we prepared a list of detailed questions in a questionnaire related to footwear and shoe products through Google Form. The questionnaire consists of a total of 16 questions including demographic questions. The questionnaire was distributed to 156 respondents, representing a diverse range of demographics. The demographic questions that are included in the questionnaire are age, gender, occupation, salary, and place of residence. Other than demographic questions, the remaining questions were focused on footwear. We focused on various aspects of footwear, including the preferred shoe size in UK size, preferable cost to buy shoe, types of shoe, favourite brands and factors that influence brand preference. Additionally, we also came up with questions to ask the respondents on the frequency of shoe purchase, reason for buying a new shoe, preferred shoe colour and material and common sources of information regarding new arrivals and promotions. Some questions are provided with multiple choices for each question to allow participants to select their preferred option.

3.2 Data Compilation

After the data collection stage, a total of 156 respondents have been recorded and we first split the work into 5 people to manually create 5 datasets in SAS and insert the data using the DATA step. The reason for splitting is that it is more efficient and it would save a lot of time during this stage. Below are the steps to create the first 20% of the questionnaire responses, the rest of the responses will be recorded using the same steps as below.

<pre>/* Step 1 : Define the Dataset */ data dataset1;</pre>	The 'data' keyword is used to define a new dataset in SAS. This line of code creates a new dataset called dataset1 where the first 20% of the questionnaire responses will be stored. We will record the rest of the responses using dataset2, dataset3, dataset4, and dataset5.
<pre>/* Step 2 : Define the Variables */ length Age \$10. Gender \$8. Region \$50. Area \$10. Occupation \$20. Salary \$20. Size \$10. Cost \$15. Type \$20. Brand \$25. Factors \$200. Frequency \$20. Reason \$40. Colour \$15. Material \$15. Advertisement \$25. ;</pre>	The 'length' statement defines the length and type of each variable. The variables prefixed with a '\$' symbol are character variables and the number after the '\$' symbol represents the maximum length of the string. For example, the 'Age' is defined as a string due to the '\$' symbol and it can hold up to 10 characters.

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<pre>/* Step 3 : Input Data */ input Age Gender Region Area Occupation Salary Size Cost Type Brand Factors Frequency Reason Colour Material Advertisement ;</pre>	The ‘input’ statement lists all the variables that will be read in from the data. This statement is to inform SAS to expect the values in this order and assign them to their corresponding variable once the actual data has been entered.
<pre>/* Step 4 : Datalines or Infile Statement */ datalines; 15 to 20 Female Klang City Student Below RM1500 4 to 6 RM201 - RM400 Casual shoes Sketchers Cheap, Frequent Discount, Comfort, 15 to 20 Female Damansara City Student Below RM1500 7 to 9 RM201 - RM200 Sport shoes Nike Cheap, Comfort, Durability Once a year 15 to 20 Female Kajang City Student Below RM1500 4 to 6 RM201 - RM200 Casual shoes Puma Frequent Discount, Loyal to Brand, Social 15 to 20 Female Kuala Lumpur City Center (KLCC) City Student Below RM1500 4 to 6 RM201 - RM200 Casual shoes Adidas Social 21 to 30 Male Klang Village Student Below RM1500 7 to 9 RM201 - RM200 Casual shoes Adidas Comfort, Environmental Requirement Once 15 to 20 Male Setapak Village Student Below RM1500 7 to 9 RM201 - RM200 Sport shoes Nike Comfort, Environmental Requirement Once 41 to 50 Female Petaling Jaya City Working Adult Above RM3000 4 to 6 RM201 - RM200 Sport shoes Adidas Frequent Discount, Comfort, 41 to 50 Female Kajang City Unemployed RM3000 - RM4000 4 to 6 RM201 - RM200 Sport shoes Sketchers Comfort, Health, Functionality 41 to 50 Female Cheras City Working Adult RM3000 - RM4000 4 to 6 RM201 - RM200 Formal shoes Sketchers Comfort, Functionality 21 to 30 Female Bukit Bintang City Student Below RM1500 4 to 6 RM201 - RM200 Casual shoes Sketchers Comfort, Health, Functionality 41 to 50 Female Sarawak City Working Adult RM3000 - RM4000 4 to 6 RM201 - RM200 Casual shoes Adidas Frequent Discount, Comfort 15 to 20 Female Petaling Jaya City Student Below RM1500 4 to 6 RM201 - RM200 Casual shoes Nike Frequent Discount, Comfort Once 15 to 20 Female no Rural Student Below RM1500 7 to 9 Below RM200 Sandals not branded Cheap, Frequent Discount, Comfort, Functi 15 to 20 Female Petaling Jaya City Student Below RM1500 7 to 9 RM201 - RM200 Casual shoes Sketchers Comfort, Durability Once a 15 to 20 Female Petaling Jaya City Student Below RM1500 7 to 9 RM201 - RM200 Casual shoes Nike Comfort, Loyal to Brand, Social 15 to 20 Male Taman Melawati City Student Below RM1500 18 to 22 Below RM100 Sport shoes Adidas Cheap, Comfort, Durability Once a 21 to 30 Male Klang City Working Adult RM3000 - RM4000 7 to 9 RM201 - RM200 Sport shoes Adidas Cheap, Frequent Discount, Comfort 15 to 20 Male Bangsar South City Working Adult RM3000 - RM4000 18 to 22 RM201 - RM200 Sport shoes Nike Cheap, Frequent Discou 41 to 50 Female Rushing City Working Adult RM1500 - RM3000 4 to 6 RM201 - RM200 Casual shoes Sketchers Frequent Discount, 15 to 20 Female Petaling Jaya City Student Below RM1500 - RM3000 3 to 5 RM201 - RM200 Sport shoes Sketchers Comfort, Durability, Loyal</pre>	The ‘datalines’ statement is used to directly input the data into a dataset within the program itself
<pre>/* Step 5 : Run the Data Step */ run;</pre>	The ‘run’ statement is used to mark the end of the DATA step. It will inform SAS to execute the preceding code before moving on to the next lines of code.
<pre>/* Step 6 : Set the Title and View the Output */ title "First Dataset"; proc print data=dataset1; run;</pre>	The ‘title’ statement is used to create a title for the dataset. While the ‘proc print data=dataset1’ is used to display the contents of the dataset name dataset1 with a title. The ‘run’ statement is used to execute the code above.
Once the other datasets have been created, we will combine all the datasets into one unified dataset by using the ‘set’ statement to concatenate multiple datasets.	

```
* merge dataset;  
data full;  
set dataset1  
    dataset2  
    dataset3  
    dataset4  
    dataset5;  
run;  
  
proc print data = full;  
run;
```

3.3 Data Cleaning

The next step is data cleaning. The purpose of data cleaning is to ensure that the data will be using the same format and does not have any sort of error so that it is accurate, consistent and reliable before it can be used for analysis purposes. Data cleaning involves identifying and correcting errors, inconsistencies, inaccuracies or potential missing values in the dataset to ensure valid, high quality results for analyses.

<pre><code>* Data cleaning; data cleaned_data; set full;</code></pre>	A new dataset named cleaned_data was created. SET statement used to read the existing dataset “full”.
Cleaning Region: <pre><code>* Region; if Region = 'Bahau' then Region = 'Southern'; if Region = 'Johor Bahru' then Region = 'Southern'; if Region = 'Petaling Jaya' then Region = 'Central'; if Region = 'Klang' then Region = 'Central'; if Region = 'Kepong' then Region = 'Central'; if Region = 'Cheras' then Region = 'Central'; if Region = 'Setapak' then Region = 'Central'; if Region = 'Damansara' then Region = 'Central'; if Region = 'Kuching' then Region = 'Sarawak'; if Region = 'Kuala Lumpur City Center (KLCC)' then Region = 'Central'; if Region = 'Bukit Bintang' then Region = 'Central'; if Region = 'Taman Melawati' then Region = 'Central'; if Region = 'Bangsar South' then Region = 'Central'; if Region = 'Subang Jaya' then Region = 'Central'; if Region = 'no' then delete; if Region = 'Singapore' then delete;</code></pre>	The if-then statement is used for standardizing and removing irrelevant data from the region variable. We standardize the region name to ensure that different places are grouped into broader categories such as ‘Southern’, ‘Central’, ‘Northern’ and ‘Eastern’. This can help to aggregate the granular regions or city names into larger geographical regions for easier analysis. Besides, we also remove invalid or unnecessary entries. We delete the data where the Region is ‘no’ or ‘Singapore’. This can simplify the dataset by reducing the numbers of unique values in the Region column.
Cleaning Brand:	The if-then statement is used for

<pre> * Brand; if Brand = 'Sketchers' then Brand = 'Skechers'; if Brand = 'new balance' then Brand = 'New Balance'; if Brand = 'New balance' then Brand = 'New Balance'; if Brand = 'lining' then Brand = 'Lining'; if Brand = 'asics' then Brand = 'Asics'; if Brand = 'Under armour' then Brand = 'Under Armour'; if Brand = 'LINE SEVEN' then Brand = 'Line Seven'; if Brand = 'Cotton on' then Brand = 'Cotton On'; if Brand = 'Support local brand' then delete; if Brand = 'not branded' then delete; if Brand = 'No specific brand' then delete; if Brand = 'Depends type of shoes' then delete; if Brand = 'Off branded' then delete; if Brand = 'Taobao' then delete; if Brand = 'no brand' then delete; if Brand = "Don't have specific brand" then delete; </pre>	standardizing brand names, while the if-then-delete statement is applied to remove rows with undesirable values. We standardized brand names to eliminate discrepancies caused by typos, case differences, or variations in naming. For example, we corrected the spelling of “Skechers” due to a typographical error, ensuring consistency in the brand name. We standardized the capitalization of brand names to maintain uniformity across the dataset, including “New Balance”, “Lining”, “Asics”, “Under Armour”, “Line Seven” and “Cotton On”. We removed entries where the brand is listed as “Support local brand”, “not branded”, “No specific brand”, “Depends type of shoes”, “Off branded”, “Taobao”, “no brand” and “Don’t have specific brand”. This process refines the dataset and ensures the focus is on recognized brands.
Cleaning Frequency: <pre> * Frequency; if Frequency = 'Once a few year' then Frequency = 'Once a few years'; </pre>	The if-then statement is used for standardizing the frequency of shoe purchases. We standardize terms in the frequency variable to improve clarity and consistency, especially when analyzing how often consumers make purchases. For instance, the phrase “Once a few year” is corrected to “Once a few years”. This ensures that frequency categories are uniform, making it easier to summarize

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	and categorize purchasing behavior.
Cleaning Reason: <pre>* Reason; if Reason = 'Got discounts!!!' then Reason = 'Got discounts'; if Reason = 'just want to buy a new one' then Reason = 'Interested to Purchase'; if Reason = 'just interested to buy' then Reason = 'Interested to Purchase'; if Reason = 'New shoes for new year' then Reason = 'Occasions';</pre>	The if-then statement is used for standardizing the reasons for purchasing new shoes, thereby enhancing the clarity of insights into consumer motivations. For instance, variations such as “Got discounts!!!” are simplified to “Got discounts”, removing extraneous punctuation and ensuring consistency. The phrases like “just want to buy a new one” and “just interested to buy” are both consolidated into the more generalized category “Interested to Purchase,” allowing for a unified understanding of consumer intent. The phrase “New shoes for new year” is standardized to “Occasions”, effectively capturing the motivation behind seasonal purchases. This process of normalization reduces the noise created by inconsistent phrasing, making it easier to analyze and interpret customer motivations accurately and effectively.
Cleaning Type: <pre>* Types; if Type = 'sneakers' then Type = 'Sneakers'; if Type = 'Rarely buy' then delete;</pre>	The if-then statement is used for standardizing the types of shoe, while the if-then-delete statement is applied to remove rows with undesirable values. We standardize the type variable to ensure

	that all entries are recognized as the same category, which is important for accurate analysis of types of shoes being purchased. We standardized the capitalization of the type of shoe to maintain uniformity across the dataset, like “Sneakers”. We removed entries where the type is “Rarely buy”. By removing these entries, we ensure the dataset focuses only on relevant purchases, leading to a clearer understanding of consumer behavior related to shoe purchases.
Cleaning Advertisement: <pre>* Advertisement; if Advertisement = 'shopping physical' then Advertisement = 'Physical store'; if Advertisement = 'Mall' then Advertisement = 'Physical store'; if Advertisement = 'In-store' then Advertisement = 'Physical store'; if Advertisement = 'SHOP' then Advertisement = 'Physical store'; if Advertisement = 'During shopping' then Advertisement = 'Physical store'; if Advertisement = 'physically' then Advertisement = 'Physical store'; if Advertisement = 'Retail shops' then Advertisement = 'Physical store';</pre>	The if-then statement is used for standardizing the common sources of information regarding new arrivals and promotions, thereby enhancing the clarity of insights into consumer motivations. The phrases like “shopping physical”, “Mall”, “In-store”, “SHOP”, “During shopping”, “physically” and “Retail shops” are consolidated into the more generalized category “Physical store”. This consolidation ensures consistency in the dataset by grouping various expressions that refer to the same concept of purchasing in a physical location, allowing for more accurate analysis of shopping behaviors related to physical retail environments.

```
proc print data = cleaned_data;  
run;
```

View the output of the cleaned dataset.

3.4 Data Encoded

Data encoding is the crucial step in the data preprocessing pipeline that follows data cleaning. During this phase, all categorical variables are transformed into numerical formats suitable for analysis. The primary purpose of data encoding is to convert categorical and textual data, which cannot be directly processed by algorithms, into numerical representations. Encoded data enables us to apply various statistical techniques and algorithms for more efficient analysis.

<pre><code>*Encoded data; data encoded_data; set cleaned_data;</code></pre>	A new dataset named encoded_data was created. SET statement used to load the existing dataset cleaned_data.
Encoding Age: <pre><code>* Age; if Age = '15 to 20' then encoded_age = 1; else if Age = '21 to 30' then encoded_age = 2; else if Age = '31 to 40' then encoded_age = 3; else if Age = '41 to 50' then encoded_age = 4; else if Age = '51 to 60' then encoded_age = 5; else if Age = 'Above 60' then encoded_age = 6;</code></pre>	The If-Then-Else-If statement is used to encode the age by their category where: Age 15 to 20 = 1 Age 21 to 30 = 2 Age 31 to 40 = 3 Age 41 to 50 = 4 Age 51 to 60 = 5 Age above 60 = 6
Encoding Gender:	The If-Then-Else-If statement is used to encode the gender where:

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<pre>*Gender; if Gender = 'Female' then encoded_gender = 1; else if Gender = 'Male' then encoded_gender = 2;</pre>	Female = 1 Male = 2
Encoding Region: <pre>* Region; if Region = 'Central' then encoded_region = 1; else if Region = 'Northern' then encoded_region = 2; else if Region = 'Southern' then encoded_region = 3; else if Region = 'Eastern' then encoded_region = 4; else if Region = 'Western' then encoded_region = 5; else if Region = 'Sabah' then encoded_region = 6; else if Region = 'Sarawak' then encoded_region = 7;</pre>	The If-Then-Else-If statement is used to encode the living region where: Central = 1 Northern = 2 Southern = 3 Eastern = 4 Western = 5 Sabah = 6 Sarawak = 7
Encoding Area: <pre>* Area; if Area = 'City' then encoded_area = 1; else if Area = 'Village' then encoded_area = 2; else if Area = 'Urban' then encoded_area = 3; else if Area = 'Rural' then encoded_area = 4;</pre>	The If-Then-Else-If statement is used to encode the living area where: City = 1 Village = 2 Urban = 3 Rural = 4
Encoding Occupation:	The If-Then-Else-If statement is used to encode the occupation

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<pre> * Occupation; if Occupation = 'Student' then encoded_occ = 1; else if Occupation = 'Working Adult' then encoded_occ = 2; else if Occupation = 'Unemployed' then encoded_occ = 3; else if Occupation = 'Retired' then encoded_occ = 4; </pre>	where: Student = 1 Working Adult = 2 Unemployed = 3 Retired = 4
Encoding Salary: <pre> * Salary; if Salary = 'Below RM1500' then encoded_salary = 1; else if Salary = 'RM1500 ~ RM3000' then encoded_salary = 2; else if Salary = 'RM3000 ~ RM6000' then encoded_salary = 3; else if Salary = 'RM6000 ~ RM10000' then encoded_salary = 4; else if Salary = 'Above RM10000' then encoded_salary = 5; </pre>	The If-Then-Else-If statement is used to encode the salary range where: Below RM1500 = 1 RM1500 ~ RM3000 = 2 RM3000 ~ RM6000 = 3 RM6000 ~ RM10000 = 4 Above RM10000 = 5
Encoding Size: <pre> *Size; if Size = '1 to 3' then encoded_size = 1; else if Size = '4 to 6' then encoded_size = 2; else if Size = '7 to 9' then encoded_size = 3; else if Size = '10 to 12' then encoded_size = 4; else if Size = 'Above 13' then encoded_size = 5; </pre>	The If-Then-Else-If statement is used to encode the preferred shoe size in UK size where: Size 1 to 3 = 1 Size 4 to 6 = 2 Size 7 to 9 = 3 Size 10 to 12 = 4

	Above 13 = 5
Encoding Cost: <pre> * Cost; if Cost = 'Below RM100' then encoded_cost = 1; else if Cost = 'RM101 ~ RM200' then encoded_cost = 2; else if Cost = 'RM201 ~ RM300' then encoded_cost = 3; else if Cost = 'RM301 ~ RM400' then encoded_cost = 4; else if Cost = 'RM401 ~ RM500' then encoded_cost = 5; else if Cost = 'Above RM500' then encoded_cost = 6; </pre>	The If-Then-Else-If statement is used to encode the preferable cost range for buying shoe where: Below RM100 = 1 RM101 ~ RM200 = 2 RM201 ~ RM300 = 3 RM301 ~ RM400 = 4 RM401 ~ RM500 = 5 Above RM500 = 6
Encoding Type: <pre> * Type; if Type = 'Casual shoes' then encoded_type = 1; else if Type = 'Formal shoes' then encoded_type = 2; else if Type = 'Sport shoes' then encoded_type = 3; else if Type = 'Sandals' then encoded_type = 4; else if Type = 'Boots' then encoded_type = 5; else if Type = 'Sneakers' then encoded_type = 6; </pre>	The If-Then-Else-If statement is used to encode the types of shoe where: Casual shoes = 1 Formal shoes = 2 Sport shoes = 3 Sandals = 4 Boots = 5 Sneakers = 6

Encoding Brand: <pre> * Brand; if Brand = 'Nike' then encoded_brand = 1; else if Brand = 'Adidas' then encoded_brand = 2; else if Brand = 'Puma' then encoded_brand = 3; else if Brand = 'Vans' then encoded_brand = 4; else if Brand = 'Skechers' then encoded_brand = 5; else if Brand = 'Bata' then encoded_brand = 6; else if Brand = 'New Balance' then encoded_brand = 7; else if Brand = 'Lining' then encoded_brand = 8; else if Brand = 'Asics' then encoded_brand = 9; else if Brand = 'Under Armour' then encoded_brand = 10; else if Brand = 'The Ballerina' then encoded_brand = 11; else if Brand = 'Converse' then encoded_brand = 12; else if Brand = 'Line Seven' then encoded_brand = 13; else if Brand = 'Cotton On' then encoded_brand = 14; else if Brand = 'Vincci' then encoded_brand = 15; else if Brand = '361°' then encoded_brand = 16; else if Brand = 'B.U.M' then encoded_brand = 17; </pre>	The If-Then-Else-If statement is used to encode the favourite brands where: Nike = 1 Adidas = 2 Puma = 3 Vans = 4 Skechers = 5 Bata = 6 New Balance = 7 Lining = 8 Asics = 9 Under Armour = 10 The Ballerina = 11 Converse = 12 Line Seven = 13 Cotton On = 14 Vincci = 15 361° = 16 B.U.M = 17
Encoding Factors:	The Index and the If-Then-Else-If

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```
*Factors;
if index(Factors, 'Cheap') > 0 then en_factors_cheap = 1;
else en_factors_cheap = 0;
if index(Factors, 'Frequent Discount') > 0 then en_factors_discount = 1;
else en_factors_discount = 0;
if index(Factors, 'Comfort') > 0 then en_factors_comfort = 1;
else en_factors_comfort = 0;
if index(Factors, 'Health') > 0 then en_factors_health = 1;
else en_factors_health = 0;
if index(Factors, 'Environmental Requirement') > 0 then en_factors_environment = 1;
else en_factors_environment = 0;
if index(Factors, 'Functionality') > 0 then en_factors_functionality = 1;
else en_factors_functionality = 0;
if index(Factors, 'Durability') > 0 then en_factors_durability = 1;
else en_factors_durability = 0;
if index(Factors, 'Loyal to Brand') > 0 then en_factors_brand = 1;
else en_factors_brand = 0;
if index(Factors, 'Social Influence') > 0 then en_factors_social = 1;
else en_factors_social = 0;
if index(Factors, 'Appearance') > 0 then en_factors_others = 1;
else if index(Factors, 'Nice design') > 0 then en_factors_others = 1;
else if index(Factors, 'Fashion and Suitable for Self') > 0 then en_factors_others = 1;
else if index(Factors, 'Nice looking') > 0 then en_factors_others = 1;
else if index(Factors, 'Stylish') > 0 then en_factors_others = 1;
else en_factors_others = 0;
```

statement is used to encode the factors that influence brand preference where:

If the 'Cheap' exists, then assigns en_factors_cheap = 1

If the 'Frequent Discount' exists, then assigns en_factors_discount = 1

If the 'Comfort' exists, then assign en_factors_comfort = 1

If the 'Health' exists, then assign en_factors_health = 1

If the 'Environment Requirement' exists, then assign en_factors_environment = 1

If the 'Functionality' exists, then assign en_factors_functionality = 1

If the 'Durability' exists, then assign en_factors_durability = 1

	If the 'Loyal to Brand' exists, then assign en_factors_ brand = 1 If the 'Social Influence' exists, then assign en_factors_ social = 1 If the 'Appearance', 'Nice design', 'Fashion and Suitable', 'Nice looking', 'Stylish' exist, then assign en_factors_ others = 1
Encoding Frequency: <pre>* Frequency; if Frequency = 'Once a few years' then encoded_freq = 1; else if Frequency = 'Once a year' then encoded_freq = 2; else if Frequency = 'Once a month' then encoded_freq = 3; else if Frequency = 'Once a week' then encoded_freq = 4;</pre>	The If-Then-Else-If statement is used to encode the frequency of shoe purchase where: Once a few years = 1 Once a year = 2 Once a month = 3 Once a week = 4
Encoding Reason: <pre>* Reason; if Reason = 'Shoe is damaged due to certain reasons' then encoded_reason = 1; else if Reason = 'Out of fashion' then encoded_reason = 2; else if Reason = 'Dislike the brand image' then encoded_reason = 3; else if Reason = 'Got discounts' then encoded_reason = 4; else if Reason = 'Interested to Purchase' then encoded_reason = 5; else if Reason = 'Occasions' then encoded_reason = 6; else encoded_reason = 7;</pre>	The If-Then-Else-If statement is used to encode the reason for buying a new shoe where: Shoe is damaged due to certain

	reasons = 1 Out of fashion = 2 Dislike the brand image = 3 Got discounts = 4 Interested to Purchase = 5 Occasions = 6 Others(else) = 7
Encoding Colour: <pre>* Colour; if Colour = 'Dark Theme' then encoded_colour = 1; else if Colour = 'Light Theme' then encoded_colour = 2;</pre>	The If-Then-Else-If statement is used to encode the preferred shoe colour where: Dark Theme = 1 Light Theme = 2
Encoding Material: <pre>* Material; if Material = 'Rubber' then encoded_material = 1; else if Material = 'Fabric' then encoded_material = 2; else if Material = 'Leather' then encoded_material = 3; else if Material = 'Textiles' then encoded_material = 4; else if Material = 'Cotton canvas' then encoded_material = 5; else encoded_material = 6;</pre>	The If-Then-Else-If statement is used to encode the preferred shoe material where: Rubber = 1 Fabric = 2 Leather = 3 Textiles = 4 Cotton canvas = 5

	Others(else) = 6
Encoding Advertisement: <pre> * Advertisement; if Advertisement = 'Social Media' then encoded_ad = 1; else if Advertisement = 'Advertisement' then encoded_ad = 2; else if Advertisement = 'Sponsors by Live Stream' then encoded_ad = 3; else if Advertisement = 'Celebrities' then encoded_ad = 4; else if Advertisement = 'Physical store' then encoded_ad = 5; else if Advertisement = 'Official WebSite' then encoded_ad = 6; else if Advertisement = 'Lazada' then encoded_ad = 7; </pre>	The If-Then-Else-If statement is used to encode the common sources of information regarding new arrivals and promotions where: Social Media = 1 Advertisement = 2 Sponsors by Live Stream = 3 Celebrities = 4 Physical store = 5 Official WebSite = 6 Lazada = 7
<pre> proc print data=encoded_data; run; </pre>	View the output of the encoded dataset.
Explanation of the code: An if-else statement is a conditional structure used to check conditions and control the flow of a program. The "if" checks a condition and runs its block if true. The "else if" checks another condition if the first is false. The "else" provides a fallback if none of the conditions are true. To encode the factors, which are multiple choice, we use the index() function to search for specific words or phrases within the Factors variable. Binary encoding is then applied	

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by assigning the value 1 to a corresponding variable (e.g., `en_factors_cheap`) if the keyword is found in the Factors string; otherwise, the value 0 is assigned.

4.0 Data analytics and visualization

4.1 Hypothesis : The latest discovery about the new footwear will affect the frequency of customer buying footwear

The purpose of this study is to find out if the latest discovery about the new footwear is influencing the frequency of customers buying footwear. The independent variable will be the new footwear discovery while the dependent variable is the frequency of customer purchases in footwear. We will use the Chi-square test for hypothesis testing.

4.2 Hypothesis Test

Null hypothesis(H_0) and alternative hypothesis(H_1) for the Chi-Square test :

H_0 = The new footwear discovery does not affect the frequency of customer purchases.

H_1 = The new footwear discovery does affect the frequency of customer purchases.

Chi-Square Test

Code :

```
* Chi-square Test;
proc freq data=encoded_data;
  tables encoded_ad * encoded_freq / chisq;
run;
```

Output :

The FREQ Procedure

Table of encoded_ad by encoded_freq						
encoded_ad	encoded_freq				Total	
	1	2	3	4		
1	42	46	7	0	95	65.07
	28.77	31.51	4.79	0.00		
	44.21	48.42	7.37	0.00		
	67.74	64.79	58.33	0.00		
2	13	20	3	1	37	25.34
	8.90	13.70	2.05	0.88		
	35.14	54.05	8.11	2.70		
	20.97	28.17	25.00	100.00		
3	0	1	2	0	3	2.05
	0.00	0.68	1.37	0.00		
	0.00	33.33	66.67	0.00		
	0.00	1.41	16.67	0.00		
4	3	0	0	0	3	2.05
	2.05	0.00	0.00	0.00		
	100.00	0.00	0.00	0.00		
	4.84	0.00	0.00	0.00		
5	3	3	0	0	6	4.11
	2.05	2.05	0.00	0.00		
	50.00	50.00	0.00	0.00		
	4.84	4.23	0.00	0.00		
6	0	1	0	0	1	0.68
	0.00	0.68	0.00	0.00		
	0.00	100.00	0.00	0.00		
	0.00	1.41	0.00	0.00		
7	1	0	0	0	1	0.68
	0.68	0.00	0.00	0.00		
	100.00	0.00	0.00	0.00		
	1.61	0.00	0.00	0.00		
Total	62	71	12	1	146	100.00
	42.47	48.63	8.22	0.68		

Statistics for Table of encoded_ad by encoded_freq			
Statistic	DF	Value	Prob
Chi-Square	18	24.6954	0.1335
Likelihood Ratio Chi-Square	18	20.4804	0.3064
Mantel-Haenszel Chi-Square	1	0.0739	0.7857
Phi Coefficient		0.4113	
Contingency Coefficient		0.3804	
Cramer's V		0.2374	
WARNING: 82% of the cells have expected counts less than 5. Chi-Square may not be a valid test.			
Sample Size = 146			

Table 4.1 : Chi-Square Table for Latest discovery and Frequency

After conducting the chi-square based on Table 4.1, the p-value that we obtained is 0.1335 which is greater than 0.05. With that being said, it failed to reject the null hypothesis for chi-square test and there is not sufficient evidence to conclude that the new footwear discovery significantly affects the frequency of customer purchases at the 5% significance level.

4.3 Data Visualization

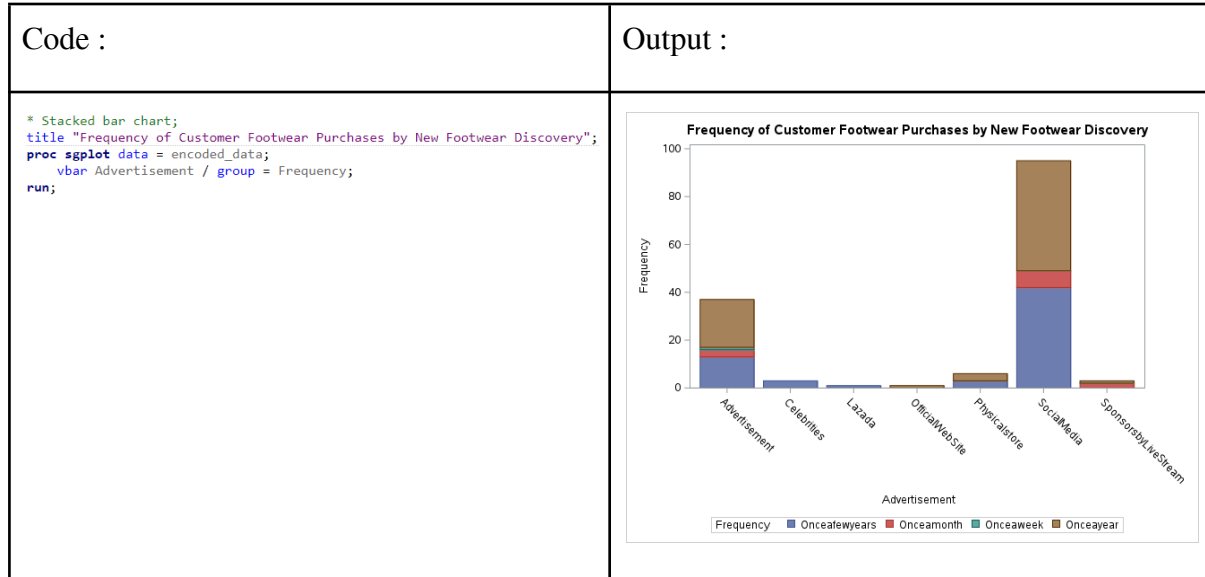


Figure 4.2.1 Frequency of Customer Footwear Purchases by New Footwear Discovery

Based on the Figure 4.2.1, the stacked bar chart shows that social media dominates the competition where most of the respondents, especially those who do not buy footwear frequently would take notice when discovering new footwear. On the other hand, advertisement, which is one of the most traditional methods for reaching customers and influencing their footwear purchases, is less dominant compared to social media. A new

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insight has been gained where social media plays the biggest role right after traditional advertising in introducing new footwear to customers and influencing them into purchasing these footwear.



Figure 4.2.2 Age and Occupation of Customer by New Footwear Discovery

Based on Figure 4.2.2, the stacked bar chart shows that the majority of the respondents who purchased footwear after hearing about the new discovery were in the younger age groups and most of them are students. We can ensure that social media is still the most effective channel for reaching the majority of customers. Majority of customers aged from 15 to 20 and 21 to 30 rely on social media as the dominant source of information to discover new footwear. Moreover, the majority of them are students or working adults and they heavily relied on social media as the first source of info in knowing the latest details about a new footwear. Overall, the impact of the new discovery seems to have been greater for younger customers who were more likely to purchase footwear after hearing about it.

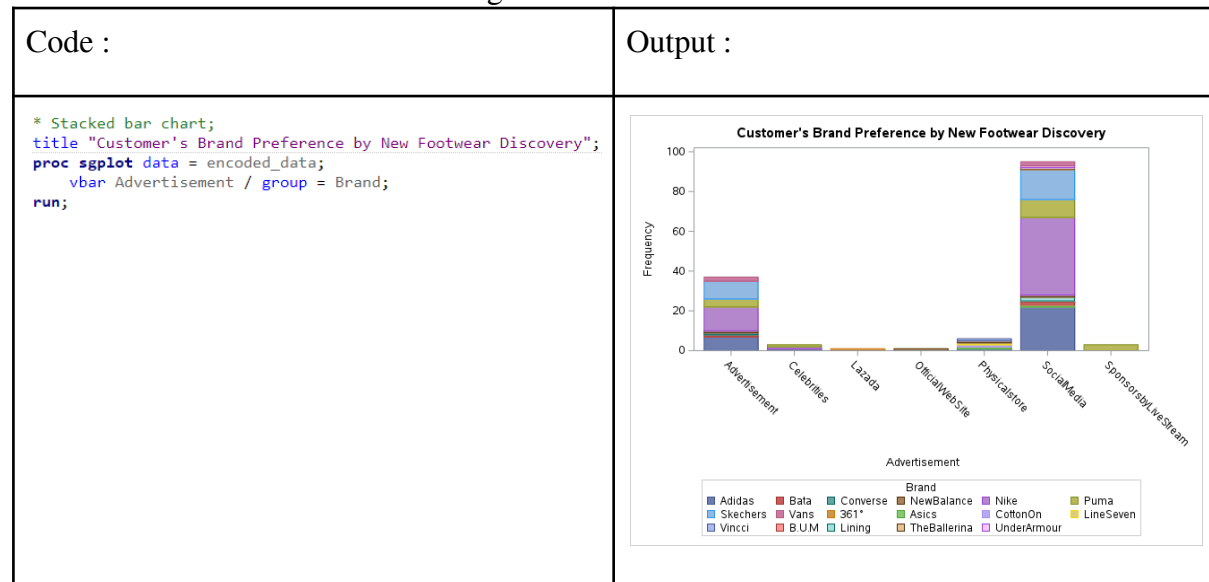


Figure 4.2.3 Customer's Brand Preference by New Footwear Discovery

Based on Figure 4.2.3, the brand Nike, Adidas and Skechers are the top 3 brands that fully utilise social media and advertisement to promote brand awareness. This goes to show how the top 3 brands still maintain their dominance in the market strongly correlated to their strategic emphasis on digital platforms where they would use it to engage directly with their target audience. With this strategy, other than allowing them to reach a global audience but also tailor their messages to resonate with various customer segments. This use of targeted advertising and social media engagement plays a vital role in establishing brand loyalty and driving repeat purchases.

5.0 Conclusion

Based on my hypothesis, the p-value obtained from the chi-square test is 0.1335 which is greater than 0.05. the null hypothesis is failed to be rejected hence there is no sufficient evidence to conclude that the new footwear discovery significantly affects the frequency of customer purchase at the 5% significance level.

Through the deep analysis that was conducted right after the chi-square test, the data from Figure 4.2.1, Figure 4.2.2 and Figure 4.2.3 clearly demonstrates the overwhelming impact of social media on consumer behavior and brand awareness in the footwear market, particularly among younger consumers. Social media has surpassed advertising in influence especially for infrequent buyers and those in the 15 to 30 age group who rely heavily on digital platforms to discover new footwear. Leading brands such as Nike, Adidas and Skechers have taken advantage of this by decisively using social media to maintain their market dominance, foster brand loyalty and also encourage repeat purchases. This emphasizes the importance of digital engagement in the future of footwear marketing.

6.0 Suggestion

The other brands can follow the top 3 brands footsteps by increasing social media engagement. They can leverage popular platforms that are famous and most recently used by consumers under the 15 to 30 age group such as TikTok and Instagram in order to engage directly with younger audiences. Other than picking the right platforms, the brands could strategize targeted advertising by tailoring their advertising strategies to specific customer segments. This would ensure the right message reaches the right audience at the right time to maximize the impact of advertising campaigns. In addition, brands should blend digital and traditional methods creatively to reach broader audiences while retaining the appeal to older demographics due to the fact that the traditional advertisements still have its value even if social media is dominant.

4.0 Data analytics and visualization

4.1 Hypothesis : The factors will affect the customer's footwear brand preference

In this study, we aim to test there is a significant association between the factors and customer's preference for footwear brands. The independent variable is factors while the dependent variable is customer's preference for footwear brands. In hypothesis testing, we are using the Chi-square test to test due to the reason that both variables are considered categorical variables.

4.2 Hypothesis Test

Null hypothesis(H_0) and alternative hypothesis(H_1) for the Chi-Square test :

H_0 = There is no significant association between factors and customer's footwear brand preference

H_1 = There is a significant association between factors and customer's footwear brand preference

The chi square test is to determine the p-value of the hypothesis. The reason that we test the factors individually is because each factor is encoded and since our factor question is multiple choice, it is difficult to perform a chi-square test for the whole factors.

Chi-square test

Code:

```
/* Chi-square test */
proc freq data=encoded_data;
  tables en_factors_cheap*encoded_brand / chisq;
  tables en_factors_discount*encoded_brand / chisq;
  tables en_factors_comfort*encoded_brand / chisq;
  tables en_factors_health*encoded_brand / chisq;
  tables en_factors_environment*encoded_brand / chisq;
  tables en_factors_functionality*encoded_brand / chisq;
  tables en_factors_durability*encoded_brand / chisq;
  tables en_factors_brand*encoded_brand / chisq;
  tables en_factors_social*encoded_brand / chisq;
  tables en_factors_others*encoded_brand / chisq;
  title 'Chi-square Test for Brand Preference and Each Factor';
run;
```

Output:

Frequency Percent Row Pct Col Pct	Table of en_factors_cheap by encoded_brand																	
	en_factors_cheap																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total
0	35	10	12	4	21	1	3	2	2	0	1	1	1	0	0	1	0	104
	24.85	13.01	12.22	2.74	14.35	0.68	2.05	1.37	1.37	0.00	0.68	0.68	0.68	0.00	0.00	0.68	0.00	71.23
	34.82	18.27	11.94	3.85	20.19	0.96	2.88	1.82	1.82	0.96	0.96	0.96	0.96	0.00	0.00	0.96	0.00	
	60.23	61.29	70.59	100.00	87.50	60.00	79.50	100.00	100.00	100.00	100.00	100.00	100.00	0.00	0.00	100.00	0.00	
1	10	12	5	0	3	1	1	0	0	0	0	0	1	1	0	1	0	42
	10.99	8.22	3.42	0.00	2.05	0.68	0.68	0.00	0.00	0.00	0.00	0.00	0.85	0.68	0.00	0.85	0.85	28.77
	38.10	28.67	11.60	0.00	7.14	2.38	2.38	0.00	0.00	0.00	0.00	0.00	2.38	2.38	0.00	2.38	2.38	
	30.77	38.71	28.41	0.00	12.50	50.00	25.00	0.00	0.00	0.00	0.00	0.00	100.00	100.00	0.00	100.00	100.00	
Total	45	22	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	145
	35.62	21.23	11.94	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of en_factors_cheap by encoded_brand

Statistic	DF	Value	Prob
Chi-Square	16	10.9185	0.2239
Likelihood Ratio Chi-Square	16	23.8883	0.0985
Mantel-Haenszel Chi-Square	1	0.0542	0.8140
Phi Coefficient		0.2694	
Contingency Coefficient		0.3485	
Cramer's V		0.3584	

WARNING: 79% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 145

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Frequency Percent Row Pct Col Pct	Table of en_factors_discount by encoded_brand																		
	en_factors_discount	encoded_brand																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total
0	36	20	9	2	16	2	3	2	1	1	1	1	1	1	0	0	1	0	67
	25.03	13.70	5.15	1.37	10.27	1.37	2.05	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.00	0.00	0.68	0.00	65.44
	39.16	20.92	9.26	2.08	15.44	2.06	3.00	2.06	1.03	1.03	1.03	1.03	1.03	1.03	0.00	0.00	1.03	0.00	102.50
	73.08	64.52	52.64	50.00	62.50	100.00	75.00	100.00	50.00	100.00	100.00	100.00	100.00	100.00	0.00	0.00	100.00	0.00	100.00
1	14	11	8	2	9	0	1	0	1	0	0	0	0	0	1	1	0	1	49
	6.59	7.83	4.84	1.37	6.16	0.00	0.68	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.68	0.68	0.00	0.68	33.56
	28.47	22.45	15.33	4.08	18.37	0.00	2.54	0.00	2.04	0.00	0.00	0.00	0.00	0.00	2.04	2.04	0.00	2.04	65.44
	25.02	25.48	47.05	50.00	37.50	0.00	25.00	0.00	50.00	0.00	0.00	0.00	0.00	0.00	100.00	100.00	0.00	100.00	100.00
Total	50	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	1	146
	35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of en_factors_discount by encoded_brand			
Statistic	DF	Value	Prob
Chi-Square	16	13.9738	0.6003
Likelihood Ratio Chi-Square	16	11.3378	0.2641
Mantel-Haenszel Chi-Square	1	0.5036	0.4752
Phi Coefficient		0.304	
Contingency Coefficient		0.265	
Cramer's V		0.304	

WARNING: 76% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

Frequency Percent Row Pct Col Pct	Table of en_factors_comfort by encoded_brand																		
	en_factors_comfort	encoded_brand																	Total
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
0	13	7	6	3	2	0	1	1	0	0	0	0	1	0	1	1	0	0	36
	8.90	4.79	4.11	2.05	1.37	0.00	0.68	0.68	0.00	0.00	0.00	0.00	0.68	0.00	0.68	0.68	0.00	0.00	24.06
	36.11	19.44	16.67	8.33	5.56	0.00	2.78	2.78	0.00	0.00	0.00	0.00	2.78	0.00	2.78	2.78	0.00	0.00	102.50
	25.00	22.66	25.29	75.00	8.33	0.00	25.00	50.00	0.00	0.00	0.00	100.00	0.00	100.00	100.00	100.00	0.00	0.00	100.00
1	36	24	11	1	22	2	3	1	2	1	1	0	1	0	0	0	1	1	110
	25.71	16.44	7.53	0.83	15.07	1.37	2.05	0.68	1.37	0.68	0.68	0.00	0.68	0.00	0.00	0.00	0.68	0.68	79.34
	35.45	21.62	10.50	0.91	20.00	1.62	2.73	0.91	6.25	9.09	9.09	0.00	9.09	0.00	0.00	0.00	9.09	9.09	110.00
	75.00	77.42	54.71	25.00	91.67	100.00	75.00	50.00	100.00	100.00	100.00	0.00	100.00	0.00	0.00	0.00	100.00	100.00	100.00
Total	50	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	1	146
	35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of en_factors_comfort by encoded_brand

Statistic	DF	Value	Prob
Chi-Square	16	22.8138	0.1188
Likelihood Ratio Chi-Square	16	23.8928	0.0921
Mantel-Haenszel Chi-Square	1	0.0591	0.8128
Phi Coefficient		0.3683	
Contingency Coefficient		0.3676	
Cramer's V		0.3683	

WARNING: 79% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

Frequency Percent Row Pct Col Pct	Table of en_factors_health by encoded_brand																		
	en_factors_health	encoded_brand																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total
0	48	29	15	3	17	2	4	2	2	1	0	1	1	1	1	1	1	1	129
	33.88	19.96	10.27	2.05	11.64	1.37	2.74	1.37	1.37	0.68	0.00	0.68	0.68	0.68	0.68	0.68	0.68	0.68	89.36
	37.21	22.48	11.63	2.33	13.18	1.55	3.10	1.55	1.55	0.78	0.00	0.78	0.78	0.78	0.78	0.78	0.78	0.78	116.44
	42.31	49.55	88.24	75.00	70.83	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
1	4	2	0	1	7	0	0	0	0	0	0	0	0	0	0	0	0	0	17
	2.74	1.37	1.37	0.68	4.79	0.00	0.00	0.00	0.00	0.00	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.00	11.64
	23.53	11.76	11.76	5.89	41.16	0.00	0.00	0.00	0.00	0.00	0.00	6.59	0.00	0.00	0.00	0.00	0.00	0.00	102.50
	7.69	6.45	11.76	25.00	29.17	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
Total	52	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	1	146
	35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of en_factors_health by encoded_brand

Statistic	DF	Value	Prob
Chi-Square	16	19.2052	0.2541
Likelihood Ratio Chi-Square	16	19.2287	0.4371
Mantel-Haenszel Chi-Square	1	0.4001	0.5271
Phi Coefficient		0.3635	
Contingency Coefficient		0.3419	
Cramer's V		0.3635	

WARNING: 85% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

Frequency Percent Row Pct Col Pct	Table of en_factors_environment by encoded_brand																		
	en_factors_environment	encoded_brand																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total
0	50	29	13	3	20	2	4	2	2	1	0	1	1	1	1	1	1	0	131
	34.06	16.89	8.90	2.00	13.70	1.37	2.74	1.37	1.37	0.68	0.00	0.68	0.68	0.68	0.68	0.68	0.68	0.00	98.73
	35.17	22.14	13.62	3.33	15.53	1.62	3.33	1.62	1.62	0.81	0.00	0.81	0.81	0.81	0.81	0.81	0.81	0.00	102.50
	65.15	65.99	75.47	75.00	63.33	100.00	100.00	100.00	100.00	100.00	0.00	100.00	100.00	100.00	100.00	100.00	100.00	0.00	100.00
1	2	3	4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	15
	1.37	1.37	2.74	0.68	2.74	0.00	0.00	0.00	0.00	0.00	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.68	10.27
	10.33	12.33	28.67	6.87	28.67	0.00	0.00	0.00	0.00	0.00	0.00	6.87	0.00	0.00	0.00	0.00	0.00	6.87	102.50
	3.88	6.45	23.53	25.00	16.67	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	100.00
Total	52	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	1	146
	35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of en_factors_environment by encoded_brand

Statistic	DF	Value	Prob
Chi-Square	16	27.3959	0.0376
Likelihood Ratio Chi-Square	16	20.2082	0.2110
Mantel-Haenszel Chi-Square	1	3.0764	0.0793
Phi Coefficient		0.4329	
Contingency Coefficient		0.3873	
Cramer's V		0.4329	

WARNING: 85% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

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Frequency Percent Row Pct Col Pct	Table of en_factors_functionality by encoded_brand																		
	en_factors_functionality	encoded_brand																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total
	0	36	22	11	3	12	3	0	0	0	0	0	1	0	0	1	1	1	84
		24.68	15.07	7.53	2.05	8.22	0.68	2.05	0.00	0.00	0.00	0.00	0.68	0.68	0.68	0.68	0.68	0.68	54.38
		38.35	23.40	11.70	3.19	12.77	1.08	3.19	0.00	0.00	0.00	0.00	1.08	1.08	1.08	1.08	1.08	1.08	84.38
		69.23	70.67	64.71	75.00	80.00	80.00	75.00	0.00	0.00	0.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	1	18	9	8	1	12	1	1	2	2	1	1	0	0	0	0	0	0	52
		10.95	9.16	4.11	0.68	8.22	0.68	0.68	1.37	1.37	0.68	0.68	0.00	0.00	0.00	0.00	0.00	0.00	35.62
		30.77	17.31	11.54	1.62	23.08	1.62	1.62	3.85	3.85	1.92	1.92	0.00	0.00	0.00	0.00	0.00	0.00	69.23
		30.77	29.03	35.29	25.00	50.00	50.00	25.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
	Total	52	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	146
		35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of en_factors_functionality by encoded_brand

Statistic	DF	Value	Prob
Chi-Square	16	18.0239	0.3225
Likelihood Ratio Chi-Square	16	21.4845	0.1808
Mantel-Haenszel Chi-Square	1	0.4272	0.5134
Phi Coefficient		0.3514	
Contingency Coefficient		0.3315	
Cramer's V		0.3514	

WARNING: 76% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

Frequency Percent Row Pct Col Pct	Table of en_factors_durability by encoded_brand																		
	en_factors_durability	encoded_brand																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total
	0	31	13	12	2	12	0	0	3	2	0	0	0	1	0	1	1	1	81
		21.23	8.90	8.22	1.37	8.22	0.00	0.00	2.05	1.37	0.68	0.00	0.00	0.68	0.00	0.68	0.68	0.68	55.48
		38.27	16.08	14.81	2.47	14.81	0.00	0.00	3.70	2.47	1.23	0.00	0.00	2.23	0.00	1.23	1.23	1.23	81.00
		59.82	41.64	70.59	50.00	50.00	0.00	0.00	75.00	100.00	50.00	0.00	0.00	100.00	0.00	100.00	100.00	100.00	100.00
	1	21	18	5	2	12	2	1	0	0	1	1	0	1	0	0	0	0	65
		14.38	12.33	3.42	1.37	8.22	1.37	0.68	0.00	0.68	0.68	0.68	0.00	0.68	0.00	0.00	0.00	0.00	44.52
		32.31	27.68	7.80	3.08	16.46	3.08	1.54	0.00	1.54	1.54	1.54	0.00	1.54	0.00	0.00	0.00	0.00	65.00
		40.38	35.08	29.41	50.00	50.00	100.00	25.00	0.00	50.00	100.00	100.00	100.00	0.00	100.00	0.00	0.00	0.00	100.00
	Total	52	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	146
		35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of en_factors_durability by encoded_brand

Statistic	DF	Value	Prob
Chi-Square	16	17.0824	0.3815
Likelihood Ratio Chi-Square	16	21.9401	0.1552
Mantel-Haenszel Chi-Square	1	3.4701	0.4629
Phi Coefficient		0.3419	
Contingency Coefficient		0.3235	
Cramer's V		0.3419	

WARNING: 76% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

Frequency Percent Row Pct Col Pct	Table of en_factors_brand by encoded_brand																		
	en_factors_brand	encoded_brand																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total
	0	41	27	14	3	16	2	3	2	2	1	1	1	1	1	1	1	1	120
		28.08	16.49	8.59	2.05	12.33	1.37	2.05	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	82.19
		34.17	22.50	11.67	2.50	15.00	1.87	2.50	1.87	1.87	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00
		78.86	87.10	82.35	75.00	75.00	100.00	75.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	1	11	4	3	1	6	0	1	0	0	0	0	0	0	0	0	0	0	26
		7.83	2.74	2.05	0.68	4.11	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.81
		42.31	15.38	11.54	3.85	23.08	0.00	3.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	65.00
		21.19	12.90	17.65	25.00	25.00	0.00	25.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
	Total	52	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	146
		35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of en_factors_brand by encoded_brand

Statistic	DF	Value	Prob
Chi-Square	16	5.0710	0.9954
Likelihood Ratio Chi-Square	16	7.4593	0.9634
Mantel-Haenszel Chi-Square	1	1.3510	0.2450
Phi Coefficient		0.1894	
Contingency Coefficient		0.1832	
Cramer's V		0.1894	

WARNING: 82% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

Frequency Percent Row Pct Col Pct	Table of en_factors_social by encoded_brand																		
	encoded_brand																		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total	
	0	43	27	14	4	23	2	4	2	2	1	1	1	1	1	1	1	1	134
		32.88	16.49	8.59	2.74	15.76	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	91.78
		35.62	20.15	10.45	2.69	17.16	1.46	2.69	1.46	1.46	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	100.00
		62.31	87.10	82.35	100.00	86.83	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	1	4	4	3	0	1	0	0	0	0	0	0	0	0	0	0	0	0	12
		2.74	2.74	2.05	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.22
		33.33	33.33	25.00	0.00	8.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	65.00
		7.89	12.90	17.65	0.00	4.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
	Total	52	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	146
		35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00
Statistics for Table of en_factors_social by encoded_brand																			
Statistic		DF	Value	Prob															
Chi-Square		16	5.4105	0.9933															
Likelihood Ratio Chi-Square		16	6.7511	0.9778															
Mantel-Haenszel Chi-Square		1	1.7001	0.1923															
Phi Coefficient			0.1929																
Contingency Coefficient			0.1891																
Cramer's V			0.1929																
WARNING: 88% of the cells have expected counts less than 5. Chi-Square may not be a valid test.																			
Sample Size = 146																			

BMDS 2113-Data Visualization Assignment

Frequency Percent Row Pct Col Pct	Table of en_factors_others by encoded_brand																	
	en_factors_others	encoded_brand																
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
	0	80	31	17	4	24	2	4	2	2	1	1	0	1	1	1	0	1
		34.28	21.23	11.84	2.74	18.44	1.37	2.74	1.37	1.37	0.88	0.88	0.00	0.88	0.88	0.88	0.00	0.88
		35.21	21.83	11.87	2.82	18.80	1.41	2.82	1.41	1.41	0.70	0.70	0.00	0.70	0.70	0.70	0.00	0.70
		85.15	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	0.00	100.00	100.00	100.00	0.00	100.00
	1	2	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0
		1.37	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.88	0.00	0.00	0.00	0.88	0.00
		50.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.00	0.00	0.00	0.00	25.00	0.00
		3.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	100.00	0.00
	Total	82	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	145
		35.82	21.23	11.84	2.74	18.44	1.37	2.74	1.37	1.37	0.88	0.88	0.88	0.88	0.88	0.88	0.88	100.00

Statistics for Table of en_factors_others by encoded_brand			
Statistic	DF	Value	Prob
Chi-Square	16	73.8304	<.0001
Likelihood Ratio Chi-Square	16	19.7154	0.2334
Linear-by-Linear Chi-Square	1	7.2108	0.0072
Phi Coefficient		0.7111	
Contingency Coefficient		0.5795	
Cramer's V		0.7111	

WARNING: 88% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 145

Table 1 : Chi-square test for each factors and brand

Based on the analysis that show in Table 1, since the p-value of en_factors_cheap, en_factors_comfort, en_factors_health, en_factors_functionality, en_factors_durability, en_factors_brand and en_factors_social are greater than 0.05. So it fails to reject the null hypothesis which means that there is no significant association between factors and customer's footwear brand preference. While the p-value of en_factors_environment and en_factors_others are less than 0.05, the null hypothesis is rejected which means that there is a significant association between factors and customer's footwear preference. Overall, most of the factors on customer preference failed to reach the significance level. Therefore, we can conclude that the factors and brands have no significant association with each other.

BMDS 2113-Data Visualization Assignment

4.3 Data Visualization

Code :

```
proc sql;
  create table Total_factors as
  select
    sum(en_factors_cheap) as Total_Cheap,
    sum(en_factors_discount) as Total_Discount,
    sum(en_factors_comfort) as Total_Comfort,
    sum(en_factors_health) as Total_Health,
    sum(en_factors_environment) as Total_Environment,
    sum(en_factors_functionality) as Total_Functionality,
    sum(en_factors_durability) as Total_Durability,
    sum(en_factors_brand) as Total_Brand,
    sum(en_factors_social) as Total_Social,
    sum(en_factors_others) as Total_Others
  from most_factors;
quit;

proc print data=Total_factors;
run;

proc sgplot data=Total_factors_barchart;
  vbar Factor_Type / response=Total_Value datalabel;
  title "Total Number of Factors";
run;
```

Output :

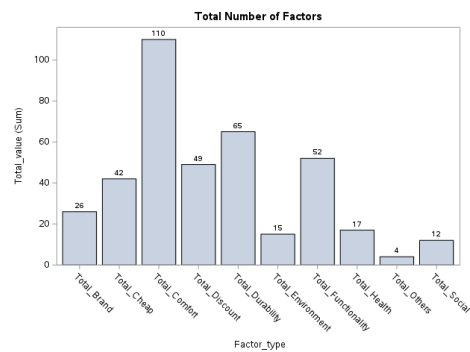


Figure 4.3.1: Total Number of Factors

```
data most_brand;
set encoded_data;

proc sql;
  create table Total_brand as
  select encoded_brand, count(*) as Total
  from most_brand
  group by encoded_brand;
quit;
run;

proc sgplot data=brand_counts;
  vbar encoded_brand / response=count datalabel;
  xaxis label='Encoded Brand';
  yaxis label='Total Count';
  title "Counts of Encoded Brands Selected with Comfort";
run;
```

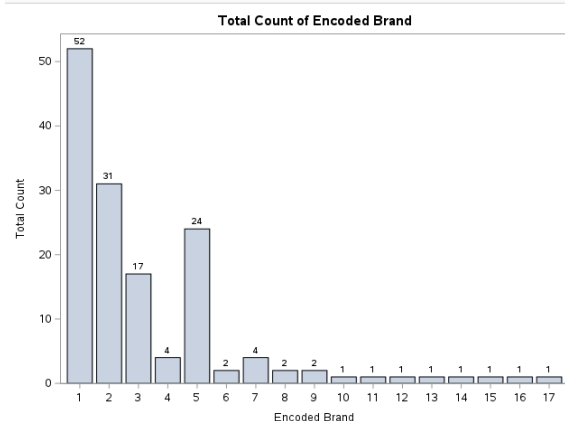


Figure 4.3.2 : Total Count of Encoded Brand

```
data ori_brand;
set encoded_data;

proc sql;
  create table Total_brandName as
  select Brand, count(*) as Total
  from most_brand
  group by Brand;
quit;
run;

proc sgplot data=original_brand_counts;
  vbar Brand / response=count datalabel;
  xaxis label='Brand';
  yaxis label='Total Count';
  title "Counts of Brands Selected with Comfort";
run;
```

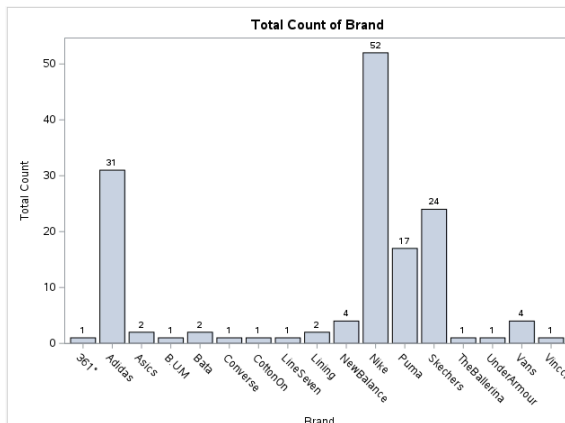


Figure 4.3.3 : Total Count of Brand

Explanation:

Based on the analysis in *Figure 4.3.1*, we can find that most of the factors that customers choose is comfort. At the same time, we found that most of the encoded brands that customer preference is 1 in *Figure 4.3.2*. All the encoded_brand that show in *Figure 4.3.3* include Nike[1], Adidas [2], Puma[3], Vans [4], Skechers [5], Bata [6], New Balance [7], Lining [8], Asics [9], Under Armour [10], The Ballerina [11], Converse [12], Line Seven[13], Cotton On [14], Vincci [15], 361°[16] and B.U.M [17]. Therefore through the *Figure 4.3.3*, we know that the most brand customer preference is Nike.

Code :

```
proc freq data=comfort_brands;
  tables encoded_brand / out=brand_counts;
  /*The out=brand_counts option creates an output dataset with the counts.*/
  title "Counts of Encoded Brands Selected with Comfort";
run;

proc sgplot data=brand_counts;
  vbar encoded_brand / response=count datalabel;
  xaxis label='Encoded Brand';
  yaxis label='Total Count';
  title "Counts of Encoded Brands Selected with Comfort";
run;
```

Output :

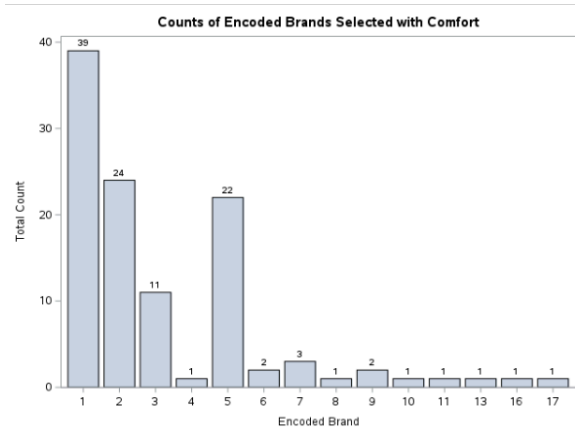


Figure 4.3.4 : Counts of Encoded Brand Selected with comfort

```
proc freq data=comfort_brands;
  tables Brand / out=original_brand_counts ;
  /*The out=brand_counts option creates an output dataset with the counts.*/
  title "Counts of Brands Selected with Comfort";
run;

proc sgplot data=original_brand_counts;
  vbar Brand / response=count datalabel;
  xaxis label='Brand';
  yaxis label='Total Count';
  title "Counts of Brands Selected with Comfort";
run;
```

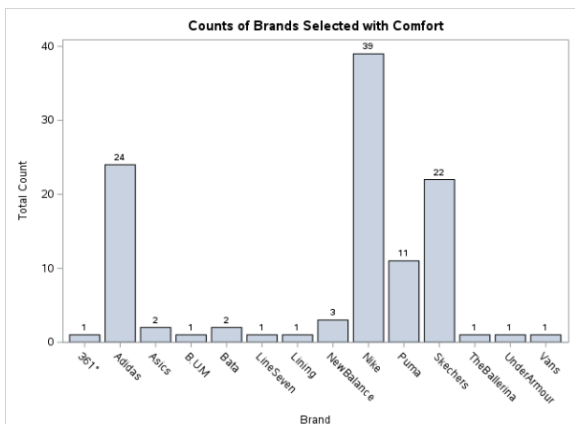


Figure 4.3.5 : Counts of Brand Selected with comfort

Explanation :

Based on the analysis in *Figure 4.3.1* and *Figure 4.3.2*, comfort is the primary consideration for most consumers when choosing footwear. At the same time, the Nike brand also shows a high preference among consumers. Based on the result, it has been prompted to conduct an in-depth analysis of this relationship between the two variables. So based on *Figure 4.3.4*, the result shows that the customer chooses the encoded_brand 1 as the factor of comfort is highest. Based on *Figure 4.3.5*, we know that the most customer preferred brand is Nike because of the comfort factor.

Code :

```
proc freq data=nike_comfort_data;
  tables encoded_type / out=type_counts;
  /*The out=brand_counts option creates an output dataset with the counts.*/
  title "Counts of Encoded Type Selected In Nike";
run;

proc sgplot data=type_counts;
  vbar encoded_type / response=count datalabel;
  xaxis label='Encoded Type';
  yaxis label='Total Count';
  title "Counts of Encoded Type Selected with Comfort in Nike";
run;
```

Output :

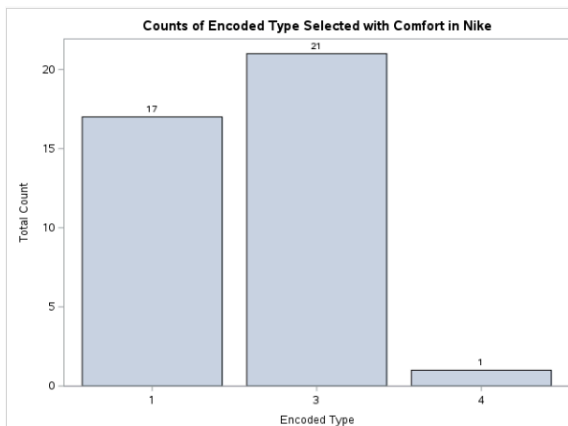


Figure 4.3.6 : Counts of Encoded Type Selected with Comfort in Nike

```
proc freq data=nike_comfort_data;
  tables Type / out=original_type_counts;
  /*The out=brand_counts option creates an output dataset with the counts.*/
  title "Counts of Type Selected In Nike";
run;

proc sgplot data=original_type_counts;
  vbar Type / response=count datalabel;
  xaxis label='Type';
  yaxis label='Total Count';
  title "Counts of Type Selected with Comfort in Nike";
run;
```

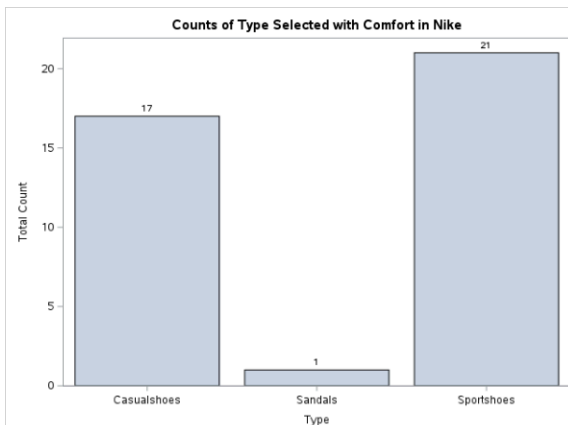


Figure 4.3.7 : Counts of Type Selected with Comfort in Nike

Explanation :

After we found that the customer chose the Nike brand the most because of the comfort factor, we conducted a deep analysis. Next we need to analyse which kind of Nike shoes customers prefer to buy. The analysis in *Figure 4.3.6* shows that there are 21 customers who prefer to buy the encoded_type 3, 17 customers prefer to buy encode_type 1 and only 1 customer prefers to buy encoded_type 4. The most customers prefer to buy is encoded_type 3 and the encoded_type 3 is the sports shoe that shows in *Figure 4.3.7*. It is concluded that most customers think that the Nike sport shoes are comfortable.

Code :

```
proc freq data=nike_comfort_sport_data;
  tables encoded_material / out=material_counts;
  title "Counts of Material Selected In Nike Sport";
run;

proc sgplot data=material_counts;
  vbar encoded_material / response=count datalabel;
  xaxis label='Encoded Material';
  yaxis label='Total Count';
  title "Counts of Encoded Material Selected In Nike Sport";
run;
```

Output :

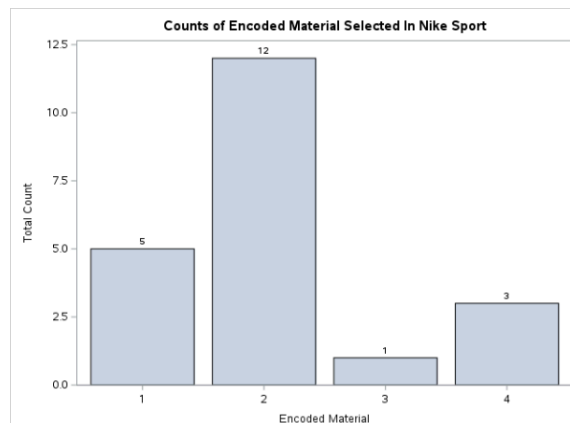


Figure 4.3.8 : Counts of Encoded Material Selected in Nike Sport

```
proc freq data=nike_comfort_sport_data;
  tables Material / out=ori_material_counts;
  title "Counts of Encoded Material Selected In Nike Sport";
run;

proc sgplot data=ori_material_counts;
  vbar Material / response=count datalabel;
  xaxis label='Material';
  yaxis label='Total Count';
  title "Counts of Material Selected In Nike Sport";
run;
```

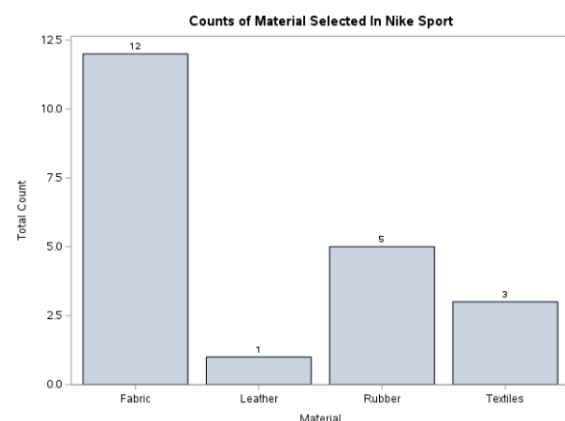


Figure 4.3.9 : Counts of Material Selected in Nike Sport

Explanation :

We found that the customer prefers to buy the comfortable sport shoes Nike. Next, we will analyze the preferences for different materials in depth to determine which material customers like best and conclude that the material is perceived as comfortable, so customers prefer to choose sport shoes using this material. In *Figure 4.3.8*, it shows that the encoded material 2 is the most preferred that have 12 customers choose, 5 customers choose encoded_material 1, 3 customers choose encoded_material 4 and only 1 customer choose encoded_material 3. Based on the *Figure 4.3.9* shows that the encoded_material of 2 is fabric, which means that most of the customers prefer to buy the fabric sport shoe in Nike due to the comfort.

Code :

```
proc freq data=nike_comfort_sport_fabric_data;
  tables encoded_colour / out=colour_counts;
  title "Counts of Encoded Colour Selected In Nike Sport Fabric";
run;

proc sgplot data=colour_counts;
  vbar encoded_colour / response=count datalabel;
  xaxis label='Encoded Colour';
  yaxis label='Total Count';
  title "Counts of Encoded Colour Selected In Nike Sport Fabric";
run;
```

Output :

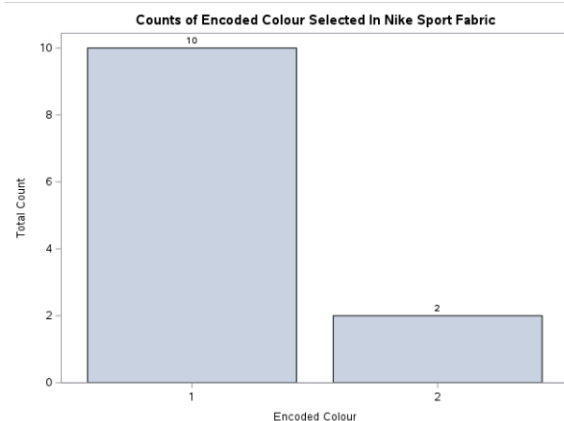


Figure 4.3.10 : Counts of Encoded Colour Selected in Nike Sport Fabric

```
proc freq data=nike_comfort_sport_fabric_data;
  tables Colour / out=or_colour_counts;
  title "Counts of Colour Selected In Nike Sport Fabric";
run;

proc sgplot data=or_colour_counts;
  vbar Colour / response=count datalabel;
  xaxis label='Colour';
  yaxis label='Total Count';
  title "Counts of Colour Selected In Nike Sport Fabric";
run;
```

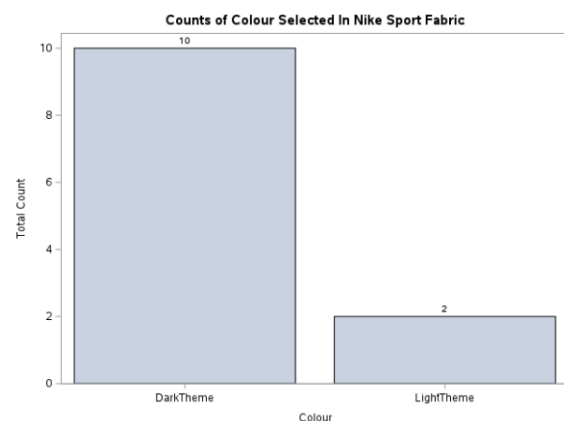


Figure 4.3.11 : Counts of Colour Selected in Nike Sport Fabric

Explanation :

In addition, after we found that most of the customers prefer Nike sport shoes, the material is fabric. We still continue deep analysis to find which theme is most popular with customers. Based on the *Figure 4.3.10*, we found that the highest encoded colour is 1 which has 10 customers selected and only 2 customers select the light theme. The theme corresponding to encoded colour 1 is the dark theme shown in *Figure 4.3.11*, which means that most of the customers prefer to buy the dark theme Nike sport shoes.

4.4 Techniques Used

The technique that we use to create all the bar charts to visualize the impact of different factors on customer preferences is using the proc sgplot. We use the vbar statement to display the total number of responses for each variable and add the datalabel to the bars for enhanced readability. The x-axis usually represents the different categorical variables, while the y-axis represents the total number of corresponding choices. These charts show the customer preferences in terms of brand, colour, material, etc, which helps to understand customer purchasing behaviour and trends.

5.0 Conclusion

Based on my hypothesis, analytics at the factors will affect the customer brand preference, since most of the chi-square test p-value is greater than 0.05, which means that the null hypothesis fails to be rejected. It considers as there is no significant association between factors and customer's footwear brand preference.

When we conducted the deep analysis, we found that the biggest factors and customer's brand preference is comfort and Nike. Our survey shows that 39 out of 110 customers choose Nike brand footwear mainly because of comfort. Based on Figure 4.3.6 and Figure 4.3.7, the most popular type of Nike footwear purchased by customers is sport shoes. The result shows that 21 out of 39 customers prefer sports shoes. It means that most of the customers think that the Nike sport shoes are comfortable. Next, we aim to find which material of Nike sport shoe most customers prefer to buy. The Figure 4.3.8 and Figure 4.3.9 shows that the material preference of Nike sport shoes that most people will prefer is 2 which is fabric material. Of the 21 customers who choose Nike sport shoes, 12 customers choose fabric. Therefore, it can be concluded that most customers think that fabric is more comfortable in Nike sport shoes. During our analysis, we also found that the dark theme is the colour that most people prefer in the Nike sport shoe fabric with 10 out of 12 customers selected. This may be related to the fact that dark theme sport shoes are less likely to get dirty and easier to maintain, so they are popular with customers.

Although the analysis failed to reject the null hypothesis, by doing deep analysis, we can conclude that the factor, comfort, is the most important factor that contributes to the fact that most people choose the Nike brand as their preferred brand. This analysis can provide a basis for future market strategies, and it is recommended that brands pay more attention to the use of comfort and material.

6.0 Suggestion

Most people prefer to buy Nike brand sport shoes because it is comfortable.

Suggestion : Most customers prefer the comfort of Nike sport shoes, especially those made of fabric. Therefore, other brands are suggested to put their focus more on sports shoes made of fabric as well. This is because comfort is an important factor in persuading customer purchases, indicating that customers generally think that fabric materials are more comfortable to wear. In addition, the analysis also shows that footwear with dark themes are particularly popular among Nike sport shoes. So that other brands can consider adding dark theme options in product design to meet the customer preference.

3. Ethel Ng Yi Yan

4.0 Data Analytics and Visualization

4.1 Hypothesis : Customer's salary will affect customer's footwear brand preference

Independent variable of the hypothesis is the “salary variable” which refers to the customer's salary while the dependent variable of the hypothesis is footwear brand preference which refers to the “brand” variable in the dataset

4.2 Hypothesis Test

Null Hypothesis (H0) and Alternative Hypothesis (H1) for Chi-Square Test

H0 : There is not a significant association among customer's footwear brand preferences and salary.

H1 :There is a significant association among customer's footwear brand preferences and salary.

Chi-Square Test

Code:

```
data assignmentEthel;
set encoded_data;

* Chi-Square Test;
proc freq data = assignmentEthel;
tables encoded_salary * encoded_brand /chisq;
run;
```

Output:

The FREQ Procedure

Table of encoded_salary by encoded_brand

Frequency Percent Row Pct Col Pct	encoded_salary	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total
1	36	18	11	0	13	2	3	2	1	0	0	0	0	0	1	0	1	1	91
24.66	12.33	7.63	1.37	8.96	1.37	2.05	1.37	0.68	0.00	0.00	0.00	0.00	0.00	0.68	0.00	0.68	0.68	0.68	62.33
30.06	13.70	12.59	2.20	14.29	2.20	3.30	2.20	1.10	0.00	0.00	0.00	0.00	0.00	1.10	0.00	1.10	1.10	1.10	91
69.23	58.06	64.71	50.00	54.17	100.00	75.00	100.00	50.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	100.00	0.00	100.00	100.00
2	6	3	1	1	4	0	0	0	0	1	1	1	1	0	0	0	0	0	22
6.16	2.95	0.68	0.68	2.74	0.00	0.00	0.00	0.00	0.68	0.68	0.68	0.68	0.68	0.00	0.00	0.00	0.00	0.00	15.07
40.91	13.64	4.55	4.55	18.18	0.00	0.00	0.00	0.00	4.55	4.55	4.55	4.55	4.55	0.00	0.00	0.00	0.00	0.00	13.01
17.31	5.68	5.68	25.00	16.67	0.00	0.00	0.00	0.00	100.00	100.00	100.00	100.00	100.00	0.00	0.00	0.00	0.00	0.00	2.00
3	5	6	1	0	6	0	0	0	1	0	0	0	0	0	0	0	0	0	19
3.42	4.11	0.68	0.00	4.11	0.00	0.00	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.01
26.32	31.68	5.36	0.00	31.68	0.00	0.00	0.00	5.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.01
9.62	19.35	5.88	0.00	25.00	0.00	0.00	0.00	50.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6
4	0	3	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	6
0.00	2.95	0.68	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.68	0.00	0.00	4.11
0.00	50.00	16.67	0.00	16.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.67	0.00	0.00	13.01
0.00	5.68	5.68	0.00	4.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	2.00
5	2	1	3	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	8
1.37	0.68	2.95	0.68	0.00	0.00	0.00	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.48
25.00	12.50	37.50	12.50	0.00	0.00	0.00	12.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.01
3.85	3.23	17.65	0.00	0.00	0.00	0.00	25.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00
Total	62	31	17	4	24	2	4	2	2	1	1	1	1	1	1	1	1	1	146
	35.62	21.23	11.64	2.74	16.44	1.37	2.74	1.37	1.37	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	100.00

Statistics for Table of encoded_salary by encoded_brand

Statistic	DF	Value	Prob
Chi-Square	64	79.0634	0.0973
Likelihood Ratio Chi-Square	64	57.0623	0.7182
Mantel-Haenszel Chi-Square	1	0.3188	0.5723
Phi Coefficient		0.7359	
Contingency Coefficient		0.5927	
Cramer's V		0.3679	

WARNING: 93% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

Table 4.1 : Chi-Square Table for Brand and Salary

Based on the chi-square that was done at Table 4.1, the p-value between encoded_salary and encoded_brand is 0.0973. As the p-value is greater than 0.05, it declared that it failed to reject the null hypothesis for chi-square test and there is not enough evidence to support

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that there is a significant association or relationship between customer's footwear brand preferences and salary. Therefore, we can say that the variable salary and brand does not have any dependency on each other. The salary range does not really influence the customer's footwear brand preference.

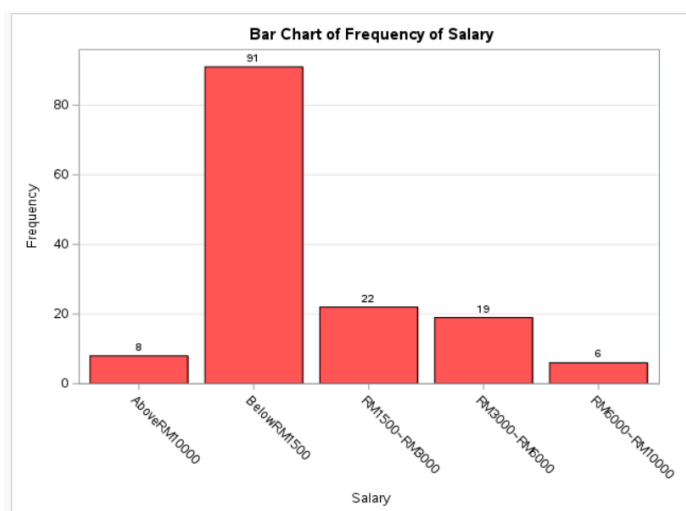
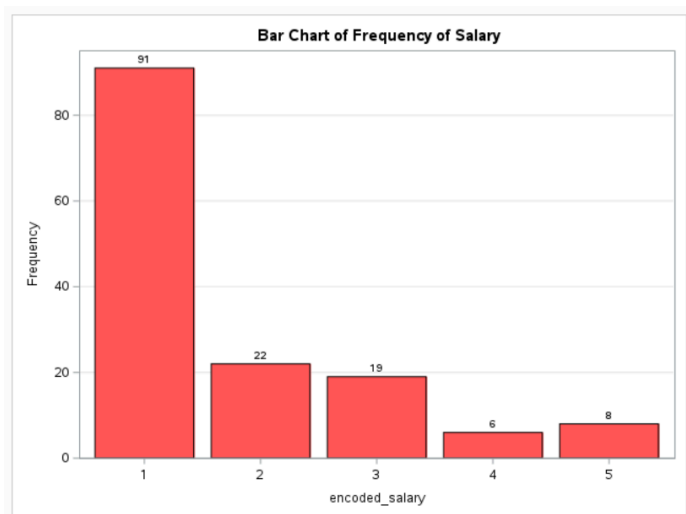
4.3 Data Visualisation

4.3.1 Bar Chart of Frequency of Salary

Code :

```
* plot graph;  
title " Bar Chart of Frequency of Salary";  
proc sgplot data = encoded_data;  
vbar salary / datalabel fillattrs=(color= lightred);  
yaxis grid;  
run;  
  
* plot graph;  
title " Bar Chart of Frequency of Salary";  
proc sgplot data = encoded_data;  
vbar encoded_salary / datalabel fillattrs=(color= lightred);  
yaxis grid;  
run;
```

Graph :



Explanation :

The bar chart above shows the frequency distribution of salary range among respondents. The x-axis of the graph represents the salary ranges while the y-axis represents the count of respondents with the respective salary ranges. As from the graph, we are able to identify that the

highest number of the samples collected from respondents have a salary or pocket money that is below RM1500, which contributes 91 pupils among 146 pupils. Besides that, there are 22 respondents who have a salary that is between RM1500 to RM 3000. Salaries ranging from RM3000-RM5000 also contribute a significant amount in the information collected from pupils which are 19 pupils among 146 pupils. Last but not least, there are 6 respondents among the 146 respondents who have a salary range of RM5000 to RM 10000 and 8 respondents that have a salary range that is above RM10000 which are considered high salary ranges. Therefore, after we visualise and analyze the bar chart, we can say that the respondents that have a salary range below Rm1500 will be the main focus for analysis.

The encoded_salary values at the graph above represents :

encoded_salary = 1 : Salary range below RM1500

encoded_salary = 2 : Salary range between RM1500 - RM3000

encoded_salary = 3 : Salary range between RM3000 - RM5000

encoded_salary = 4 : Salary range between RM5000 - RM10000

encoded_salary = 5 : Salary range above RM10000

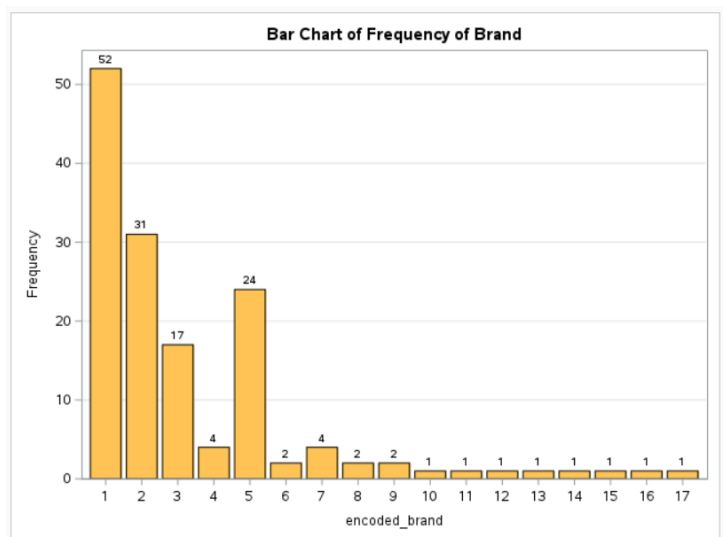
4.3.2 Bar Chart of Frequency of Brand

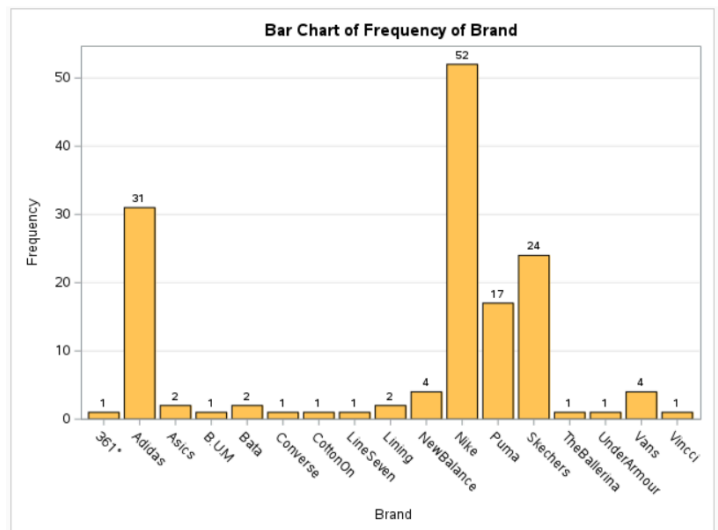
Code :

```
title "Bar Chart of Frequency of Brand";
proc sgplot data = encoded_data;
vbar brand / datalabel fillattrs=(color = lightorange);
yaxis grid;
run;

title "Bar Chart of Frequency of Brand";
proc sgplot data = encoded_data;
vbar encoded_brand / datalabel fillattrs=(color = lightorange);
yaxis grid;
run;
```

Graph :





Explanation :

Bar Chart above illustrates the frequency or count of each brand that is preferred by respondents. The x-axis of the graph represents the brand of the footwear while the y-axis represents the count of each brand that has been chosen by respondents. The brand, Nike has the highest frequency.

There are 52 respondents who have chosen Nike as their preference or favourite shoe brand, followed by the footwear brand Adidas and Skechers which consist of 31 frequency and 24 frequency. Moreover, Puma comes with 17 mentions which are placed at the fourth position.

Other brands such as New Balance and Vans show significant low frequency which only contributes 4 instances each. Lastly, several less frequent brands including Asics, Lining, Converse and others appear infrequently with count ranging 1 to 2 respondents for each brand. Therefore, we can conclude that the brands Nike, Adidas and Skechers are more favourable by the data that we have collected with the respondent.

The encoded_brand values at the graph above represents :

encoded_brand = 1 : Nike

encoded_brand = 2 : Adidas

encoded_brand = 3 : Puma

encoded_brand = 4 : Vans

encoded_brand = 5 : Skechers

encoded_brand = 6 : Bata

encoded_brand = 7 : New Balance

encoded_brand = 8 : Lining

encoded_brand = 9 : Asics

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encoded_brand = 10 : Under Armour

encoded_brand = 11 : The Ballerina

encoded_brand = 12 : Converse

encoded_brand = 13 : Line Seven

encoded_brand = 14 : Cotton On

encoded_brand = 15 : Vincci

encoded_brand = 16 : 361°

encoded_brand = 17 : B.U.M

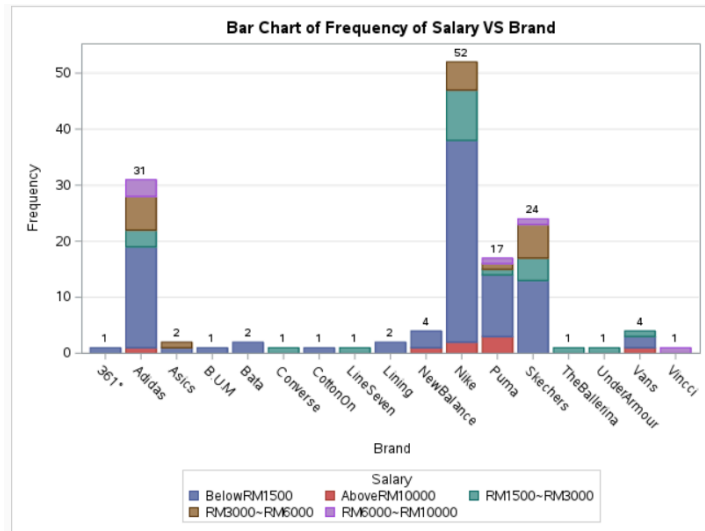
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4.3.3 Bar Chart of Frequency of Salary VS Brand

Code :

```
title "Bar Chart of Frequency of Salary VS Brand ";
proc sgplot data = encoded_data;
vbar brand / group = salary groupdisplay= stack datalabel;
yaxis grid;
run;
```

Graph :



Explanation :

The bar chart above illustrates the frequency of salary ranges across footwear's brand with the use of grouping method stack. The x-axis represents different footwear's brand while the y-axis represents the frequency of each shoe brand. Meanwhile, every bar in the bar chart has been divided into different segments while each segment represents different salary ranges. For instance, the blue segments represents salary ranges that is lower than RM1500, the green segments represents salary ranges of RM1500 - RM3000, the brown segments represents salary ranges of RM3000 - RM5000, the purple segments represents salary ranges of RM5000 - RM10000 and lastly the red segments represents the salary ranges which is above RM10000. As from the graph above, we can clearly identify that the brand Nike is most preferable by 52 respondents, followed by Adidas with 31 frequency and Skechers with 24 preferable respondents. Most respondents that have a salary range below RM1500 prefers the shoe brand Nike, which contributes a significant portion to its frequency followed by salary ranges Rm 1500 - RM 3000 compared to other brands, which means that respondents with a considered low range of salary prefer to buy Nike brand. Since the brand Nike holds a significant amount among different brands, therefore we are going to do deeper analysis in order to identify the factors that affect the popularity of a brand.

4.3.4 Bar Chart of Frequency of Nike Footwear Purchase by Salary Ranges

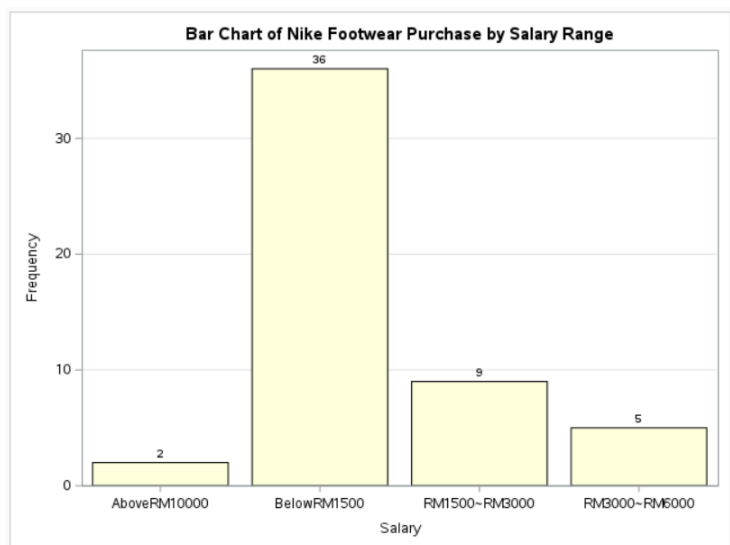
Code :

```
proc sql;
  create table brandbysalary as
  select *
  from assignmentEthel
  where encoded_brand = 1;
quit;

proc print data = brandbysalary;
run;

title "Bar Chart of Nike Footwear Purchase
by Salary Range";
proc sgplot data = brandbysalary;
vbar salary / datalabel fillattrs=(color = lightyellow);
yaxis grid;
run;
```

Graph :



Explanation :

The bar chart above shows the frequency of shoe brand Nike by salary range. The x-axis represents the salary ranges about respondents who preferred Nike brand and the y-axis represents the frequency of the respondents' salary ranges. As we can observe from the graph, the highest frequency of salary range is below RM1500 which consists of 36 respondents which Nike brand is their favourite shoe brand. Besides that, the second highest frequency of salary range with their favoured brand Nike are between RM1500 - RM3000 which consists of 9 mentions. This means that the brand Nike is particularly popular among lower-income earners. Besides that, there are only 5 respondents who earn the salary range among RM3000 to Rm 5000 preferred Nike brand which can be said less commonly preferred by respondents in this salary range. Lastly, there are only 2 respondents in the highest salary range whose salary range above Rm10000 purchase footwear with Nike brand, which shows that Nike brand can be said as not frequently chosen by high income earners according to the dataset that we have collected. We can conclude that lower-income individuals prioritize Nike for its brand appeal despite their financial constraints while higher income individuals might prefer other brands or luxury alternatives. Therefore, the salary ranges below Rm1500 which Nike prefers will be the next focus in the in-depth analysis.

Below RM1500.

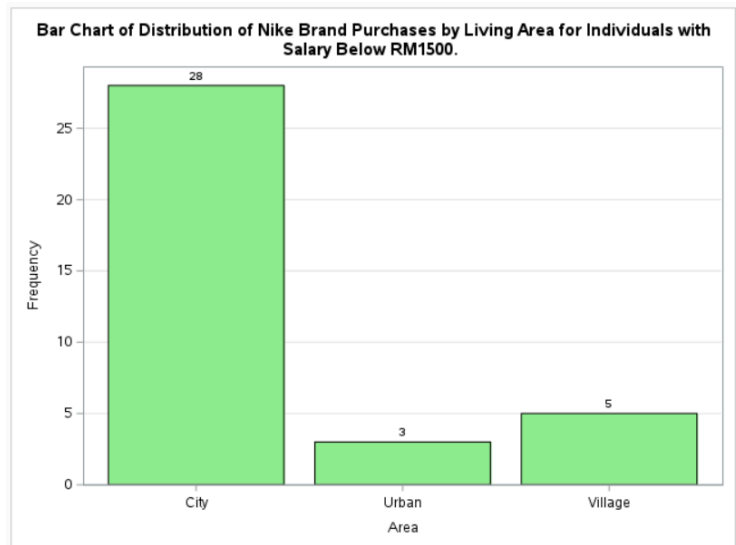
Code :

```
proc sql;
  create table brandbyarea as
  select *
  from assignmentEthel
  where encoded_brand = 1 and encoded_salary = 1;
quit;

proc print data = brandbyarea;
run;

title "Bar Chart of Distribution of Nike Brand Purchases by
Living Area for Individuals with Salary Below RM1500.";
proc sgplot data = brandbyarea;
  vbar Area / datalabel fillattrs=(color = lightgreen);
  yaxis grid;
run;
```

Graph :



Explanation :

The bar chart above shows the distribution of respondents which preferred Nike brand by living area with salaries below RM1500. The x-axis represents the living area for respondents which preferred Nike brand with salaries below RM1500 with three different components which are the city area, urban area and village area while the y-axis of the graph represents the frequency of living area. There are 28 respondents among 36 respondents who favoured Nike brand with low salary living in the city area. In addition, the frequency of respondents living in the village area drops to 5, suggesting that Nike brand footwear are less commonly purchased for respondents who live in the village area, followed by the respondents living in the urban area which only consists of 2 respondents. Therefore, in order to continue the in-depth analysis, we can say that city area will be the main objective for analysis to find the most popular footwear and we can conclude that most city area respondents with salary below RM1500 preferred Nike brand footwear compared to other respondents who live in the other living area with similar salary ranges.

with Salary below RM1500

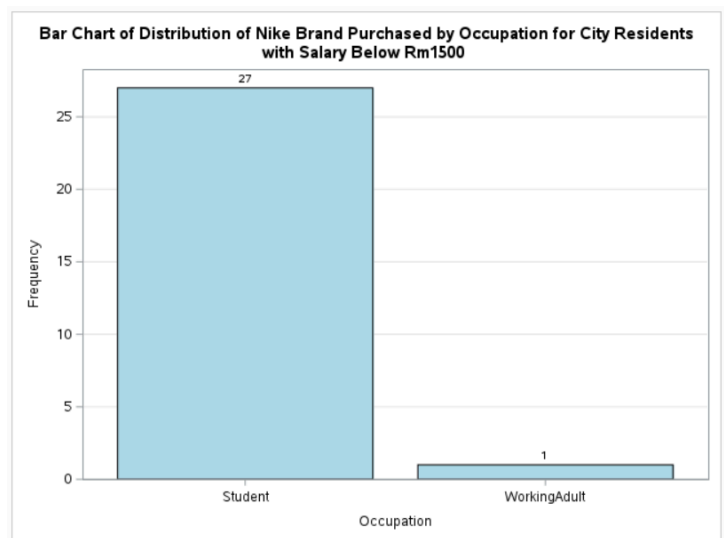
Code :

```
proc sql;
  create table brandbyocc as
  select *
  from assignmentEthe1
  where encoded_brand = 1 and encoded_salary = 1
  and encoded_area = 1;
quit;

proc print data = brandbyocc;
run;

title "Bar Chart of Distribution of Nike Brand
Purchased by Occupation for City Residents with
Salary Below Rm1500";
proc sgplot data = brandbyocc;
vbar Occupation / datalabel fillattrs=(color = lightblue);
yaxis grid;
run;
```

Graph :



Explanation :

The bar chart above illustrates the frequency distribution of respondents' occupation which preferred Nike brand with salaries range below Rm1500 and living in the city area. The x-axis represents the occupations of the respondents while the y-axis represents the frequency distribution of each of the occupations. For respondents who favoured the footwear brand Nike that is living in the city area with salaries range below Rm1500, there are only two types of occupations, which are students and working adults. For students who have a salary range below RM1500 can be their salary for a part time job and we also consider their pocket money as their salary, while for working adults that have a salary below Rm1500 range, the respondent might be a fresh graduate from university or a freelancer. As we can observe from the graph, 27 respondents who are living in the city area with salary range below RM1500 and preferred Nike brand are occupations as a student while only 1 respondent from the graph is a working adult. From the bar chart, we can conclude that the most common occupation among the respondents is student which will become one of the focus in the future analysis.

with Salary Below RM1500

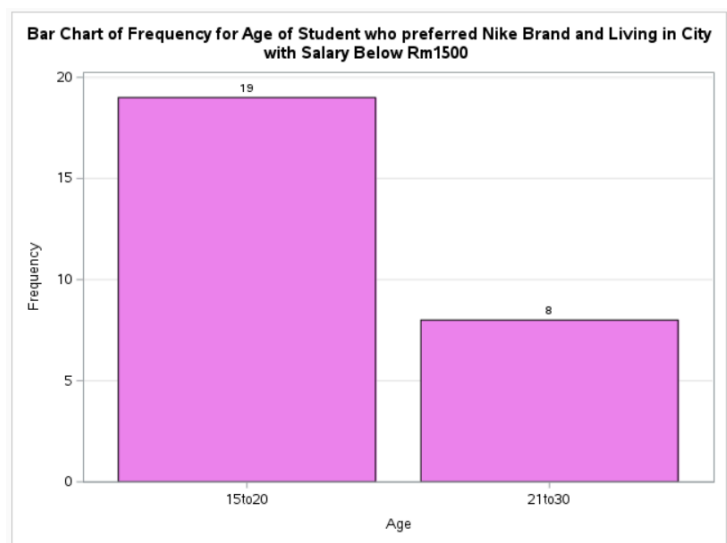
Code :

```
proc sql;
  create table brandbyage as
  select *
  from assignmentEthel
  where encoded_brand = 1 and encoded_salary = 1
  and encoded_area = 1
  and encoded_occ = 1;
quit;

proc print data = brandbyage;
run;

title "Bar Chart of Distribution of
Nike Brand Preference by
Age Among City Students
with Salary Below RM1500 ";
proc sgplot data = brandbyage;
vbar age / datalabel fillattrs=(color = violet);
yaxis grid;
run;
```

Graph :



Explanation :

The bar chart above displays the frequency distribution of age of students who preferred Nike brand footwear and living in the city area with a salary range below RM1500. The x-axis of the graph represents the age range of the students respondents who live in the city area, whose favoured brand is Nike and salary below RM1500. The age is segmented into two age groups which are age range 15-20 years old and age range 21-30 years old. The y-axis of the graph represents the frequency or count of respondents of each age range. As we are able to observe from the graph, respondents with an age range 15-20 years old hold significant components which is 19 respondents among the total number of 27 student respondents who preferred Nike brand with low salary and live in the city area. Besides that, student respondents who are aged between 21-30 consist of 8 frequencies. This can be said that the Nike brand is more popular among younger students and there can be said that is more likely to fit the economic demographic described. Therefore, students with the age group from 15 - 20 who live in the city area with below RM1500 salary ranges and desirable footwear brand, Nike will be the emphasis for detailed assessment.

4.3.8 Bar Chart of Nike Brand Preference by Gender Among City Students Aged 15-20

with salary below RM1500

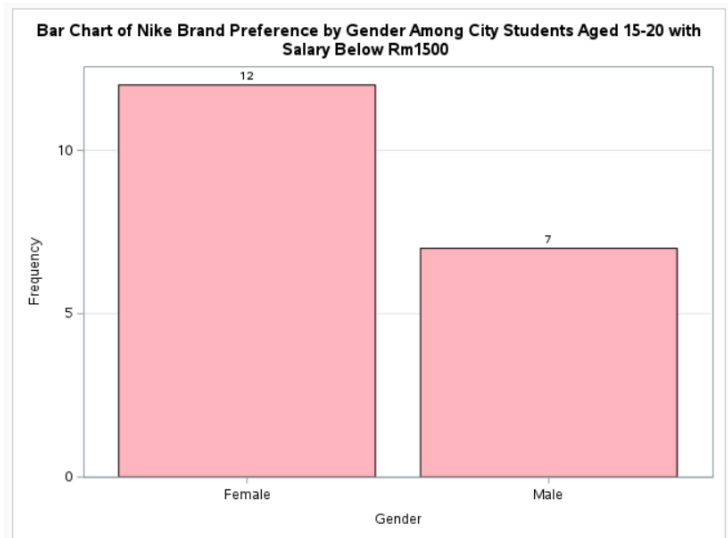
Code :

```
proc sql;
  create table brandbygender as
  select *
  from assignmentEthel
  where encoded_brand = 1 and encoded_salary = 1
  and encoded_area = 1
  and encoded_occ = 1
  and encoded_age = 1;
quit;

proc print data = brandbygender;
run;

title "Bar Chart of Nike Brand Preference by Gender
Among City Students Aged 15-20
with Salary Below Rm1500";
proc sgplot data = brandbygender;
vbar gender / datalabel fillattrs=(color = lightpink);
yaxis grid;
run;
```

Graph :



Explanation :

The bar chart above illustrates the frequency of gender of students aged 15-20 who live in the city area, preferred Nike brand footwear with salary ranges below Rm1500. The x-axis of the graph represents the gender of the respondents which consists of two segments, female and male while the y-axis represents the frequency of each gender who lived in the city area student respondent preferred Nike brand with salary range below Rm1500. From the observation from the barchart, there are 12 female students in the age group of 15 - 20 years old who prefer Nike , live in the city and earn less than Rm1500. Besides that, male students consist of 7 among the total number of 19 students with the same preferences and salary ranges. As the distribution of gender among the students aged 15-20 have are more numerous compared to the other distribution from the dataset that has been collected from respondents, therefore female student age 15 -20 who preferred Nike brand for their footwear preferences, living in the city with a salary range below RM1500 will be the highlights for in-depth evaluation.

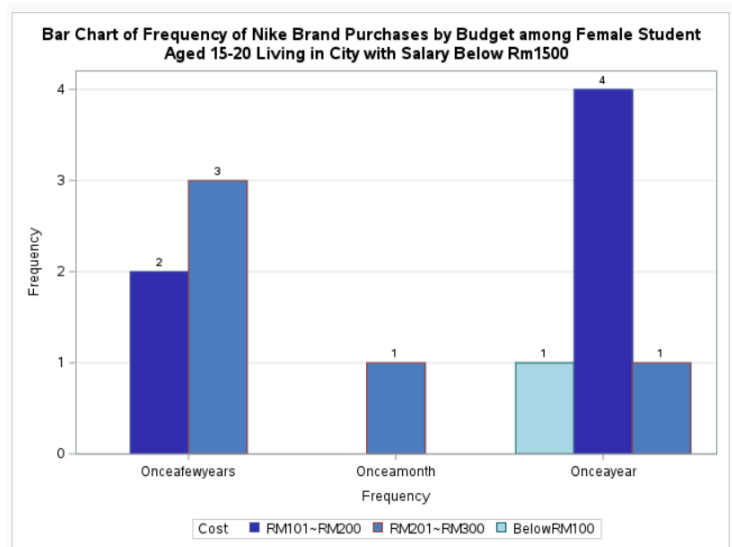
Code :

```
proc sql;
  create table brandbyfreqcost as
  select *
  from assignmentEthe1
  where encoded_brand = 1 and encoded_salary = 1
  and encoded_area = 1 and encoded_occ = 1
  and encoded_age = 1 and encoded_gender = 1;
quit;

proc print data = brandbyfreqcost;
run;

title "Bar Chart of Frequency of Nike Brand Purchases
by Budget among Female Student Aged 15-20 Living in City
with Salary Below Rm1500";
proc sgplot data = brandbyfreqcost;
  vbar frequency / group = cost groupdisplay=cluster datalabel ;
  styleattrs datacolors=(bib bigb lightblue);
  yaxis grid;
run;
```

Graph :



Explanation :

The bar chart above displays the cluster of buying frequency and budget of female students aged 15 - 20 who preferred Nike brand, living in the City area with salary range below Rm1500. The x-axis represents the purchase frequency of footwear which are categorized as once a month, once a year and once a few years. As the cluster bar chart is used to represent subcategory of the variables, therefore we have clustered the costs which also can be said as the budget of buying a new footwear, segmented into few segments which are footwear costs below Rm100, footwear costs range Rm101-Rm200 and footwear cost RM201 - Rm300. The y-axis represents the frequency of buying frequency and costs of Nike brand footwear of female students aged 15 -20 years old with salary range below Rm1500, living in the city area. Different color codes have been used in order to differentiate the cluster according to the costs. From the graph above, we are able to identify that there are most female students who usually purchase shoes at least once a year or once a few years. It is rare for female students to purchase footwear once a month as it only contributes 1 frequency among the total of 12 frequencies. As seen in the graph, most of the students purchase footwear which costs higher, costs range between RM201 - Rm300, especially when they are purchasing footwear once a few years. However, when female students' purchasing frequency is increased to once a year, they are more likely to buy cheap shoes which range from Rm101 - Rm200.

4.4 Techniques Used

The technique I have used is the “*proc sgplot*” which is used to plot a chart or graph for visualisation. The “*vbar*” statement is used to create a vertical bar chart followed by the variable at the x-axis to visualise the total number of respondents of each of the variables. The statement “*title*” is used to provide a title of the graph while the “*fillattrs=(colors=)*” is used to define the colours of the bar chart. I have also used the “*datalabels*” functions and “*y-axis grid*” functions in order to provide labels of each bar to show the frequency of counts as well as add grid lines to the y-axis for better readability. In certain bar charts, the “*group*” option is used to group bars with other variables while the “*groupdisplay*” function allows users to choose whether to present it as a stacked bar graph or a clustered bar graph. The “*styleattrs datacolors = ()*” is used to fill in the bar chart colour especially for a stacked bar chart or clustered bar chart.

For further analysis, I have also used the technique *proc sql* before plotting the bar chart in order to make a subset from the full dataset that has been collected for easier visualization and understanding. I have selected all types of variables using the function “*select*” with asterisks (*) from the dataset “*assignmentEthel*” table. In order to perform certain conditions, I have also used the “*where*” clause. For example, in order to find the salary ranges of respondents with preference Nike brand, therefore the conditions will be *encoded_brand = 1* , where *encoded_brand = 1* equals Nike brand.

5.0 Conclusion

Based on the hypothesis that has been created, analytics at the customer's salary will affect customer's footwear brand preference and returns the failure in rejection of null hypothesis as the chi-square test p-value is 0.0973 which is greater or larger than the p-value of 0.05. It concludes that there is no significant association or relationship between customer's salary and customer's footwear brand preference.

In the initial analysis, we discovered that the most common salary ranges and most preferable brands among the respondents are earning salary ranges below Rm1500. 91 among 146 of them fall within this salary range and 52 of 146 respondents preferred Nike footwear brand. For the second layer analysis, we have found out that there are only 36 respondents out of 52 respondents who preferred Nike brand footwear, which are the main components among all the salary ranges of respondents who favoured Nike brand footwear and it will be the focus of the third layer analysis. In the third layer analysis, we have analysis about the living area to determine whether the living area will affect the respondents options. From the analysis at Figure 4.2.5 above, we can conclude that most of the respondents who live in the city area with salary ranges below RM1500 will prefer to buy Nike branded footwear compared to other areas which stand 28 frequencies respectively in the total of 36 respondents. In the fourth layered analysis, we have done an analysis about the occupation of respondents who live in the city area with salary ranges below RM1500, preferred Nike brand. From the figure 4.2.6 above, we can conclude that almost every respondent is a student which stands 27 respondents over the total of 28 respondents and the other 1 respondents are working adults. This means that the main focus of the further analysis will be the students who live in the city area and have a salary range below RM1500, choosing Nike as their favourite footwear brand. In the fifth layer analysis, most of the students who preferred Nike branded falls in the aged group from 15 - 20 which means the Nike brand is more popular in the youngsters with the salary ranges below RM1500 and stays in the city area. From the sixth layered analysis, we can see that the frequency of female students is larger or greater than the male which we can conclude that the Nike brand is more welcomed in the female students grouped with salary ranges of RM1500 and living in the city area. Lastly, in the eighth layer of analysis, which is also the last layer of analysis, we have an analysis about the frequency of purchasing shoes and costs that a female student will spend when they have only a salary below Rm1500. From the figure 4.2.9 above, we can clearly determine that most of the female students will

decrease their budgets when they frequently purchase footwear and they are more likely to increase their budgets if they only purchase footwear once a few years.

Although the analysis failed to reject the null hypothesis, which means that the salary ranges and the brand preference does not really have a significant relationship between them. However, in my opinion, the salary ranges could still influence a customer's footwear brand preference. Higher-income individuals have more ability to purchase higher-priced footwear as they have greater spending power. On the other hand, individuals with lower incomes tend to prioritize affordability, which more often obtaining more budget friendly footwear. This difference in purchasing behaviour suggests that income levels can subtly shape brand preference, even if the data did not conclusive prove in this instance.

6.0 Suggestion

Most female students aged 15 - 20 years old who stay in the city area with salary ranges below RM1500 preferred to buy Nike brand footwear.

Suggestion : Most students, especially female students aged 15 to 20 years old living in city areas with salaries below RM1500 prefer Nike brand footwear due to the reasonable or affordable shoes price. As we can observe from Figure 4.2.9 the costs of a pair of footwear will affect the purchasing frequency. Therefore, other brands should focus on producing footwear with lower prices and better quality in order to attract more customers and increase sales. Besides that, other brands can focus on adjusting the footwear price or increase promotional activities in order to boost their sales revenue. This is because students, especially within the age of 15 - 20 years old, do not really have a high salary or pocket money which makes it difficult to purchase footwear with expensive prices. Therefore, by adjusting the prices of footwear, it can encourage customers to increase their purchasing power and purchasing frequency of footwear.

4. Gan Khai Li**4.0 Data analytics and visualization**

4.1 Hypothesis : Customer's justification to get a new footwear will affect the customer's footwear preference

Based on the hypothesis, we can identify the “Reason” variable in the dataset, which represents the customer's justification for purchasing new footwear, as the independent variable (IV). In contrast, the “Type” variable, which represents the customer’s footwear preference, serves as the dependent variable (DV).

4.2 Hypothesis Test

Null Hypothesis (H0) and Alternative Hypothesis (H1) for Chi-Square test

H0: There is no significant association between the customer's justification to get a new footwear and their footwear preference.

H1: There is a significant association between the customer's justification to get a new footwear and their footwear preference.

Chi-Square Test

Code:

```
data hypothesis_GKL;
set encoded_data;

*Chi-Square Test;
proc freq data=hypothesis_GKL;
    tables encoded_reason * encoded_type / chisq;
run;
```

Output:

Table of encoded_reason by encoded_type							
encoded_reason	encoded_type						Total
	1	2	3	4	5	6	
1	41 28.08 35.65 70.69	5 3.42 4.35 71.43	57 39.04 49.57 89.06	11 7.53 9.57 78.57	0 0.00 0.00 0.00	1 0.68 0.87 50.00	115 78.77
2	10 6.85 52.63 17.24	0 0.00 0.00 0.00	5 3.42 26.32 7.81	3 2.05 15.79 21.43	1 0.68 5.26 100.00	0 0.00 0.00 0.00	19 13.01
3	1 0.68 25.00 1.72	2 1.37 50.00 28.57	1 0.68 25.00 1.56	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	4 2.74
4	1 0.68 100.00 1.72	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	1 0.68
5	2 1.37 100.00 3.45	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	2 1.37
6	1 0.68 100.00 1.72	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	0 0.00 0.00 0.00	1 0.68
7	2 1.37 50.00 3.45	0 0.00 0.00 0.00	1 0.68 25.00 1.56	0 0.00 0.00 0.00	0 0.00 0.00 0.00	1 0.68 25.00 50.00	4 2.74
Total	58 39.73	7 4.79	64 43.84	14 9.59	1 0.68	2 1.37	146 100.00

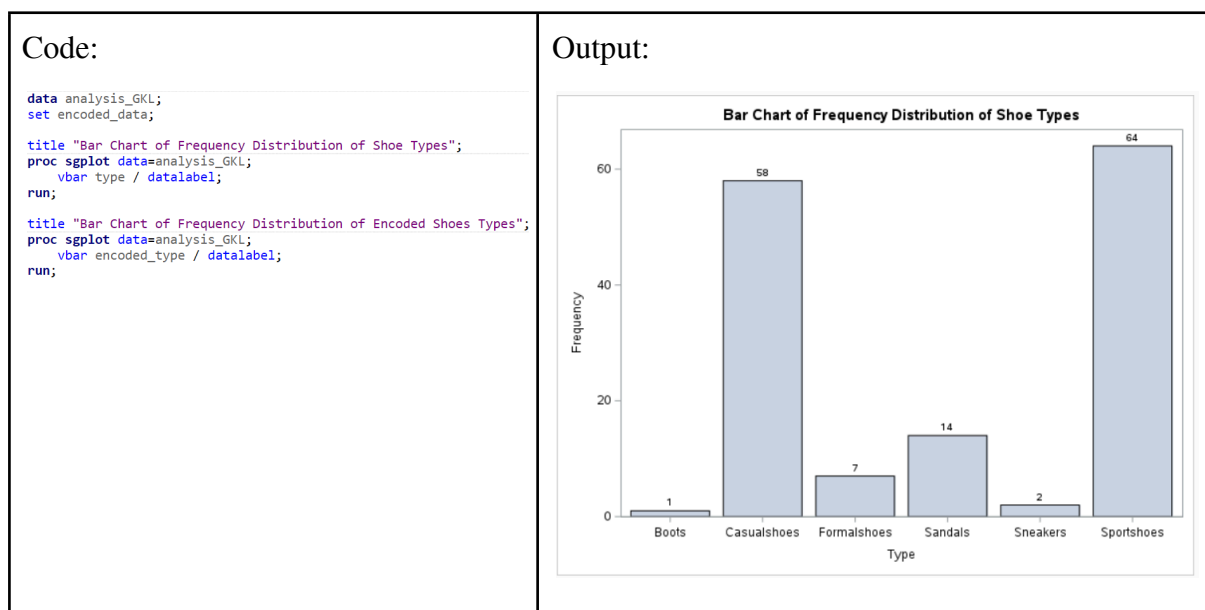
Statistics for Table of encoded_reason by encoded_type			
Statistic	DF	Value	Prob
Chi-Square	30	53.7250	0.0049
Likelihood Ratio Chi-Square	30	31.1038	0.4103
Mantel-Haenszel Chi-Square	1	0.7082	0.4001
Phi Coefficient		0.6066	
Contingency Coefficient		0.5186	
Cramer's V		0.2713	
WARNING: 86% of the cells have expected counts less than 5. Chi-Square may not be a valid test.			
Sample Size = 146			

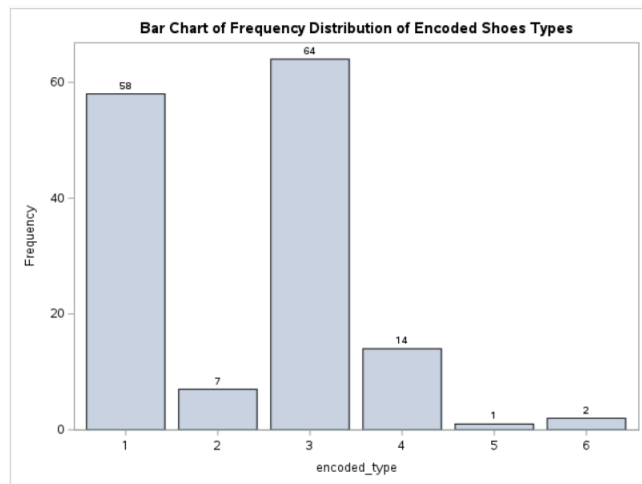
Table 4.1 : Chi-Square Table for Type and Reason

According to Table 4.1, the chi-square test is used to examine the hypothesis that a customer's justification to get a new footwear will affect the customer's footwear preference. The p-value between the encoded reason and encoded type, as shown in Table 4.1, is 0.0049. Since this p-value is less than 0.05, we reject the null hypothesis at a significance level of 0.05. This indicates that there is statistically significant evidence to suggest an association between the customer's justification for getting new footwear and their footwear preference.

4.3 Data Visualization

4.3.1 Bar Chart of Frequency Distribution of Shoe Types & Encoded Shoes Types



**Explanation:**

The bar chart presents the frequency distribution of various shoe types, showing the number of occurrences for each type in the dataset. Based on the bar chart, sports shoes, with 64 respondents, and casual shoes, with 58 respondents, have the highest frequencies, indicating that they are the most commonly represented types. Additionally, sandals with 14 respondents and formal shoes with 7 respondents have moderate representation. Other types, such as sneaker has 2 respondents and boots has 1 respondent are the least common shoe types in this dataset, indicating low popularity or limited availability. Through this bar chart, we can see that each respondent has a preferred type of shoe they would like to buy.

Based on the bar chart showing the frequency distribution of different shoe types, the encoding is as follows:

- 1 represents casual shoes
- 2 represents formal shoes
- 3 represents sports shoes
- 4 represents sandals
- 5 represents boots
- 6 represents sneakers

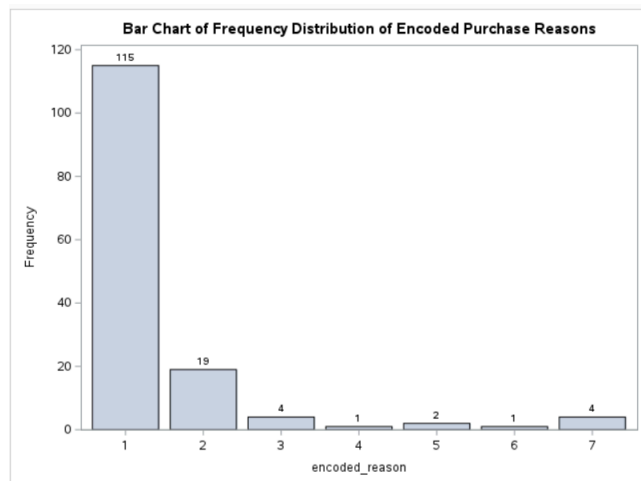
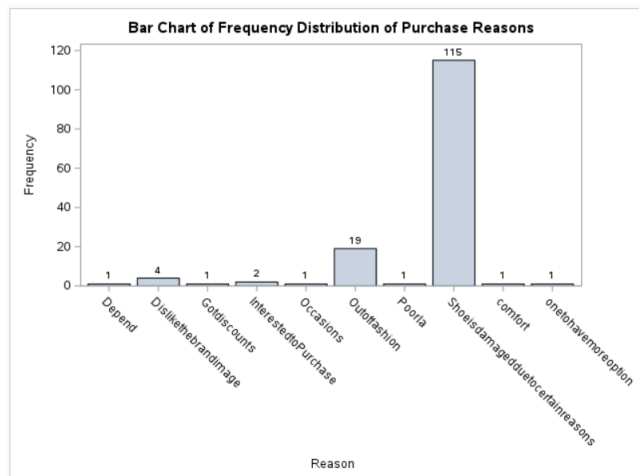
Code:

```
data analysis_GKL;
set encoded_data;

title "Bar Chart of Frequency Distribution of Purchase Reasons";
proc sgplot data=analysis_GKL;
vbar reason / datalabel;
run;

title "Bar Chart of Frequency Distribution of Encoded Purchase Reasons";
proc sgplot data=analysis_GKL;
vbar encoded_reason / datalabel;
run;
```

Output:



Explanation:

The bar chart presents the frequency distribution of purchase reasons, showing the number of occurrences for each reason in the dataset. From the chart, it is evident that most people make a purchase because “Shoe is damaged due to certain reasons,” with 115 respondents supporting this reason. The second most common reason is that the shoes are “Out of fashion,” with 19 respondents. A few respondents cited “Interested to purchase,” with 2 respondents, and “Dislike the brand image” with 4 respondents. The remaining reasons, each supported by 1 respondent, include “Depend”, “Got discounts”, “Occasions”, “Poorla”, “Comfort” and “One to have more options”. From this bar chart, we can know that each respondent has their own reasons for buying shoes.

Based on the bar chart showing the frequency distribution of different purchase reason, the encoding is as follows:

- 1 for “Shoe is damaged due to certain reasons”
- 2 for “Out of fashion”
- 3 for “Dislike the brand image”
- 4 for “Got discounts”
- 5 for “Interested to Purchase”
- 6 for “Occasions”
- 7 for the combination of the “Depend”, “Poorla”, “Comfort” and “One to have more options”

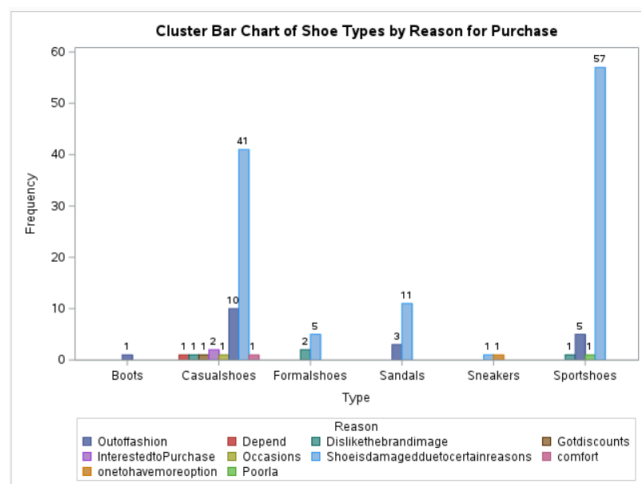
4.3.3 Cluster Bar Chart of Shoe Types by Reason for Purchase

Code:

```
data analysis_GKL;
set encoded_data;

title "Cluster Bar Chart of Shoe Types by Reason for Purchase";
proc sgplot data=analysis_GKL;
vbar type / group=reason groupdisplay=cluster
datalabel datalabelfitpolicy=none;
run;
```

Output:



Explanation:

The Clustered Bar Chart illustrates the comparison of shoe types against various reasons for purchase. The different types of shoes are displayed on the x-axis, and each bar in the clusters is colour-coded to represent specific reasons for purchase. The frequency of each reason for choosing the shoe type is shown at the top of each bar.

From the chart, we can observe that most people choose the type of shoe due to damage for various reasons. The types of shoes included in this category are sports shoes with 57 respondents, casual shoes with 41 respondents, formal shoes with 5 respondents, sandals with 11 respondents, and sneakers with 1 respondent. Additionally, being out of fashion is also a reason for purchasing new shoes, with 5 respondents for sports shoes, 10 for casual shoes, 3 for sandals, and 1 for boots. Furthermore, there are 2 respondents who purchased formal shoes, 1 respondent who purchased sports shoes, and 1 respondent who purchased

casual shoes because they dislike the brand image. Moreover, two respondents indicated that their interest in purchasing was the reason for buying casual shoes. Through this bar chart, we can see that each respondent has specific reasons for choosing the type of shoes they buy.

Based on the bar chart, we can see that sports shoes have the highest frequency among different types of footwear. Therefore, we will conduct a detailed analysis to understand the reason that sports shoes are the most popular choice among respondents.

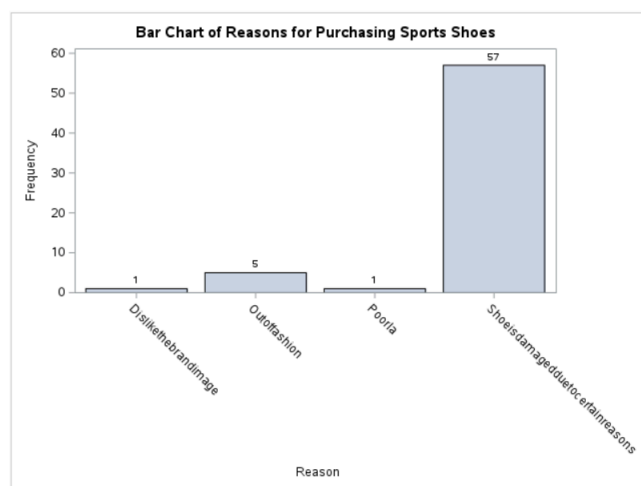
4.3.4 Bar Chart of Reasons for Purchasing Sports Shoes

Code:

```
*Step 1-Reason for buying a Sport Shoes;
data depth_analysis_GKL;
set encoded_data;
where Type= "Sport shoes";
run;

title "Bar Chart of Reasons for Purchasing Sports Shoes";
proc sgplot data=depth_analysis_GKL;
vbar reason / datalabel;
run;
```

Output:



Explanation:

This is the **first** layer of the in-depth analysis, focusing on:

1. “Sports shoes” as the selected type

The bar chart displays the frequency of the reasons for purchasing sports shoes along with their corresponding frequencies. The primary reason respondents purchase sports shoes is that the shoe is damaged due to certain reasons, with 57 respondents supporting this reason. The secondary reason is that the shoes are out of fashion, which 5 respondents identified. Other reasons include a dislike for the brand image and poor quality, each of which was supported by 1 respondent.

From the bar chart, the primary reason for buying the sports shoe is that the shoe is

damaged due to certain reasons, making it the next focus for in-depth analysis.

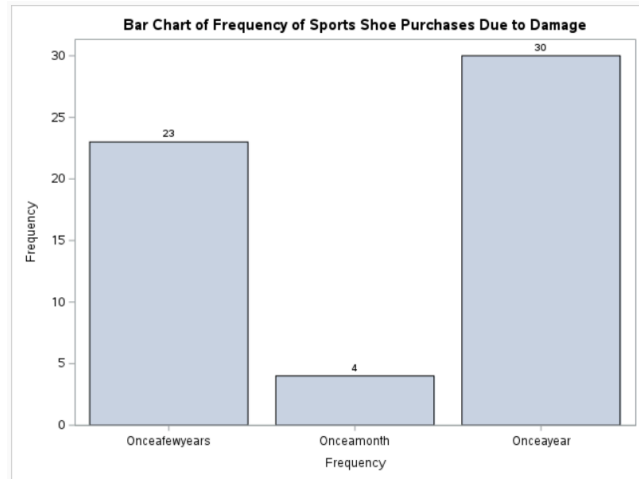
4.3.5 Bar Chart of Frequency of Sports Shoe Purchases Due to Damage

Code:

```
*Step 2-Frequency of shoe purchase;
proc sql;
create table depth_analysis_GKL as
select *
from encoded_data
where type = "Sport shoes"
and reason = "Shoe is damaged due to certain reasons";
;
quit;

title "Bar Chart of Frequency of Sports Shoe Purchases Due to Damage";
proc sgplot data=depth_analysis_GKL;
vbar frequency / datalabel;
run;
```

Output:



Explanation:

This is the **second** layer of the in-depth analysis, focusing on:

1. “Sports shoes” as the selected type
2. “Shoe is damaged due to certain reasons” as the chosen reason for purchasing

The bar chart illustrates the frequency of sports shoes purchases due to damage. According to the bar chart, 30 respondents purchase shoes once a year, while 23 respondents buy shoes every few years. Additionally, 4 respondents buy shoes once a month.

From the bar chart, the most common frequency of respondents purchasing sports shoes is once a year. This condition will become the next focus of the in-depth analysis.

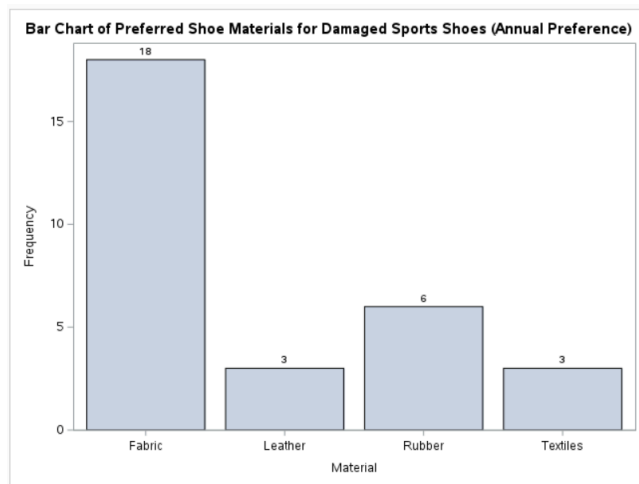
4.3.6 Bar Chart of Preferred Shoe Materials for Damaged Sports Shoes (Annual Preference)

Code:

```
*Step 3-Preferred shoe material ;
proc sql;
create table depth_analysis_GKL as
select *
from encoded_data
where type = "Sport shoes"
and reason = "Shoe is damaged due to certain reasons"
and Frequency = 'Once a year'
;
quit;

title "Bar Chart of Preferred Shoe Materials for
Damaged Sports Shoes (Annual Preference)";
proc sgplot data=depth_analysis_GKL;
vbar material / datalabel;
run;
```

Output:



Explanation:

This is the **third** layer of the in-depth analysis, focusing on:

1. “Sports shoes” as the selected type
2. “Shoe is damaged due to certain reasons” as the chosen reason for purchasing
3. “Once a year” as the frequency of shoe purchases

The bar chart illustrates the frequency of preferred shoe materials for damaged sports shoes (annual preference). Based on the bar chart, 18 respondents choose fabric as their preferred shoe material, while 6 respondents select rubber. Leather and textiles are each chosen by 3 respondents.

From the bar chart, the most common preferred shoe materials among respondents is fabric. This condition will become the next focus of the in-depth analysis.

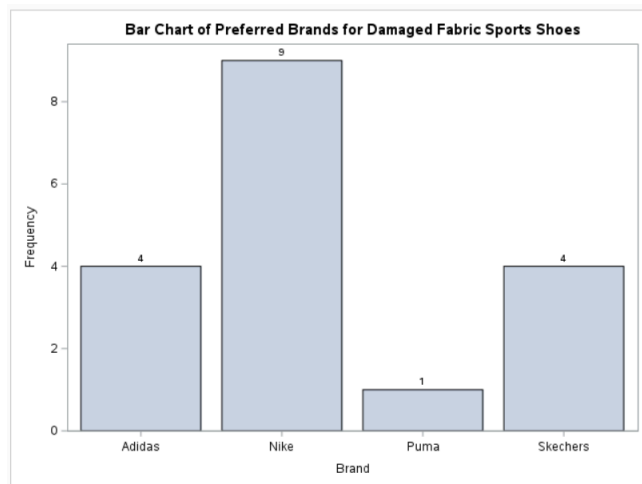
4.3.7 Bar Chart of Preferred Brands for Damaged Fabric Sports Shoes

Code:

```
*Step 4-Favourite brands;
proc sql;
create table depth_analysis_GKL as
select *
from encoded_data
where type = "Sport shoes"
and reason = "Shoe is damaged due to certain reasons"
and Frequency = 'Once a year'
and Material = 'Fabric'
;
quit;

title "Bar Chart of Preferred Brands for Damaged Fabric Sports Shoes";
proc sgplot data=depth_analysis_GKL;
vbar brand / datalabel;
run;
```

Output:



Explanation:

This is the **fourth** layer of the in-depth analysis, focusing on:

1. "Sports shoes" as the selected type
2. "Shoe is damaged due to certain reasons" as the chosen reason for purchasing
3. "Once a year" as the frequency of shoe purchases
4. "Fabric" as the selected preferred material

The bar chart illustrates the frequency of preferred brands for damaged fabric sports shoes. According to the bar chart, Nike is the most commonly preferred brand, with 9 respondents. Both Adidas and Skechers are supported by 4 respondents each, while Puma has 1 respondent purchasing fabric sports shoes.

From the bar chart, the most common preferred brand among respondents is Nike. This condition will become the next focus of the in-depth analysis.

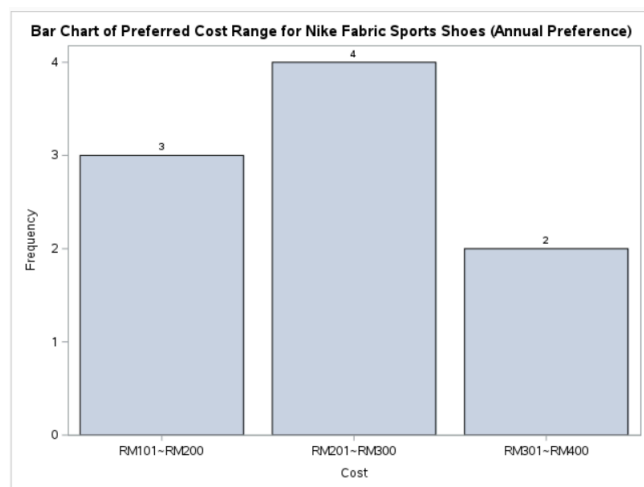
4.3.8 Bar Chart of Preferred Cost Range for Nike Fabric Sports Shoes (Annual Preference)

Code:

```
*Step 5-Preferable cost to buy shoe;
proc sql;
create table depth_analysis_GKL as
select *
from encoded_data
where type = "Sport shoes"
and reason = "Shoe is damaged due to certain reasons"
and Frequency = 'Once a year'
and Material = 'Fabric'
and Brand = 'Nike'
;
quit;

title "Bar Chart of Preferred Cost Range for Nike Fabric Sports Shoes (Annual Preference)";
proc sgplot data=depth_analysis_GKL;
vbar cost / datalabel;
run;
```

Output:



Explanation:

This is the **fifth** layer of the in-depth analysis, focusing on:

1. “Sports shoes” as the selected type
2. “Shoe is damaged due to certain reasons” as the chosen reason for purchasing
3. “Once a year” as the frequency of shoe purchases
4. “Fabric” as the selected preferred material
5. “Nike” as the selected favourite brand

The bar chart illustrates the frequency of the preferred cost range for Nike fabric sports shoes (Annual Preference). According to the bar chart, 4 respondents choose a preferred cost range of between RM201 and RM300 for their purchases. The preferred cost range of between RM101 and RM200 has 3 respondents supporting it. Between RM301 and RM400 is the preferred cost range for 2 respondents.

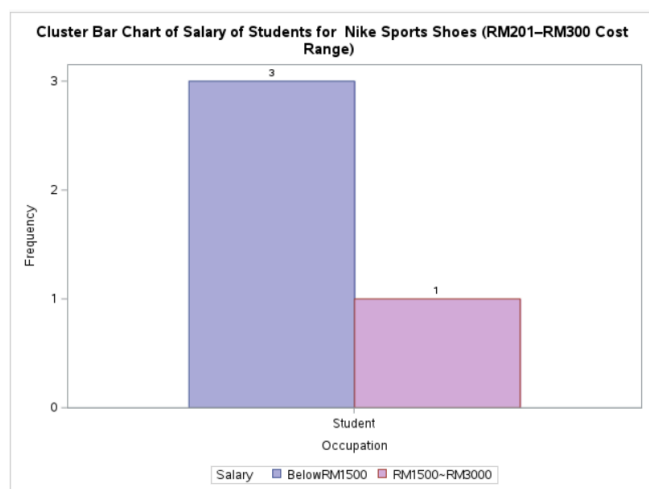
From the bar chart, the most common preferred cost range among respondents is between RM201 and RM300. This condition will become the next focus of the in-depth analysis.

Code:

```
*Step 6-Occupation;
proc sql;
create table depth_analysis_GKL as
select *
from encoded_data
where type = "Sport shoes"
and reason = "Shoe is damaged due to certain reasons"
and Frequency = 'Once a year'
and Material = 'Fabric'
and Brand = 'Nike'
and Cost = 'RM201 ~ RM300'
;
quit;

title "Cluster Bar Chart of Salary of Students for
Nike Sports Shoes (RM201-RM300 Cost Range)";
proc sgplot data=depth_analysis_GKL;
vbar occupation / group=salary groupdisplay=cluster
datalabel datalabelfitpolicy=none;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

This is the **sixth** layer of the in-depth analysis, focusing on:

1. “Sports shoes” as the selected type
2. “Shoe is damaged due to certain reasons” as the chosen reason for purchasing
3. “Once a year” as the frequency of shoe purchases
4. “Fabric” as the selected preferred material
5. “Nike” as the selected favourite brand
6. “RM201 - RM300” as the selected cost range

The cluster bar chart illustrates the salary of Students for Nike Sports Shoes (RM201-RM300 Cost Range). Based on the bar chart, the respondent’s occupations include students. There are 3 respondents with a salary range below RM1500, and 1 respondent falls within the salary range of between RM1500 and RM3000.

From the bar chart, the most common preferred salary range among respondents is below RM1500 and the predominant occupation is student. This condition will become the next focus of the in-depth analysis.

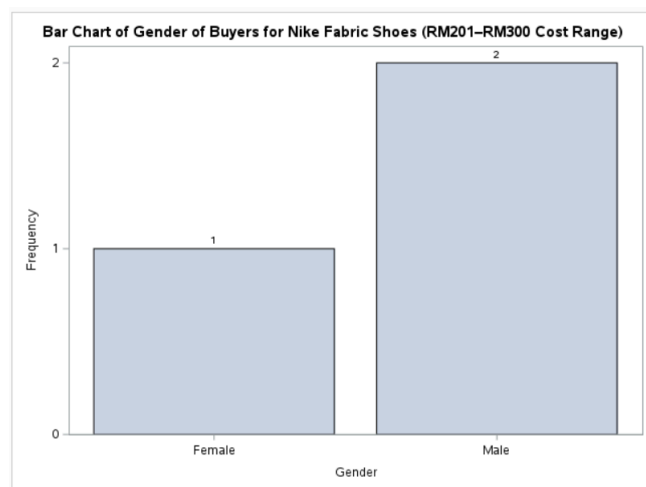
4.3.10 Bar Chart of Gender of Buyers for Nike Fabric Shoes (RM201-RM300 Cost Range)

Code:

```
*Step 7-Gender;
proc sql;
create table depth_analysis_GKL as
select *
from encoded_data
where type = "Sport shoes"
and reason = "Shoe is damaged due to certain reasons"
and Frequency = 'Once a year'
and Material = 'Fabric'
and Brand = 'Nike'
and Cost = 'RM201 ~ RM300'
and Salary = 'Below RM1500'
and Occupation = 'Student'
;
quit;

title "Bar Chart of Gender of Buyers for Nike Fabric
Shoes (RM201-RM300 Cost Range)";
proc sgplot data=depth_analysis_GKL;
vbar gender / datalabel;
run;
```

Output:



Explanation:

This is the **seventh(last)** layer of the in-depth analysis, focusing on:

1. “Sports shoes” as the selected type
2. “Shoe is damaged due to certain reasons” as the chosen reason for purchasing
3. “Once a year” as the frequency of shoe purchases
4. “Fabric” as the selected preferred material
5. “Nike” as the selected favourite brand
6. “RM201 - RM300” as the selected cost range
7. “Below RM1500” as selected salary range and “Student” as the selected occupation

The bar chart illustrates the gender of buyers for Nike fabric shoes (RM201-RM300 Cost Range). Based on the bar chart, there are 2 male respondents who show the highest support for purchasing sports shoes, highlighting the selected condition. Additionally, there is 1 female respondent who also purchased sports shoes, further emphasizing the selected condition.

According to the bar chart, the most common group among those who purchase sports shoes is male. This concludes the in-depth analysis.

4.4 Technique Used

The data visualisation technique I used to represent the data is a bar chart. I used the title statement to add a title to the chart. Additionally, I used the proc “sgplot” procedure to create the bar chart and the “vbar” statement to generate a vertical bar chart for the specified variable. The “group” option is used to group the bars by another variable, making it easier to compare the categories, and the groupdisplay=cluster option displays the grouped bars in a clustered format for clearer comparison. The datalabel option is used to add data labels on top of the bars, and datalabelfitpolicy=none ensures that the labels are not adjusted for fit, making sure all data labels are visible. “styleattrs datacolors=(vpab vpap)” is used to specify the colours of the bars, where vpab stands for ‘very pale blue’ and vpap stands for ‘very pale purple’.

To perform a depth analysis, I used proc sql to start an SQL procedure in SAS. As an example of using the create table statement in SQL, I applied it to create a table named depth_analysis_GKL in the SAS environment. The select * statement is used to select all columns from the encoded_data table, while the “from encoded_data” clause specifies the source table from which the data is being retrieved. I used the WHERE clause to filter the rows, and the AND statement allows for continuous filtering based on multiple conditions. The “quit;” command is used to terminate the proc sql procedure, indicating the end of the SQL code execution.

5.0 Conclusion

Based on hypothesis testing, the customer's justification to get a new footwear will affect the customer's footwear preference. The null hypothesis is rejected as the chi-square test p-value of 0.0049 is less than the significance level of 0.05. It is concluded that there is a significant association between the customer's justification for purchasing new footwear and their footwear preference.

Based on the analysis, we found that the most common shoe type is sports shoes, chosen by 64 respondents. We have selected this as the focus for the first layer of in-depth analysis. After conducting the first layer of analysis, the primary reason for purchasing sports shoes is that the shoe is damaged due to certain reasons, as indicated by 57 respondents. This finding will be further explored in the second layer of in-depth analysis, building on the initial focus. Moving on to the second layer of deep analysis, we discovered

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that the most common frequency of respondents purchasing shoes is once a year, as supported by 30 respondents. This finding will be the focus of the third layer of in-depth analysis, continuing from the initial focus. In the third layer of the deep analysis, we observed that the most commonly preferred shoe material among respondents is fabric, as supported by 18 respondents. This finding will be the focus of the fourth layer of in-depth analysis, continuing from the initial focus.

Next, in the fourth layer of the deep analysis, we discovered that the most preferred brand among respondents is Nike, chosen by 9 respondents. This finding will be further explored in the fifth layer of in-depth analysis, building on the initial focus. In the fifth layer of deep analysis, we found that the most common preferred cost range among respondents is between RM201 and RM300, as indicated by 4 respondents. This finding will be the focus of the sixth layer of deep analysis, while still maintaining the initial focus. Moving on to the sixth layer of deep analysis, we found that the most common preferred salary range among respondents is below RM1500, with the predominant occupation being students, as supported by 3 respondents. This finding will be the focus of the next layer of deep analysis, building on the initial focus. Lastly, in the seventh layer of deep analysis, we discovered that the most common group among those who purchase sports shoes is male, with 2 respondents identifying as such. This concludes our deep analysis.

While the analysis rejects the null hypothesis, it reveals that a significant number of respondents purchase sports shoes primarily because their shoe is damaged due to certain reasons. This highlights that the decision to buy new sports shoes is often driven by practical needs rather than mere preference or style considerations. This insight can help companies better understand consumer behavior and make informed decisions about improving their products.

6.0 Suggestion

Most male students with a salary below RM1500 preferred Nike sports shoes priced between RM201 and RM300, as the fabric material of the shoe is damaged due to certain reasons, prompting them to buy new pairs once a year.

Suggestion:

Many consumers purchase new sports shoes because their shoes are damaged due to certain reasons. In this case, it is recommended that the company enhance fabric durability by using better materials that are more resistant to wear and tear. Moreover, the company should conduct research to identify common damage issues and address them in future designs to improve shoe sales. Additionally, since most consumers are students with lower incomes, the company could implement targeted discounts and promotions specifically for students or individuals with limited financial means. This approach may enhance consumer loyalty, particularly among students, encouraging them to make purchases and ultimately increasing shoe sales. The company can implement a shoe recycling program that allows customers to return worn shoes in exchange for discounts on new pairs. This initiative can enhance brand loyalty and attract consumer interest.

4.0 Data analytics and visualization

4.1 Hypothesis : Customer's footwear category choices depends on the customer's gender

The independent variable in this study is the customer's gender while the dependent variable is the customer's choices in footwear categories.

4.2 Hypothesis Test

Chi Square test for Null Hypothesis (H0) and Alternative Hypothesis (H1)

Null Hypothesis (H0): There is no significant difference in footwear category choices between different genders.

Alternative Hypothesis (H1): There is a significant difference in footwear category choices between different genders.

Chi-Square Test

```
proc freq data=KCJY;
  tables encoded_gender*encoded_type / chisq;
run;
```

The FREQ Procedure

Frequency
Percent
Row Pct
Col Pct

Table of encoded_gender by encoded_type							
encoded_gender	encoded_type						Total
	1	2	3	4	5	6	
1	41	6	35	8	1	1	92
	28.08	4.11	23.97	5.48	0.68	0.68	63.01
	44.57	6.52	38.04	8.70	1.09	1.09	
	70.69	85.71	54.69	57.14	100.00	50.00	
2	17	1	29	6	0	1	54
	11.64	0.68	19.86	4.11	0.00	0.68	36.99
	31.48	1.85	53.70	11.11	0.00	1.85	
	29.31	14.29	45.31	42.86	0.00	50.00	
Total	58	7	64	14	1	2	146
	39.73	4.79	43.84	9.59	0.68	1.37	100.00

Statistics for Table of encoded_gender by encoded_type

Statistic	DF	Value	Prob
Chi-Square	5	5.8570	0.3204
Likelihood Ratio Chi-Square	5	6.4294	0.2667
Mantel-Haenszel Chi-Square	1	2.6790	0.1017
Phi Coefficient		0.2003	
Contingency Coefficient		0.1964	
Cramer's V		0.2003	

WARNING: 50% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 146

Table 4.1 Chi Square Test for Gender and Type

Based on Table 4.1 above, a chi square test was conducted to evaluate the hypothesis that a customer's gender influences their footwear category choices . At a significant level of 0.05, the p value for the relationship between the encoded gender and encoded type is 0.3204 which is greater than 0.05. Therefore, fail to reject the null hypothesis. This indicates that there is no significant evidence to support the claim that footwear category choice differs between genders at the 5% significance level.

4.3 Data Visualization

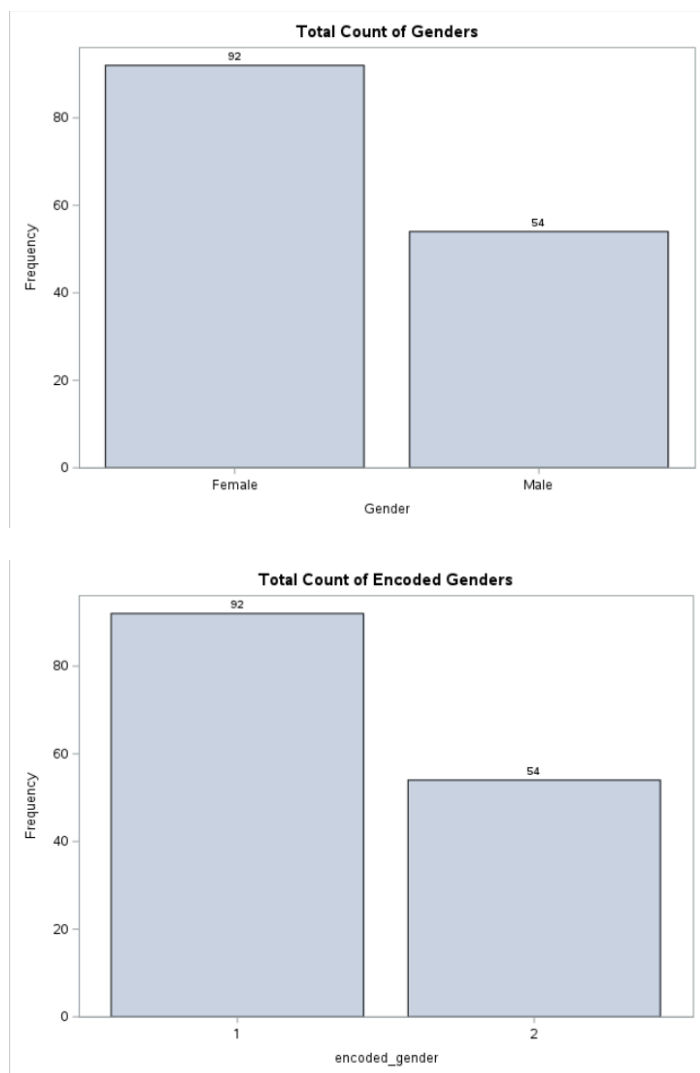
4.3.1 Total Count of Genders

Code:

```
title "Total Count of Genders";
proc sgplot data=KCJY;
  vbar gender / datalabel; *add label to display counts;
run;

title "Total Count of Encoded Genders";
proc sgplot data=KCJY;
  vbar encoded_gender / datalabel; *add label to display counts;
run;
```

Output:



Explanation:

The bar charts above illustrate the distribution of responses by gender from the dataset. In the

encoded data, 1 represents female respondents and 2 represents male respondents. The dataset includes 92 female respondents and 54 male respondents, highlighting a larger proportion of female responses compared to male responses.

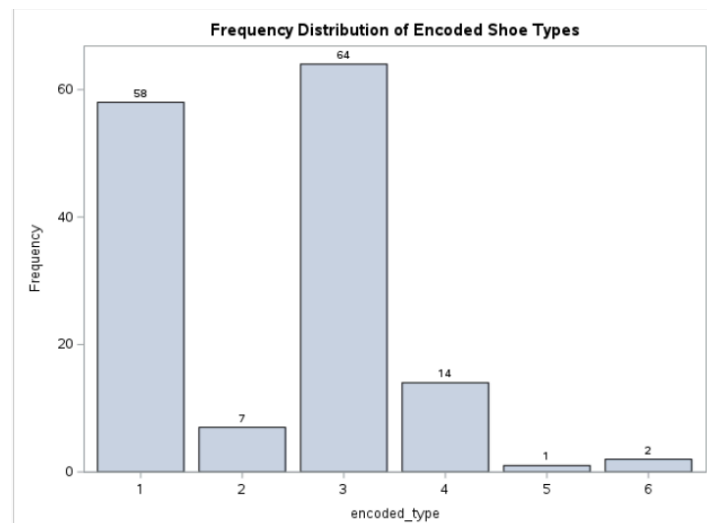
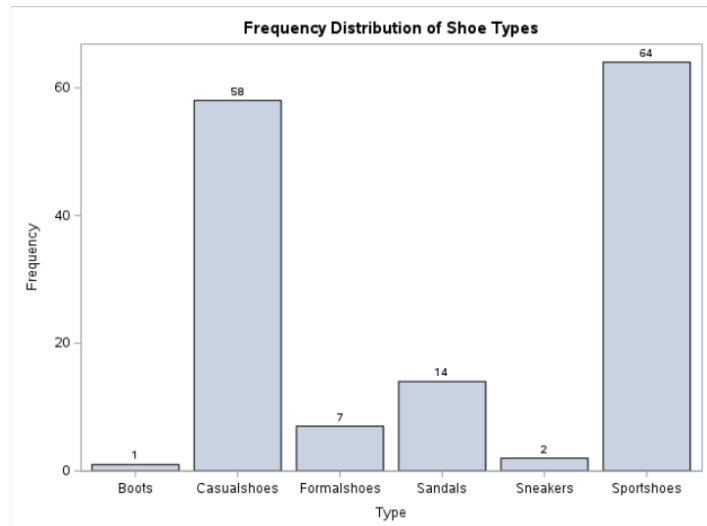
4.3.2 Frequency Distribution of Shoe Types

Code:

```
title "Frequency Distribution of Shoe Types";
proc sgplot data=KCJY;
    vbar type / datalabel;
run;

title "Frequency Distribution of Encoded Shoe Types";
proc sgplot data=KCJY;
    vbar encoded_type / datalabel;
run;
```

Output:



Explanation:

The bar chart displays the frequency of different shoe types. From the graph, Sport shoes are the most frequent with 64 occurrences, followed by Casual shoes are second most frequent type with 58 occurrences. Next, Sandals and Formal shoes have moderate frequencies, appearing 14 and 7 times respectively. After that, Sneakers are less common with only 2 occurrences and lastly, Boots are the least frequent, appearing just once.

The below graph shows the encoded bar chart of the shoe types. The encoded values are

represent:

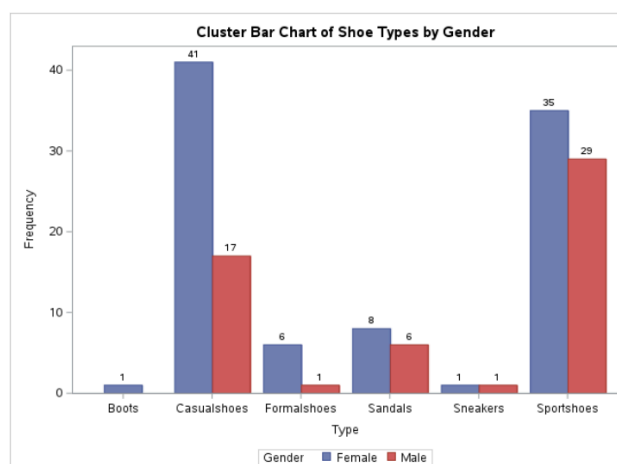
- 1: Casual shoes
- 2: Formal shoes
- 3: Sport shoes
- 4: Sandals
- 5: Boots
- 6: Sneakers

4.3.2 Cluster Bar Chart of Shoe Types by Gender

Code:

```
title "Cluster Bar Chart of Shoe Types by Gender";
proc sgplot data=KCDY;
  vbar type / group=gender groupdisplay=cluster datalabel datalabelfitpolicy=none;
run;
```

Output:



Explanation:

The bar chart provides a clear comparison of shoe type preferences between genders. The blue bars represent Female's shoe type while the red bars represent Male's shoe type. The bars are clustered together for each shoe type, allowing comparison between genders. Boots have a very low frequency, with 1 occurrence among females and none for males. Casual shoes significantly higher among females with 41 occurrences compared to males with 17 occurrences. Formal shoes appear 6 times and just 1 time for males, showing marked difference between the two genders. The Sandals is fairly balanced with 8 occurrences among females and 6 occurrences among males. Next, Sport shoes show high frequency between both genders, though females have 35 occurrences and males have 29 occurrences, making this shoe type quite common overall. In contrast, Sneakers have a very low frequency, with just 1 occurrence each for both females and males. Overall, the chart suggests that females tend to favor Casual shoes and Sport shoes significantly more than males, while other shoe types show smaller gender differences.

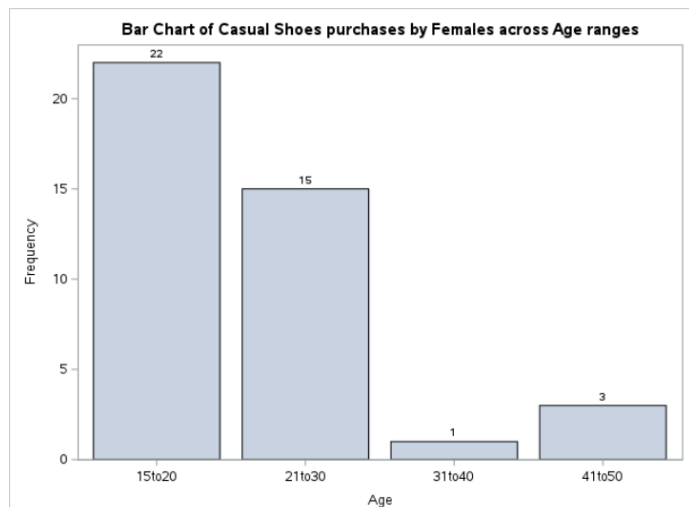
4.3.4 Bar Chart of Casual Shoes purchases by Females across Age ranges

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where type = "Casual shoes"
and gender = "Female"
;
quit;

title "Bar Chart of Casual Shoes purchases by Females across Age ranges";
proc sgplot data=KCJY;
vbar age / datalabel;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

The bar chart highlights the distribution of Casual shoes purchases among females across different age groups. The height of each bar indicates the number of Casual shoe purchases made by females in that age range. For the age group 15 to 20, it has the highest number of purchases with 22 occurrences, indicating that younger females are the most frequent buyers of Casual shoes. The second highest frequency of purchases is the age group of 21 to 30. This age group consists of 15 occurrences showing that this age group also contributes significantly to Casual shoes purchases. Next, the age group 31 to 40 shows a significant drop in purchases, with only 1 occurrence indicating a much lower interest in Casual shoes among females in this age group. Lastly is the age group 41 to 50, there is a slight increase in purchases compared to the previous age group, with 3 occurrences. However, this age group still shows a relatively low frequency of Casual shoes purchases. The bar chart demonstrates that Casual shoes purchased by females are most common in younger age groups, particularly among females aged 15 to 20, followed by those aged 21 to 30. Purchases drop significantly in older age ranges, particularly between 31 to 40, and slightly increasing in the 41 to 50 age group.

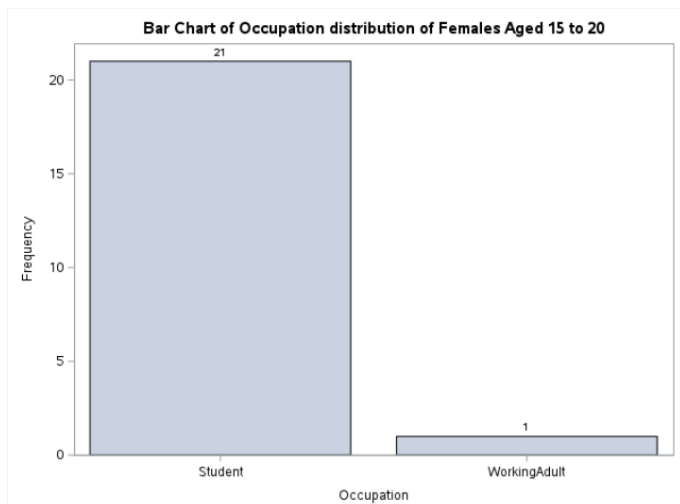
4.3.5 Bar Chart of Occupation distribution of Females Aged 15 to 20

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where age = "15 to 20"
and gender = "Female"
and type = "Casual shoes"
;
quit;

title "Bar Chart of Occupation distribution of Females Aged 15 to 20";
proc sgplot data=KCJY;
vbar occupation / datalabel;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

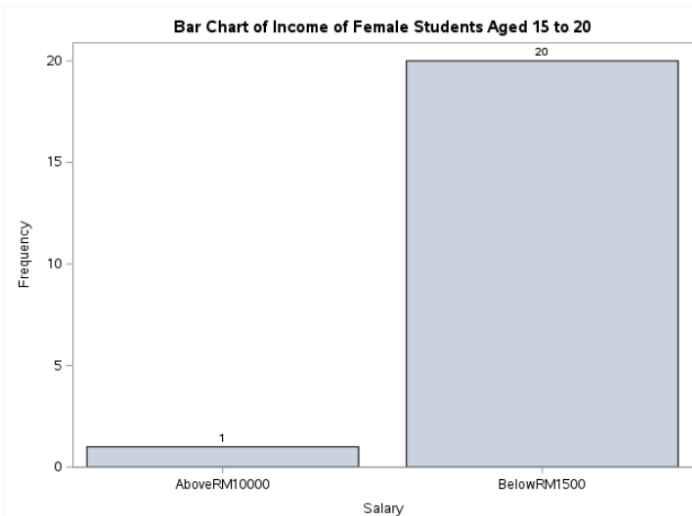
The bar chart presents the occupation distribution for females aged 15 to 20. The bars represent the categories of occupation, which are Student and Working Adult. These categories reflect the occupations of females in this specific age group within the dataset. The Student category shows the highest frequency, with 21 occurrences, indicating that the vast majority of females in this age range are students. In contrast, only 1 individual falls under the Working Adult category, highlighting that very few females in this age group are employed. The distribution clearly demonstrates that females aged 15 to 20 are predominantly students, with only a small percentage already engaged in the workforce.

4.3.6 Bar Chart of Income of Female Students Aged 15 to 20

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where age = "15 to 20"
and gender = "Female"
and type = "Casual shoes"
and Occupation = "Student"
;
quit;
title "Bar Chart of Income of Female Students Aged 15 to 20";
proc sgplot data=KCJY;
vbar salary / datalabel;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

This bar chart displays a clear visualization of the income distribution of female students aged 15 to 20. The two salary categories displayed are Above RM10000 and Below RM1500. The bar representing Above RM10000 is very short, with only 1 student in this category, indicating that only one female student in this age group earns above RM10000. In contrast, the bar for Below RM1500 is significantly taller, with a frequency of 20, showing that most students (20 out of 21) fall into this lower income range. The overwhelming majority of students earn below RM1500, with only one student earning more than RM10000. This distribution reflects the economic reality of students in this age group, where low income is common.

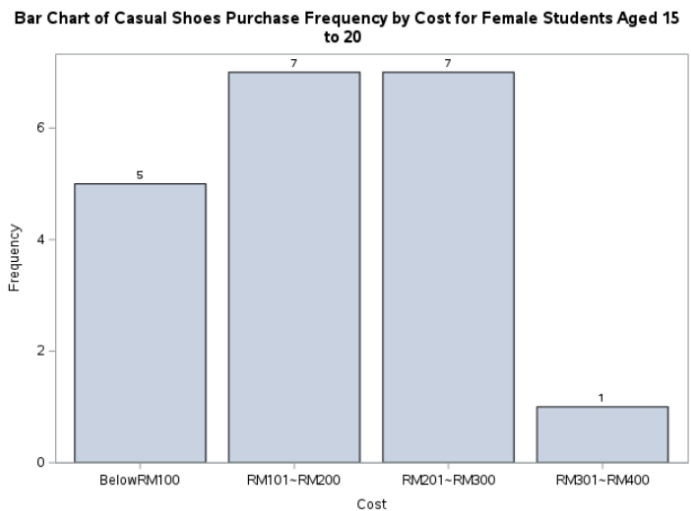
4.3.7 Bar Chart of Casual Shoes Purchase Frequency by Cost for Female Students Aged 15 to 20

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where age = "15 to 20"
and gender = "Female"
and type = "Casual shoes"
and Occupation = "Student"
and salary = "Below RM1500"
;
quit;

title "Bar Chart of Casual Shoes Purchase Frequency by
Cost for Female Students Aged 15 to 20";
proc sgplot data=KCJY;
vbar cost / datalabel;
styleattr datacolors=(vpab vpap);
run;
```

Output:



Explanation:

The chart displays the purchase frequency of Casual shoes by cost for female students aged 15 to 20 who earn below RM1500. There are four cost categories represented by bars. The Below RM100 category shows a frequency of 5 purchases, indicating that five students bought casual shoes in this price range. The cost categories RM101 to RM200 and RM201 to RM300 both have the highest frequency with the same frequency of 7 purchases, showing that these cost ranges are the most popular among the students. Finally, the RM301 to RM400 category has the lowest frequency, with only 1 purchase, showing that very few students buy shoes in this higher cost range. Overall, this bar chart effectively shows the purchasing habits of female students aged 15 to 20 with incomes below RM1500. The majority purchases fall within the RM101~RM200 and RM201~300 cost range, with a smaller but still notable number purchases for shoes priced below RM100. Very few students choose shoes priced above RM300, highlighting a clear preference for more affordable footwear options.

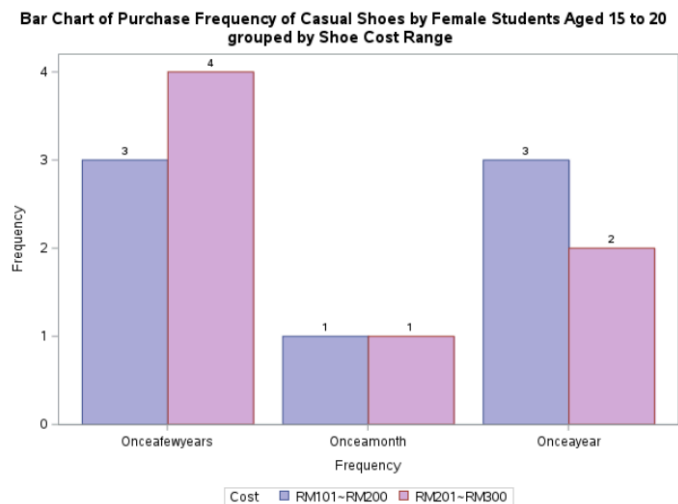
4.3.8 Bar Chart of Casual Shoes Purchase Frequency by Cost for Female Students Aged 15 to 20

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where age = "15 to 20"
and gender = "Female"
and type = "Casual shoes"
and Occupation = "Student"
and salary = "Below RM1500"
and Cost in ("RM101 ~ RM200", "RM201 ~ RM300")
;
quit;

title "Bar Chart of Purchase Frequency of Casual Shoes
by Female Students Aged 15 to 20 grouped by Shoe Cost Range";
proc sgplot data=KCJY;
vbar frequency / group=cost groupdisplay=cluster datalabel;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

The bar chart represents the frequency of Casual shoe purchases among female students aged 15 to 20, grouped by the cost range of the shoes. The blue bar represents RM101 to RM200 while the purple bar represents RM201 to RM300. There are three categories for the frequency of purchases which are Once a few years that consists of 3 students purchasing shoes in the RM101~RM200 range, while 4 students purchasing shoes in the RM201~RM300 range. Next is Once a month, it consists of only 1 student in the range RM101~RM200 and RM201~RM300 makes monthly shoe purchases. Lastly, Once a year the frequency consists of 3 students buying shoes in the RM101~RM200 range and 2 students in the RM201~RM300 range.

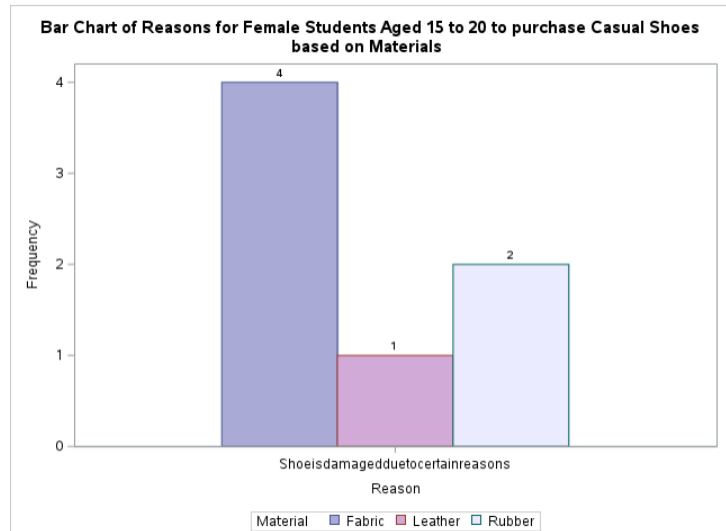
4.3.9 Bar Chart of Reasons for Female Students Aged 15 to 20 to purchase Casual Shoes based on Materials

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where age = "15 to 20"
and gender = "Female"
and type = "Casual shoes"
and Occupation = "Student"
and salary = "Below RM1500"
and Cost in ("RM101 ~ RM200", "RM201 ~ RM300")
and frequency = "Once a few years"
;
quit;

title "Bar Chart of Reasons for Female Students
Aged 15 to 20 to purchase Casual Shoes based on Materials";
proc sgplot data=KCJY;
vbar reason / group=material groupdisplay=cluster datalabel;
styleattr datacolors=(vpab vpap);
run;
```

Output:



Explanation:

The bar chart provides detailed insights into the reason for Casual shoes purchased among female students aged 15 to 20, focusing on the materials of the shoes and the primary reason for purchase. According to the graph, the main reason students purchase new shoes is due to damage. Fabric shoes seem to be the most commonly damaged material with 4 students replacing shoes made of this material. Rubber shoes are the second most frequently damaged material, with 2 students needing to replace them. Lastly, Leather shoes are the least likely to be damaged with only 1 student replacing them for this reason. This suggests that fabric shoes are less durable compared to rubber and leather shoes among this age group.

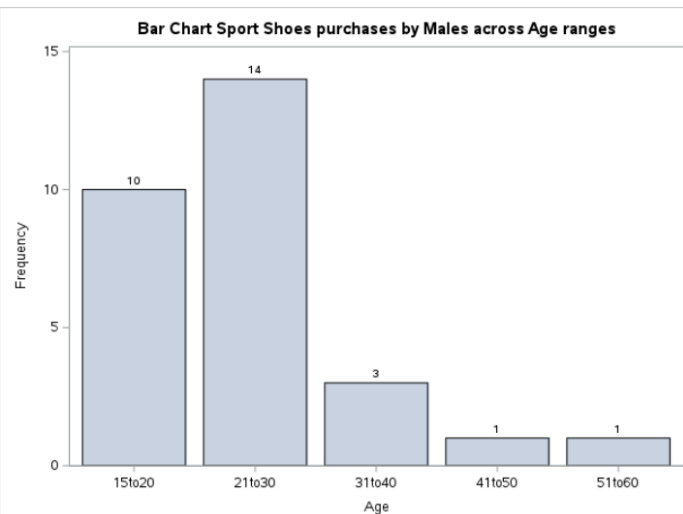
4.3.10 Bar Chart Sport Shoes purchases by Males across Age ranges

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where type = "Sport shoes"
and gender = "Male"
;
quit;

title "Bar Chart Sport Shoes purchases by Males across Age ranges";
proc sgplot data=KCJY;
vbar age / datalabel;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

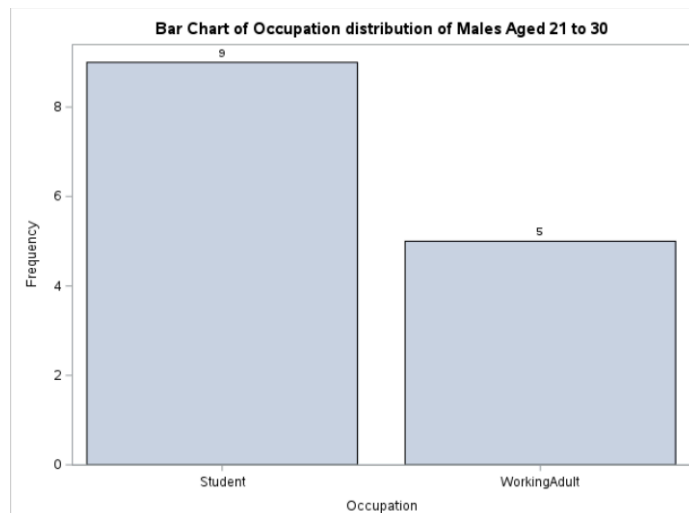
The bar chart highlights the distribution of Sport shoes purchases among males across different age groups. The height of each bar indicates the number of Casual shoe purchases made by females in that age range. For the age group 15 to 20, there is the second highest number of sport shoes purchases with 10 occurrences, indicating that younger males are significantly buyers of Sport shoes. The age group with the highest frequency of purchases is 21 to 30 with 14 occurrences showing a strong preference for sport shoes among males in this age range. For male aged 31 to 40 there is a noticeable decline, with only 3 purchases, indicating a reduced interest in sport shoes. In the older age groups, 41 to 50 and 51 to 60, the number of purchases is minimal, with just 1 occurrence each. The chart suggests that sport shoes are most popular among younger males, particularly those aged 21 to 30, followed by males aged 15 to 20. There is a significant drop in purchases as age increases, especially between 31 to 40, with the lowest interest seen in males aged 41 to 60.

4.3.11 Bar Chart of Occupation distribution of Males Aged 21 to 30

Code:

```
proc sql;  
create table KCJY as  
select * from encoded_data  
where age = "21 to 30"  
and gender = "Male"  
and type = "Sport shoes"  
;  
quit;  
  
title "Bar Chart of Occupation distribution of Males Aged 21 to 30";  
proc sgplot data=KCJY;  
vbar occupation / datalabel;  
styleattr datacolors=(vpab vpap);  
run;
```

Output:



Explanation:

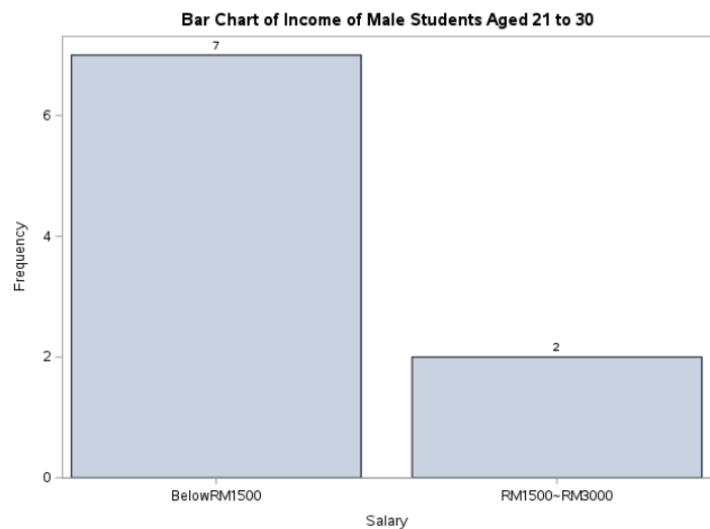
The bar chart presents the occupation distribution for males aged 21 to 30. The bars represent two categories of occupation, which are Student and Working Adult. These categories reflect the occupations of males in this specific age group within the dataset. The Student category shows the highest frequency, with 9 occurrences, indicating that the vast majority of males in this age range are students. In contrast, 5 individuals fall under the Working Adult category. The distribution clearly demonstrates that males aged 21 to 30 are predominantly students, with fewer engaged in employment.

4.3.12 Bar Chart of Income of Male Students Aged 21 to 30

Code:

```
proc sql;  
create table KCJY as  
select * from encoded_data  
where age = "21 to 30"  
and gender = "Male"  
and type = "Sport shoes"  
and Occupation = "Student"  
;  
quit;  
  
title "Bar Chart of Income of Male Students Aged 21 to 30";  
proc sgplot data=KCJY;  
vbar salary / datalabel;  
styleattrs datacolors=(vpab vpap);  
run;
```

Output:



Explanation:

This bar chart displays a clear visualization of the income distribution of male students aged 21 to 30. The two salary categories displayed are Below RM1500 and RM1500 to RM3000. The bar representing RM1500 to RM300 shows a lower frequency, with 2 students in this salary range. In contrast, the bar representing the Below RM1500 category is significantly taller, with a frequency of 7, showing that the majority of male students in this age group earn less than RM1500. The overwhelming majority of students earn below RM1500, with only a small portion earning between RM1500 to RM3000. It reflects the typical financial situation of students in this age group, where low income is more common.

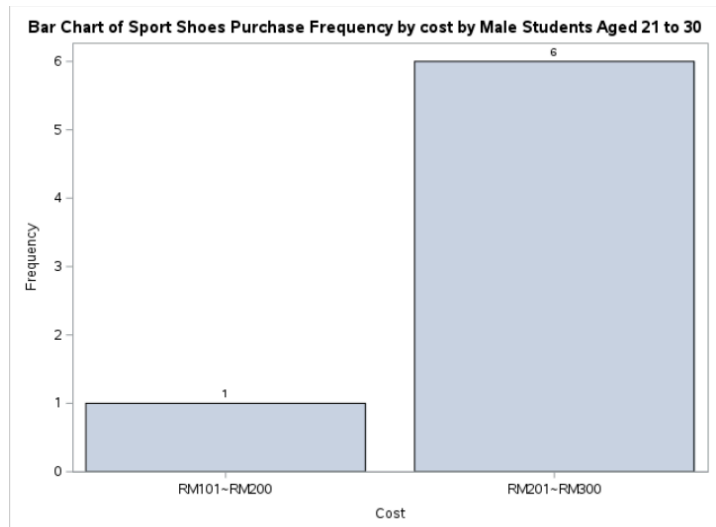
4.3.13 Bar Chart of Sport Shoes Purchase Frequency by Cost by Male Students Aged 21 to 30

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where age = "21 to 30"
and gender = "Male"
and type = "Sport shoes"
and Occupation = "Student"
and salary = "Below RM1500"
;
quit;

title "Bar Chart of Sport Shoes Purchase Frequency
by cost by Male Students Aged 21 to 30";
proc sgplot data=KCJY;
vbar cost / datalabel;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

The chart displays the purchase frequency of Sport shoes by cost for male students aged 21 to 30 who earn below RM1500. There are two cost categories represented by bars. The RM101 to RM200 shows lower frequency with only one student compared to the other, showing that very few students buy shoes in this higher cost range. In contrast, the RM201 to RM300 has the highest frequency of 6 students, indicating that these cost ranges are the most popular among the students. Overall, this bar chart effectively shows the purchasing habits of male students aged 21 to 0 with incomes below RM1500. The majority purchases fall within the RM201~RM300 cost range. Very few students choose a shoe price range between RM101~RM200, highlighting a clear preference for more affordable footwear options.

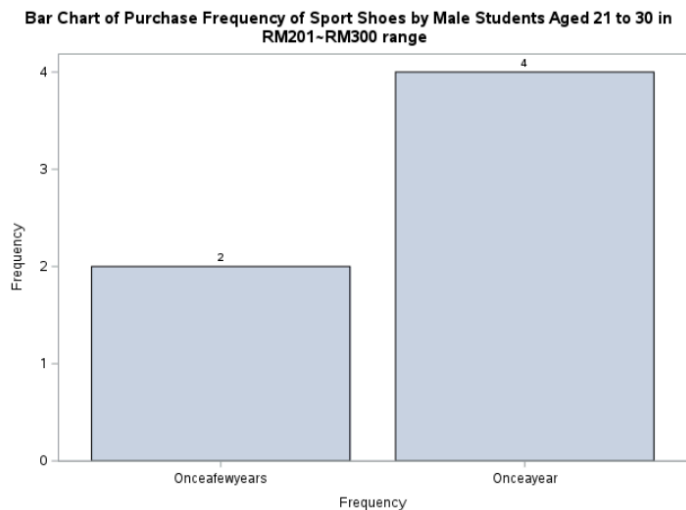
4.3.14 Bar Chart of Purchase Frequency of Sport Shoes by Male Students Aged 21 to 30 in RM201~RM300 range

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where age = "21 to 30"
and gender = "Male"
and type = "Sport shoes"
and Occupation = "Student"
and salary = "Below RM1500"
and cost = "RM201 ~ RM300"
;
quit;

title "Bar Chart of Purchase Frequency of
Sport Shoes by Male Students Aged 21 to 30 in RM201~RM300 range";
proc sgplot data=KCJY;
vbar frequency / datalabel;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

The bar chart represents the frequency of Sport shoe purchases among male students aged 21 to 30 in the cost range of RM201~RM300. There are two categories for the frequency of purchases which are Once a few years that consists of 2 students, showing that they purchase shoes less frequently. In contrast, for the Once a year category consists of 4 students indicating that more students in this aged group tend to buy shoes annually within the cost range. This distribution highlights that a larger proportion of students make yearly purchases in the RM201 to RM300 range, while a smaller number buy shoes less frequently.

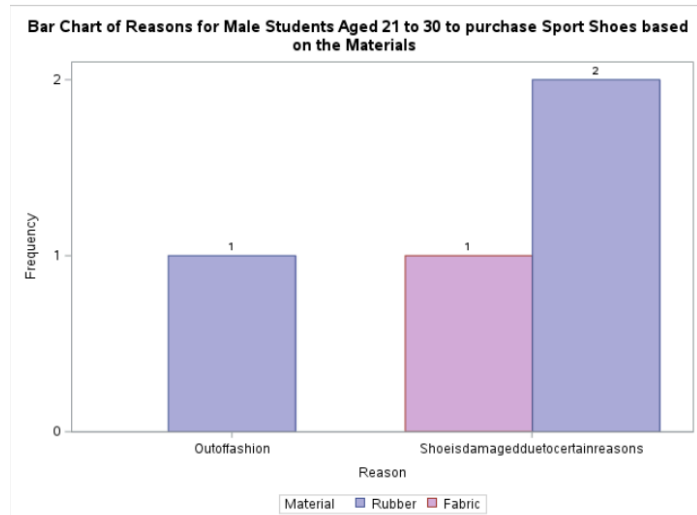
4.3.15 Bar Chart of Reasons for Male Students Aged 21 to 30 to purchase Sport Shoes based on the Materials

Code:

```
proc sql;
create table KCJY as
select * from encoded_data
where age = "21 to 30"
and gender = "Male"
and type = "Sport shoes"
and Occupation = "Student"
and salary = "Below RM1500"
and Cost = "RM201 ~ RM300"
and frequency = "Once a year"
;
quit;

title "Bar Chart of Reasons for Male Students
Aged 21 to 30 to purchase Sport Shoes based on the Materials";
proc sgplot data=KCJY;
vbar reason / group=material groupdisplay=cluster datalabel;
styleattrs datacolors=(vpab vpap);
run;
```

Output:



Explanation:

The bar chart provides insights into the reasons for Sport shoe purchases among male students aged 21 to 30. According to the graph, the main reasons for replacing the shoes are that they are either out of fashion or damaged due to certain reasons. Rubber shoes are the most frequently damaged materials, with 2 students replacing them due to damage. Fabric shoes are damaged less often with 1 student replacing them for this reason. Additionally, 1 student replaced rubber shoes because they were out of fashion. Overall, damage is the leading cause of replacement, particularly for rubber shoes, while fashion influences a smaller proportion of purchases.

4.4 Technique Used

I had to carry out data visualization to present complex data in a clear and understandable format, making it easier to identify patterns and insights. First, I use SQL code to filter and create a dataset from a larger table. The “proc sql” statement is to start a SQL procedure, allowing to run SQL queries within a SAS environment. The use of “create table KCJY as” to create a new table named KCJY. The as keyword indicates that the table will be populated with the results of the following query. The “select * from” statement is used to select all columns from the table. The “where” clause is used to filter the records based on several conditions. “And” clause is used to combine multiple conditions within the “where” clause. Last is the “quit” statement that ends the SQL procedure. It signals to SAS that the SQL commands have been completed.

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To have a better visualization of the data, I use bar charts to present my data. The “title” statement is used to set the title for the chart that will be produced. Next, “proc sgplot data” is used to create a graphical representation of the data in the table. I also use “vbar” to create a vertical bar chart based on the variable selected. “Group” statement is used when the bars are grouped by the variable, allowing comparison of the reason selected in the vertical bar within the grouped variable. To specify that the bars should be displayed side by side, “groupdisplay” statement is used. “Datalabels” statement adds data labels on top of the bars to show the exact values. To specify custom color attributes for the bars in the chart, “styleattrs datacolor= (vpab vpap)” statement can be used. vpab and vpap refer to predefined color styles. “run” executes the preceding command to generate the bar chart.

5.0 Conclusion

Based on the chi square test conducted, we can conclude that gender does not significantly influence the choice of footwear category. This conclusion is supported by the p value of 0.3204, which is higher than the significance level of 0.05. Since the p value exceeds the significance level of 0.05, we fail to reject the null hypothesis. Therefore, there is insufficient evidence to suggest a relationship between gender and footwear category preference, indicating that gender does not play a key role in determining footwear choices in this dataset

From our analysis, we observe that casual shoes are the most popular footwear choice among females, with 41 respondents selecting them. In contrast, sport shoes are favored by males, with 29 male respondents opting for them. To gain deeper insights, we focused on the most popular footwear for each gender in our initial analysis. Next, we are deep into the age of the customers. For females, casual shoes are most commonly purchased by younger age groups, especially those aged 15 to 20, which shows the highest number of purchases. On the other hand, males prefer sports shoes with the highest purchase rate in the 21 to 30 range, followed closely by the 15 to 20 group. Overall, both female and male consumers demonstrate a stronger interest in footwear when they are younger, but this interest wanes with age. Females are most likely to purchase casual shoes during their late teens and early twenties while males are more likely to buy sport shoes in their early adulthood.

After that, we carry the third depth analysis based on the finding data previously. We examined the occupation distribution of respondents. Among males aged 21 to 30 and

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females aged 15 to 20 the majority identify as students with 21 responses and only one employed. For male aged 21 to 30 are primarily students with a majority engaged in academic pursuits. Overall, students make up the bulk of footwear purchasers in both age groups. Furthermore, the fourth analysis is being carried out focused on the income of these students. We found that most of the female students aged 15 to 20 earn below RM1500 with 20 out of 21 responses. Similarly, male students also have a strong tendency towards low income with 7 of the respondents earning below RM1500.

Moving to the fifth layer of analysis, we explored the purchasing habits of female students aged 15 to 20 and male students aged 21 to 30. It has shown that female students tend to purchase casual shoes in the range of RM 101 to RM 200 and RM201 to RM300. In comparison, male students favor sport shoes primarily within the RM201 to RM300 category. Moreover, in the sixth layer of analysis, we can observe that female students aged 15 to 20 primarily purchase casual shoes less frequently. either once a few years or once a year, with only a few making monthly purchases across the two price ranges of RM101 to RM200 and RM201 and RM300 whereas male students aged 21 to 30 primarily purchase sport shoes within the RM201 to RM300 price range on an annual basis as majority of students fall into "Once a year" category. Finally in our last layer of analysis, we discovered that the primary reason female students aged 15 to 20 purchase new casual shoes is due to damage, with fabric shoes being the most commonly replaced. 4 students replaced fabric shoes, making them the most prone to damage. For the male students aged 21 to 30 primarily replace sport shoes either due to damage or because they are out of fashion. Rubber shoes are the most frequently replaced due to damage with 2 students citing this reason.

Although the hypothesis failed to be rejected, showing that there is no statistically significant relationship between the gender and the footwear category preference, there are still observable differences in the types of footwear preferred by females and males. Specifically, females tend to prefer casual shoes whereas males prefer sports shoes. This indicates that while gender may not be a strong determining factor in footwear category choice overall, there are still notable trends in how each gender gravitates toward specific types of footwear.

6.0 Suggestion

Most of the female students aged 15 to 20 with a salary below RM1500 tend to purchase fabric casual shoes at cost range in RM101 to RM200 and RM201 to RM300 primarily when the shoes are damaged due to certain reasons.

Most of the male students aged 21 to 30 with a salary below RM1500 tend to purchase rubber sports shoes at cost range in RM201 to RM300 once a year as the shoes are damaged due to certain reasons or out of fashion.

Suggestions:

Both female and male customers show a strong interest in footwear during their younger years, particularly in the 15 to 30 age range. To effectively engage this demographic, targeted marketing campaigns should leverage social media platforms like Instagram and TikTok, where younger customers are more likely to interact with product promotions and trends. Next, expanding online presence and enhancing e-commerce options is essential, as younger consumers often prefer convenient, budget-friendly shopping. Example, Investing in user experience on websites and mobile apps can make a significant difference and consider offering free shipping, easy returns and exclusive online discounts to attract more online buyers. Additionally, Integrating product reviews and style guides can also help customers make informed purchasing decisions. Moreover, expand product durability and offer warranty services. Improving the durability of key materials like fabric and rubber or introducing a warranty program offering free or discounted replacements within a certain time frame. Highlight the durability and quality of the footwear to attract more customers concerned about value for money. Lastly, it's crucial to optimize pricing, ensuring footwear is positioned within the RM101 to RM300 range to meet the budgetary constraints of this demographic. By addressing these areas, brands can better attract and retain young consumers in the competitive footwear market.

7.0 References

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8.0 Appendix

Data Visualisation

1. https://docs.google.com/forms/d/1GQg0k_QWEb8mYDLzD5PygV-hFNuuLGpD94YKybuG1uQ/edit [Google Form]
2. <https://docs.google.com/spreadsheets/d/15QCD8gTwj74A1XFXuUkEyFgHY1lcvioANDG9qfluOIs/edit?usp=sharing> [Google Sheet]

Timestamp	Question 1 : What is your age?	Question 2 : What is your gender?	Question 3 : Which region of Peninsula Malaysia do you live in?	Question 4 : Where do you live in?	Question 5 : What is your occupation?	Question 6 : What is the range of your salary?
22/07/2024 19:08:36	15 to 20	Female	Klang	City	Student	Below RM 1500
22/07/2024 19:11:43	15 to 20	Female	Damansara	City	Student	Below RM 1500
22/07/2024 19:12:43	15 to 20	Female	Kepong	City	Student	Below RM 1500
22/07/2024 19:14:15	15 to 20	Female	Kuala Lumpur City Center (KLCC)	City	Student	Below RM 1500
22/07/2024 19:23:57	21 to 30	Male	Klang	Village	Student	Below RM 1500
22/07/2024 19:29:53	15 to 20	Male	Setapak	Village	Student	Below RM 1500
22/07/2024 20:43:23	41 to 50	Female	Petaling Jaya	City	Working Adult	Above RM 10000
22/07/2024 21:01:02	41 to 50	Female	Kuching	City	Unemployed	RM 3000 ~ RM 6000
22/07/2024 21:46:10	41 to 50	Female	Cheras	City	Working Adult	RM 3000 ~ RM 6000
22/07/2024 21:52:07	21 to 30	Female	Bukit Bintang	City	Student	Below RM 1500

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22/07/2024 21:59:48	41 to 50	Female	Sarawak	City	Working Adult	RM 3000 ~ RM 6000
22/07/2024 22:12:26	15 to 20	Female	Petaling Jaya	City	Student	Below RM 1500
22/07/2024 23:13:03	15 to 20	Female	no	Rural	Student	Below RM 1500
22/07/2024 23:22:05	15 to 20	Female	Petaling Jaya	City	Student	Below RM 1500
22/07/2024 23:25:31	15 to 20	Female	Petaling Jaya	City	Student	Below RM 1500
22/07/2024 23:28:51	15 to 20	Male	Taman Melawati	City	Student	Below RM 1500
23/07/2024 00:25:50	21 to 30	Male	Klang	City	Working Adult	RM 3000 ~ RM 6000
23/07/2024 00:59:48	21 to 30	Male	Bangsar South	City	Working Adult	RM 3000 ~ RM 6000
23/07/2024 10:17:19	41 to 50	Female	Kuching	City	Working Adult	RM 1500 ~ RM 3000
23/07/2024 10:45:17	15 to 20	Female	Petaling Jaya	City	Student	RM 1500 ~ RM 3000
23/07/2024 12:06:44	15 to 20	Female	Klang	Village	Student	Below RM 1500
23/07/2024 14:52:46	15 to 20	Female	Damansara	City	Student	Below RM 1500
23/07/2024 14:53:50	15 to 20	Female	Petaling Jaya	City	Student	Below RM 1500
23/07/2024 16:24:51	31 to 40	Male	Kepong	Urban	Working Adult	RM 1500 ~ RM 3000
23/07/2024 17:54:25	31 to 40	Male	Klang	Urban	Working Adult	Above RM 10000
23/07/2024 17:59:29	31 to 40	Female	Kepong	City	Working Adult	RM 3000 ~ RM 6000

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23/07/2024 18:07:08	21 to 30	Female	Petaling Jaya	City	Working Adult	RM 6000 ~ RM 10000
23/07/2024 18:10:44	21 to 30	Female	Kepong	City	Working Adult	RM 3000 ~ RM 6000
23/07/2024 18:18:46	21 to 30	Female	Johor Bahru	City	Working Adult	RM 6000 ~ RM 10000
23/07/2024 18:25:40	21 to 30	Male	Cheras	City	Working Adult	RM 3000 ~ RM 6000
23/07/2024 18:33:37	21 to 30	Female	Bahau	Rural	Working Adult	RM 1500 ~ RM 3000
23/07/2024 18:51:15	31 to 40	Female	Cheras	City	Working Adult	Above RM 10000
23/07/2024 19:15:41	31 to 40	Female	Subang Jaya	City	Working Adult	Above RM 10000
23/07/2024 21:14:39	21 to 30	Female	Cheras	City	Working Adult	RM 1500 ~ RM 3000
23/07/2024 21:18:04	21 to 30	Female	Singapore	City	Working Adult	Above RM 10000
23/07/2024 21:18:40	21 to 30	Female	Singapore	City	Working Adult	RM 6000 ~ RM 10000
23/07/2024 23:38:07	15 to 20	Male	Setapak	City	Student	Below RM 1500
24/07/2024 15:21:04	15 to 20	Male	Setapak	City	Student	Below RM 1500
25/07/2024 13:59:04	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:00:14	21 to 30	Male	Northern	City	Student	Below RM 1500

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25/07/2024 14:01:32	15 to 20	Male	Central	City	Student	Below RM 1500
25/07/2024 14:01:57	15 to 20	Male	Central	Rural	Student	Below RM 1500
25/07/2024 14:02:02	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:02:28	21 to 30	Female	Central	City	Student	Below RM 1500
25/07/2024 14:02:44	21 to 30	Male	Central	City	Student	Below RM 1500
25/07/2024 14:03:16	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:04:25	15 to 20	Male	Central	City	Student	Below RM 1500
25/07/2024 14:04:32	41 to 50	Female	Central	Village	Retired	Above RM 10000
25/07/2024 14:05:30	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:07:27	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:07:57	21 to 30	Male	Southern	Village	Unemployed	RM 3000 ~ RM 6000
25/07/2024 14:10:42	41 to 50	Female	Sarawak	Rural	Retired	RM 6000 ~ RM 10000
25/07/2024 14:11:16	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:12:47	21 to 30	Female	Northern	Village	Student	Below RM 1500
25/07/2024 14:13:47	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:14:22	15 to 20	Male	Central	City	Student	Below RM 1500
25/07/2024 14:15:13	21 to 30	Female	Central	City	Student	Below RM 1500
25/07/2024 14:15:19	15 to 20	Male	Central	Urban	Student	Below RM 1500
25/07/2024 14:16:38	21 to 30	Male	Southern	Village	Working Adult	RM 1500 ~ RM 3000
25/07/2024 14:17:08	21 to 30	Male	Central	City	Working Adult	RM 1500 ~ RM 3000

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25/07/2024 14:17:20	21 to 30	Male	Central	City	Student	Below RM 1500
25/07/2024 14:17:43	15 to 20	Male	Central	Urban	Student	Below RM 1500
25/07/2024 14:18:42	15 to 20	Male	Central	City	Student	Below RM 1500
25/07/2024 14:19:10	15 to 20	Male	Central	City	Student	Below RM 1500
25/07/2024 14:22:25	21 to 30	Female	Sarawak	City	Student	Below RM 1500
25/07/2024 14:23:20	31 to 40	Female	Central	City	Working Adult	RM 6000 ~ RM 10000
25/07/2024 14:28:23	31 to 40	Female	Central	City	Working Adult	RM 6000 ~ RM 10000
25/07/2024 14:29:26	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:30:53	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:31:05	21 to 30	Male	Central	City	Student	Below RM 1500
25/07/2024 14:32:13	21 to 30	Female	Sarawak	City	Student	Below RM 1500
25/07/2024 14:32:39	21 to 30	Female	Central	City	Student	Below RM 1500
25/07/2024 14:34:31	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 14:36:07	21 to 30	Female	Sarawak	City	Student	Below RM 1500
25/07/2024 14:37:23	31 to 40	Male	Central	City	Working Adult	RM 3000 ~ RM 6000
25/07/2024 14:37:53	21 to 30	Female	Sarawak	City	Unemployed	Below RM 1500
25/07/2024 14:38:26	51 to 60	Female	Sarawak	City	Working Adult	RM 3000 ~ RM 6000

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25/07/2024 14:40:14	21 to 30	Male	Sarawak	City	Working Adult	RM 1500 ~ RM 3000
25/07/2024 14:40:24	21 to 30	Female	Sarawak	City	Working Adult	RM 1500 ~ RM 3000
25/07/2024 14:47:53	31 to 40	Female	Sabah	Urban	Student	Below RM 1500
25/07/2024 14:57:22	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 15:18:11	15 to 20	Female	Central	Village	Student	Below RM 1500
25/07/2024 15:36:48	21 to 30	Male	Southern	City	Student	Below RM 1500
25/07/2024 15:42:50	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 15:53:46	15 to 20	Male	Central	Urban	Student	Below RM 1500
25/07/2024 15:54:41	51 to 60	Male	Sarawak	City	Working Adult	RM 3000 ~ RM 6000
25/07/2024 15:56:16	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 15:57:24	21 to 30	Female	Sarawak	Urban	Working Adult	Below RM 1500
25/07/2024 16:14:15	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 16:23:35	15 to 20	Male	Central	City	Student	Below RM 1500
25/07/2024 16:24:59	15 to 20	Female	Central	Urban	Student	Below RM 1500
25/07/2024 16:36:26	21 to 30	Female	Central	Village	Student	Below RM 1500
25/07/2024 16:40:56	21 to 30	Male	Northern	Village	Unemployed	Below RM 1500
25/07/2024 16:42:33	21 to 30	Female	Northern	Village	Working Adult	RM 3000 ~ RM 6000

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25/07/2024 17:00:34	21 to 30	Male	Southern	City	Student	Below RM 1500
25/07/2024 17:00:56	15 to 20	Female	Southern	City	Student	Above RM 10000
25/07/2024 17:05:14	15 to 20	Female	Central	City	Working Adult	RM 1500 ~ RM 3000
25/07/2024 17:07:23	15 to 20	Female	Central	Urban	Student	Below RM 1500
25/07/2024 17:08:37	15 to 20	Female	Central	Urban	Student	Below RM 1500
25/07/2024 17:11:03	21 to 30	Male	Eastern	City	Student	Below RM 1500
25/07/2024 17:15:39	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 17:15:58	15 to 20	Male	Southern	Urban	Unemployed	RM 1500 ~ RM 3000
25/07/2024 17:17:52	41 to 50	Male	Central	City	Working Adult	Above RM 10000
25/07/2024 17:32:51	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 17:34:17	15 to 20	Male	Central	Urban	Student	Below RM 1500
25/07/2024 18:12:50	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 18:16:48	21 to 30	Female	Central	City	Working Adult	RM 3000 ~ RM 6000
25/07/2024 18:17:42	15 to 20	Male	Central	City	Student	Below RM 1500
25/07/2024 18:32:43	31 to 40	Male	Sabah	City	Working Adult	RM 1500 ~ RM 3000
25/07/2024 19:10:12	15 to 20	Male	Central	City	Student	Below RM 1500
25/07/2024 19:53:20	15 to 20	Male	Eastern	City	Student	Below RM 1500
25/07/2024 20:21:45	15 to 20	Male	Central	City	Student	Below RM 1500

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25/07/2024 22:38:39	21 to 30	Male	Southern	City	Working Adult	Above RM 10000
25/07/2024 22:47:59	21 to 30	Male	Northern	City	Working Adult	RM 3000 ~ RM 6000
25/07/2024 23:22:02	15 to 20	Female	Southern	City	Student	Below RM 1500
25/07/2024 23:25:59	15 to 20	Female	Central	City	Student	Below RM 1500
25/07/2024 23:40:00	15 to 20	Female	Sarawak	Village	Working Adult	RM 1500 ~ RM 3000
26/07/2024 08:44:32	15 to 20	Male	Central	City	Student	Below RM 1500
26/07/2024 08:52:57	21 to 30	Female	Central	Urban	Student	RM 3000 ~ RM 6000
26/07/2024 09:14:50	21 to 30	Female	Southern	Village	Student	Below RM 1500
26/07/2024 09:15:40	21 to 30	Female	Southern	Village	Student	RM 1500 ~ RM 3000
26/07/2024 09:16:55	21 to 30	Male	Northern	Village	Working Adult	RM 3000 ~ RM 6000
26/07/2024 09:18:07	21 to 30	Male	Central	Village	Student	RM 3000 ~ RM 6000
26/07/2024 09:19:14	21 to 30	Male	Sabah	Urban	Student	RM 1500 ~ RM 3000
26/07/2024 14:21:24	21 to 30	Female	Northern	Village	Student	Below RM 1500
26/07/2024 23:30:13	41 to 50	Female	Central	City	Working Adult	RM 6000 ~ RM 10000
27/07/2024 13:25:51	21 to 30	Female	Central	City	Student	Below RM 1500
27/07/2024 14:15:32	15 to 20	Female	Central	City	Working Adult	Below RM 1500

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27/07/2024 14:16:41	15 to 20	Female	Central	Urban	Working Adult	RM 1500 ~ RM 3000
27/07/2024 14:19:47	41 to 50	Female	Central	City	Working Adult	RM 3000 ~ RM 6000
27/07/2024 14:33:24	21 to 30	Female	Central	City	Student	Below RM 1500
27/07/2024 14:47:18	15 to 20	Male	Central	City	Student	Below RM 1500
27/07/2024 15:07:17	Above 60	Female	Sarawak	Village	Student	Above RM 10000
27/07/2024 15:10:42	15 to 20	Female	Central	City	Student	Below RM 1500
27/07/2024 16:15:56	15 to 20	Female	Central	City	Working Adult	RM 1500 ~ RM 3000
27/07/2024 17:28:27	15 to 20	Female	Northern	City	Working Adult	Below RM 1500
27/07/2024 17:34:01	15 to 20	Female	Central	City	Student	RM 1500 ~ RM 3000
27/07/2024 17:36:24	21 to 30	Male	Sarawak	City	Student	RM 1500 ~ RM 3000
27/07/2024 17:37:10	15 to 20	Female	Northern	City	Working Adult	Below RM 1500
27/07/2024 17:46:12	21 to 30	Male	Central	City	Student	RM 1500 ~ RM 3000
27/07/2024 17:48:45	15 to 20	Female	Central	City	Student	RM 1500 ~ RM 3000
27/07/2024 17:50:12	21 to 30	Male	Central	City	Student	Below RM 1500
27/07/2024 18:05:04	15 to 20	Male	Central	Urban	Student	Below RM 1500
27/07/2024 18:08:21	21 to 30	Female	Central	City	Working Adult	RM 1500 ~ RM 3000

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27/07/2024 18:20:49	21 to 30	Female	Central	City	Working Adult	RM 6000 ~ RM 10000
27/07/2024 18:43:05	15 to 20	Female	Northern	Village	Student	Below RM 1500
27/07/2024 19:11:59	15 to 20	Female	Central	City	Student	Below RM 1500
27/07/2024 19:18:27	15 to 20	Female	Central	Urban	Student	Below RM 1500
27/07/2024 19:26:29	15 to 20	Female	Central	City	Student	Below RM 1500
27/07/2024 19:28:31	15 to 20	Male	Sarawak	Village	Working Adult	RM 1500 ~ RM 3000
27/07/2024 22:30:35	15 to 20	Female	Central	City	Student	Below RM 1500
28/07/2024 13:05:47	21 to 30	Female	Sarawak	Urban	Unemployed	RM 1500 ~ RM 3000
01/08/2024 14:34:33	15 to 20	Male	Central	Urban	Student	Below RM 1500
01/08/2024 14:35:37	21 to 30	Female	Central	City	Student	Below RM 1500
01/08/2024 14:36:01	15 to 20	Female	Central	City	Student	Below RM 1500
01/08/2024 15:58:57	21 to 30	Female	Northern	Urban	Student	Below RM 1500
09/08/2024 18:11:55	15 to 20	Female	Central	City	Student	RM 3000 ~ RM 6000

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Question 7 : What is your shoe size in UK size?	Question 8 : What is your preferable cost when you buy shoes?	Question 9 : What type of shoes do you mostly buy?	Question 10 : What brand do you mostly buy?	Question 11 : What are the factor(s) that persuaded you into buying the shoe brand above?	Question 12 : How often do you buy shoes?
4 to 6	RM 301 ~ RM 400	Casual shoes	Sketchers	Cheap, Frequent Discount, Comfort, Health, Environmental Requirement, Functionality, Durability, Loyal to Brand, Social Influences	Once a month
7 to 9	RM 101 ~ RM 200	Sport shoes	Nike	Cheap, Comfort, Durability	Once a year
4 to 6	RM 201 ~ RM 300	Casual shoes	Puma	Frequent Discount, Loyal to Brand, Social Influences	Once a year
4 to 6	RM 101 ~ RM 200	Casual shoes	Adidas	Social Influences	Once a few years
7 to 9	RM 301 ~ RM 400	Casual shoes	Adidas	Comfort, Environmental Requirement	Once a year
7 to 9	RM 401 ~ RM 500	Sport shoes	Nike	Comfort, Environmental Requirement	Once a few years
4 to 6	RM 201 ~ RM 300	Sport shoes	Adidas	Frequent Discount, Comfort, Durability, Loyal to Brand	Once a year
4 to 6	RM 101 ~ RM 200	Sport shoes	Sketchers	Comfort, Health, Functionality	Once a year
4 to 6	RM 101 ~ RM 200	Formal shoes	Sketchers	Comfort, Functionality, Durability	Once a year
4 to 6	RM 101 ~ RM 200	Casual shoes	Sketchers	Comfort, Health, Functionality, Durability, Loyal to Brand	Once a year
4 to 6	RM 201 ~ RM 300	Casual shoes	Adidas	Frequent Discount, Comfort, Functionality, Durability	Once a year
4 to 6	RM 201 ~ RM 300	Casual shoes	Nike	Frequent Discount, Comfort	Once a month
7 to 9	Below RM 100	Sandals	not branded	Cheap, Frequent Discount, Comfort, Functionality	Once a year
7 to 9	RM 201 ~ RM 300	Casual shoes	Sketchers	Comfort, Durability	Once a few years
7 to 9	RM 201 ~ RM 300	Casual shoes	Nike	Comfort, Loyal to Brand, Social Influences, Appearance	Once a few years

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10 to 12	Below RM 100	Sport shoes	Adidas	Cheap, Comfort, Durability	Once a year
7 to 9	RM 101 ~ RM 200	Sport shoes	Adidas	Cheap, Frequent Discount, Comfort, Environmental Requirement, Durability	Once a year
10 to 12	RM 101 ~ RM 200	Sport shoes	Nike	Cheap, Frequent Discount, Functionality	Once a month
4 to 6	RM 101 ~ RM 200	Casual shoes	Sketchers	Frequent Discount, Comfort, Health, Functionality	Once a year
1 to 3	RM 101 ~ RM 200	Sport shoes	Sketchers	Comfort, Durability, Loyal to Brand	Once a year
4 to 6	RM 101 ~ RM 200	Sandals	Sketchers	Comfort, Health	Once a few years
4 to 6	RM 201 ~ RM 300	Sport shoes	new balance	Comfort	Once a month
7 to 9	RM 201 ~ RM 300	Casual shoes	Nike	Cheap, Comfort	Once a few years
7 to 9	RM 401 ~ RM 500	Sport shoes	Nike	Comfort, Durability, Loyal to Brand	Once a few years
7 to 9	Above 500	Casual shoes	New balance	Loyal to Brand	Once a year
7 to 9	Below RM 100	Sandals	Nike	Cheap, Comfort, Durability	Once a year
4 to 6	RM 201 ~ RM 300	Sport shoes	No specific brand	Social Influences, Nice design	Once a year
7 to 9	RM 201 ~ RM 300	sneakers	Nike	Fashion and Suitable for Self	Once a year
7 to 9	RM 201 ~ RM 300	Sport shoes	Sketchers	Comfort, Durability	Once a few years
4 to 6	RM 401 ~ RM 500	Casual shoes	Nike	Comfort, Environmental Requirement, Functionality, Durability	Once a year
1 to 3	RM 101 ~ RM 200	Casual shoes	Adidas	Frequent Discount, Comfort, Functionality, Durability, Social Influences	Once a year
4 to 6	RM 101 ~ RM 200	Rarely buy	Depends type of shoes	Comfort, Functionality, Durability	Once a year

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4 to 6	RM 201 ~ RM 300	Sport shoes	Puma	Loyal to Brand	Once a year
4 to 6	RM 101 ~ RM 200	Sandals	Adidas	Frequent Discount, Comfort	Once a month
4 to 6	RM 301 ~ RM 400	Sandals	Nike	Comfort	Once a year
4 to 6	RM 101 ~ RM 200	Sport shoes	Nike	Frequent Discount, Comfort	Once a year
10 to 12	RM 301 ~ RM 400	Sport shoes	lining	Functionality	Once a few years
10 to 12	RM 201 ~ RM 300	Casual shoes	Nike	Cheap, Frequent Discount, Comfort, Health	Once a few years
7 to 9	RM 201 ~ RM 300	Casual shoes	New Balance	Cheap, Comfort	Once a year
10 to 12	RM 201 ~ RM 300	Sport shoes	Adidas	Frequent Discount, Functionality	Once a year
7 to 9	RM 201 ~ RM 300	Sandals	Nike	Cheap	Once a few years
10 to 12	RM 101 ~ RM 200	Casual shoes	Adidas	Cheap, Comfort, Durability	Once a few years
7 to 9	RM 101 ~ RM 200	Sport shoes	Puma	Cheap, Frequent Discount	Once a year
7 to 9	Below RM 100	Sport shoes	Puma	Cheap, Comfort, Functionality	Once a year
7 to 9	RM 201 ~ RM 300	Sport shoes	Nike	Functionality	Once a few years
4 to 6	RM 201 ~ RM 300	Sport shoes	Sketchers	Frequent Discount, Comfort	Once a year
7 to 9	Below RM 100	Casual shoes	Bata	Cheap, Comfort, Durability	Once a year
4 to 6	Above 500	Formal shoes	Puma	Comfort, Environmental Requirement, Durability	Once a year
4 to 6	RM 201 ~ RM 300	Casual shoes	Puma	Frequent Discount, Comfort, Functionality	Once a few years
4 to 6	RM 201 ~ RM 300	Sport shoes	Nike	Cheap, Comfort	Once a year
10 to 12	RM 201 ~ RM 300	Casual shoes	Sketchers	Cheap, Frequent Discount, Comfort, Health, Environmental Requirement	Once a month
4 to 6	RM 401 ~ RM 500	Sport shoes	Puma	Frequent Discount, Environmental Requirement, Loyal to Brand	Once a year

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7 to 9	Below RM 100	Casual shoes	Puma	Cheap, Comfort, Functionality	Once a few years
7 to 9	RM 201 ~ RM 300	Casual shoes	Nike	Frequent Discount, Comfort	Once a few years
4 to 6	RM 101 ~ RM 200	Sport shoes	Nike	Comfort, Functionality, Durability, Loyal to Brand	Once a few years
7 to 9	RM 201 ~ RM 300	Sport shoes	Sketchers	Frequent Discount, Functionality, Durability, Loyal to Brand	Once a year
4 to 6	RM 101 ~ RM 200	Casual shoes	Nike	Frequent Discount, Comfort, Durability	Once a few years
10 to 12	RM 101 ~ RM 200	Casual shoes	Bata	Comfort, Functionality, Durability	Once a few years
Above 13	Below RM 100	Casual shoes	Vans	Environmental Requirement	Once a few years
10 to 12	RM 101 ~ RM 200	Sport shoes	Nike	Comfort	Once a few years
7 to 9	RM 201 ~ RM 300	Sport shoes	Nike	Comfort, Functionality, Durability	Once a year
7 to 9	RM 201 ~ RM 300	Sport shoes	Nike	Cheap, Comfort, Durability	Once a few years
7 to 9	RM 101 ~ RM 200	Sandals	Adidas	Frequent Discount, Durability	Once a few years
1 to 3	Below RM 100	Casual shoes	Nike	Cheap	Once a few years
4 to 6	RM 101 ~ RM 200	Casual shoes	Adidas	Cheap, Comfort, Functionality, Durability	Once a few years
7 to 9	RM 101 ~ RM 200	Formal shoes	Adidas	Durability	Once a year
7 to 9	RM 101 ~ RM 200	Formal shoes	Adidas	Durability	Once a year
4 to 6	RM 101 ~ RM 200	Casual shoes	Nike	Cheap, Frequent Discount, Comfort	Once a year
4 to 6	RM 101 ~ RM 200	Sport shoes	Adidas	Cheap, Comfort	Once a year
10 to 12	RM 201 ~ RM 300	Sport shoes	Adidas	Cheap, Frequent Discount, Comfort	Once a few years
4 to 6	Above 500	Sport shoes	Asics	Frequent Discount, Comfort, Functionality	Once a few years
4 to 6	Below RM 100	Casual shoes	Taobao	Cheap, Comfort	Once a few years
4 to 6	RM 101 ~ RM 200	Sport shoes	Nike	Cheap, Frequent Discount	Once a year

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7 to 9	RM 101 ~ RM 200	Casual shoes	Adidas	Comfort	Once a few years
Above 13	RM 301 ~ RM 400	Sport shoes	Nike	Comfort, Health, Durability	Once a year
1 to 3	RM 101 ~ RM 200	Casual shoes	Sketchers	Comfort	Once a few years
4 to 6	RM 101 ~ RM 200	Sandals	Sketchers	Cheap, Comfort, Functionality	Once a few years
7 to 9	RM 301 ~ RM 400	Sandals	Sketchers	Comfort, Durability	Once a few years
4 to 6	Below RM 100	Casual shoes	Nike	Comfort	Once a year
4 to 6	RM 201 ~ RM 300	Casual shoes	Vans	Frequent Discount, Durability	Once a few years
7 to 9	Below RM 100	Sport shoes	Adidas	Cheap, Frequent Discount, Comfort, Durability	Once a few years
4 to 6	RM 101 ~ RM 200	Casual shoes	Adidas	Frequent Discount, Comfort, Social Influences	Once a few years
10 to 12	RM 201 ~ RM 300	Sport shoes	Sketchers	Frequent Discount, Comfort, Health, Environmental Requirement, Functionality, Durability, Loyal to Brand	Once a year
7 to 9	RM 201 ~ RM 300	Sport shoes	Adidas	Comfort, Functionality, Loyal to Brand, Social Influences	Once a few years
7 to 9	RM 101 ~ RM 200	Casual shoes	Off branded	Cheap	Once a few years
7 to 9	RM 201 ~ RM 300	Sport shoes	asics	Comfort, Functionality, Durability	Once a month
4 to 6	RM 101 ~ RM 200	Casual shoes	Nike	Durability	Once a year
7 to 9	RM 301 ~ RM 400	Casual shoes	Nike	Functionality, Durability, Loyal to Brand, Social Influences	Once a year
7 to 9	Below RM 100	Casual shoes	361°	Cheap, Comfort, Nice looking	Once a few years
10 to 12	RM 301 ~ RM 400	Sport shoes	Nike	Cheap, Comfort, Functionality	Once a year
7 to 9	RM 101 ~ RM 200	Sport shoes	Sketchers	Comfort	Once a year

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7 to 9	RM 201 ~ RM 300	Casual shoes	Nike	Frequent Discount, Comfort, Functionality, Loyal to Brand	Once a year
Above 13	Below RM 100	Sandals	Sketchers	Comfort	Once a few years
7 to 9	RM 201 ~ RM 300	Sport shoes	Sketchers	Comfort	Once a year
10 to 12	Above 500	Casual shoes	Nike	Comfort, Durability, Social Influences	Once a year
4 to 6	Above 500	Casual shoes	Nike	Frequent Discount, Loyal to Brand	Once a month
4 to 6	RM 301 ~ RM 400	Sport shoes	Nike	Comfort, Loyal to Brand, Social Influences	Once a month
4 to 6	RM 101 ~ RM 200	Sport shoes	Adidas	Comfort, Durability	Once a year
4 to 6	RM 101 ~ RM 200	Sandals	Cotton on	Cheap, Frequent Discount	Once a year
7 to 9	RM 201 ~ RM 300	Sport shoes	Nike	Frequent Discount	Once a year
7 to 9	Below RM 100	Sandals	Nike	Durability	Once a year
7 to 9	RM 101 ~ RM 200	Casual shoes	Nike	Health, Functionality	Once a few years
7 to 9	RM 101 ~ RM 200	Sport shoes	Nike	Comfort, Durability	Once a year
7 to 9	RM 101 ~ RM 200	Casual shoes	Puma	Cheap, Frequent Discount, Comfort, Health, Environmental Requirement, Functionality	Once a year
7 to 9	RM 301 ~ RM 400	Sport shoes	Adidas	Comfort, Functionality, Durability, Loyal to Brand	Once a few years
4 to 6	Below RM 100	Sandals	Adidas	Cheap	Once a year
4 to 6	Below RM 100	Casual shoes	Adidas	Cheap, Durability	Once a year
7 to 9	RM 101 ~ RM 200	Casual shoes	Nike	Comfort, Functionality, Loyal to Brand	Once a year
7 to 9	RM 101 ~ RM 200	Sport shoes	LINE SEVEN	Cheap, Comfort, Durability	Once a year
10 to 12	RM 101 ~ RM 200	Formal shoes	Vans	Frequent Discount, Comfort, Health, Functionality, Durability	Once a few years

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7 to 9	RM 201 ~ RM 300	Sport shoes	Nike	Frequent Discount, Comfort, Loyal to Brand	Once a year
7 to 9	RM 201 ~ RM 300	Casual shoes	Puma	Comfort	Once a year
7 to 9	RM 101 ~ RM 200	Casual shoes	Puma	Comfort, Durability	Once a year
Above 13	RM 101 ~ RM 200	Sport shoes	Adidas	Cheap, Frequent Discount, Comfort, Health, Functionality, Durability	Once a few years
4 to 6	Below RM 100	Casual shoes	B.U.M	Cheap, Frequent Discount, Comfort, Environmental Requirement	Once a few years
7 to 9	Below RM 100	Casual shoes	Sketchers	Comfort, Functionality	Once a few years
10 to 12	Below RM 100	Formal shoes	no brand	Cheap, Frequent Discount, Comfort	Once a few years
10 to 12	RM 201 ~ RM 300	Casual shoes	Nike	Frequent Discount, Comfort, Functionality, Durability	Once a year
7 to 9	RM 101 ~ RM 200	Sport shoes	Adidas	Cheap, Frequent Discount, Comfort, Durability	Once a few years
7 to 9	RM 401 ~ RM 500	Sport shoes	Nike	Frequent Discount, Comfort, Loyal to Brand	Once a year
7 to 9	RM 301 ~ RM 400	Sport shoes	Nike	Comfort, Functionality, Durability	Once a few years
10 to 12	RM 201 ~ RM 300	Sport shoes	Sketchers	Frequent Discount, Functionality, Durability	Once a few years
7 to 9	RM 201 ~ RM 300	Sandals	Puma	Frequent Discount, Environmental Requirement, Durability	Once a month
7 to 9	RM 101 ~ RM 200	Sandals	Puma	Comfort, Functionality, Social Influences	Once a month
4 to 6	RM 201 ~ RM 300	Formal shoes	Adidas	Comfort	Once a year
4 to 6	RM 101 ~ RM 200	Casual shoes	Vincci	Frequent Discount	Once a year
4 to 6	RM 101 ~ RM 200	Casual shoes	Nike	Cheap, Comfort	Once a few years
4 to 6	RM 101 ~ RM 200	Sport shoes	Adidas	Comfort, Functionality, Durability	Once a few years

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1 to 3	Below RM 100	Casual shoes	Support local brand	Cheap, Frequent Discount, Comfort, Health, Environmental Requirement, Functionality	Once a year
7 to 9	RM 101 ~ RM 200	Sport shoes	Adidas	Comfort, Functionality, Durability, Loyal to Brand	Once a few years
7 to 9	Below RM 100	Sandals	Puma	Frequent Discount, Comfort	Once a few years
10 to 12	RM 301 ~ RM 400	Sport shoes	lining	Comfort, Functionality	Once a few years
Above 13	Above 500	Boots	Vans	Loyal to Brand	Once a week
4 to 6	RM 101 ~ RM 200	Casual shoes	Nike	Cheap, Comfort, Health, Durability	Once a few years
4 to 6	Below RM 100	Sport shoes	Sketchers	Comfort, Environmental Requirement, Functionality, Durability	Once a year
7 to 9	Below RM 100	Sport shoes	Puma	Comfort, Functionality, Durability	Once a few years
4 to 6	RM 101 ~ RM 200	Sport shoes	Adidas	Frequent Discount, Comfort, Health	Once a few years
10 to 12	RM 201 ~ RM 300	Sport shoes	Under armour	Comfort, Functionality, Durability	Once a few years
7 to 9	Below RM 100	Casual shoes	Nike	Comfort	Once a year
10 to 12	RM 201 ~ RM 300	Sport shoes	Nike	Comfort, Durability	Once a year
4 to 6	Below RM 100	Sport shoes	Nike	Comfort, Durability	Once a year
7 to 9	RM 101 ~ RM 200	Sport shoes	Nike	Frequent Discount, Comfort, Functionality	Once a year
7 to 9	Above 500	Casual shoes	Puma	Social Influences	Once a few years
7 to 9	RM 101 ~ RM 200	Formal shoes	The Ballerina	Comfort, Health, Environmental Requirement, Functionality, Durability	Once a few years
4 to 6	Below RM 100	Casual shoes	Adidas	Cheap, Comfort	Once a year
7 to 9	Below RM 100	Sport shoes	Sketchers	Frequent Discount, Comfort, Loyal to Brand	Once a few years

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4 to 6	Below RM 100	Casual shoes	Don't have specific brand	Comfort, Functionality, Durability	Once a year
7 to 9	RM 101 ~ RM 200	Casual shoes	Nike	Comfort, Functionality, Durability	Once a month
4 to 6	RM 201 ~ RM 300	Sport shoes	Nike	Cheap, Comfort, Durability	Once a few years
7 to 9	RM 401 ~ RM 500	Sneakers	Converse	Stylish	Once a few years
4 to 6	Below RM 100	Casual shoes	Sketchers	Cheap, Frequent Discount, Comfort, Functionality, Durability	Once a year
4 to 6	RM 201 ~ RM 300	Casual shoes	Nike	Frequent Discount, Comfort, Functionality	Once a few years
7 to 9	RM 301 ~ RM 400	Sport shoes	New Balance	Frequent Discount, Comfort, Functionality, Durability	Once a few years
1 to 3	Below RM 100	Casual shoes	Puma	Cheap, Comfort, Health, Functionality, Durability	Once a year
4 to 6	RM 101 ~ RM 200	Sport shoes	Puma	Cheap, Frequent Discount, Comfort	Once a year
7 to 9	RM 101 ~ RM 200	Casual shoes	Bata	Cheap, Comfort, Functionality, Loyal to Brand	Once a few years
Above 13	Below RM 100	Sport shoes	Nike	Cheap, Frequent Discount, Health, Environmental Requirement, Durability, Loyal to Brand	Once a few years

BMDS 2113-Data Visualization Assignment

Question 13 : Select one of the justification below as the reason to get a new shoe.	Question 14 : Which shoes colour do you prefer?	Question 15 : What is your preferable material for the shoes?	Question 16 : How did you find out about the new shoes arrival and promotions?
Got discounts ! ! !	Light Theme	Leather	Social Media
Out of fashion	Dark Theme	Rubber	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Rubber	Social Media
Out of fashion	Dark Theme	Rubber	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Textiles	Social Media
Out of fashion	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	shopping physical
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Advertisement

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Depend	Light Theme	All	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Light Theme	Leather	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Official WebSite
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Out of fashion	Light Theme	Fabric	Celebrities
one to have more option	Light Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Leather	Mall
Out of fashion	Light Theme	Fabric	Social Media
Out of fashion	Light Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Textiles	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media

BMDS 2113-Data Visualization Assignment

Out of fashion	Light Theme	Rubber	Advertisement
Out of fashion	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Textiles	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Dislike the brand image	Dark Theme	Textiles	Sponsors by Live Stream
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Out of fashion	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Leather	Advertisement
Out of fashion	Dark Theme	Leather	Celebrities
Shoe is damaged due to certain reasons	Light Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Poor la	Light Theme	Textiles	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Textiles	Advertisement
Dislike the brand image	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Celebrities

BMDS 2113-Data Visualization Assignment

Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Dislike the brand image	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Rubber	Social Media
Out of fashion	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Dislike the brand image	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Out of fashion	Light Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Physical store
Shoe is damaged due to certain reasons	Light Theme	Rubber	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Celebrities
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Textiles	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media

BMDS 2113-Data Visualization Assignment

Shoe is damaged due to certain reasons	Light Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Light Theme	Rubber	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Textiles	Social Media
Shoe is damaged due to certain reasons	Light Theme	Rubber	Social Media
Out of fashion	Light Theme	Leather	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Lazada
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
just want to buy a new one	Dark Theme	Fabric	Social Media
just interested to buy	Light Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Rubber	In-store
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Out of fashion	Dark Theme	Fabric	Advertisement

BMDS 2113-Data Visualization Assignment

Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Textiles	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Advertisement
Out of fashion	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	SHOP
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Out of fashion	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Textiles	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Out of fashion	Dark Theme	Textiles	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Sponsors by Live Stream

BMDS 2113-Data Visualization Assignment

Out of fashion	Light Theme	Fabric	Sponsors by Live Stream
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Leather	During shopping
Shoe is damaged due to certain reasons	Light Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Textiles	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Leather	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Textiles	Social Media
Out of fashion	Light Theme	Textiles	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	physically
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
comfort	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Textiles	Advertisement
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Textiles	Social Media
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media

BMDS 2113-Data Visualization Assignment

Shoe is damaged due to certain reasons	Light Theme	Fabric	Advertisement
Out of fashion	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Rubber	Social Media
Shoe is damaged due to certain reasons	Dark Theme	Cotton canvas	Advertisement
New shoes for new year	Dark Theme	Fabric	Advertisement
Shoe is damaged due to certain reasons	Light Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Light Theme	Rubber	Retail shops
Shoe is damaged due to certain reasons	Light Theme	Fabric	Social Media
Shoe is damaged due to certain reasons	Light Theme	Leather	Social Media
Shoe is damaged due to certain reasons	Light Theme	Anything	I don't.
Out of fashion	Dark Theme	Fabric	Social Media

BMDS 2113-Data Visualization Assignment

CLO2 Demonstrate ability to communicate information to the viewer using data visualization tools. (A3, PLO5)

CLO3 Produce effective data visualizations in order to provide new insights into a research question. (P4, PLO3)

Criterion	Missing or Unacceptable (0-5 mark)	Poor (6-10 marks)	Good (11-15 marks)	Excellent (16-20 marks)	Marks
Data Preparation	The data preparation step is missing or very poor. Irrelevant, inaccurate, or inappropriate information.	The data preparation steps are not listed and described clearly.	The data preparation steps are relevant, offering details about the data collection with some relevant function and techniques	The data preparation steps introduction are informative, succinct, and offer sufficiently specific details about the data collection with more relevant function and techniques	
Data Analytics	Analytical methods and results are not properly explained. The explanations are not aligned with the business intelligence.	Analytical methods and results are explained. However, the explanations are confusing, incomplete or lacked relevance to business intelligence.	Analytical methods and results are explained. The explanations are appropriate and related to the business intelligence.	Analytical methods and results are explained well. The explanations are clear, structured, appropriate and related to the business intelligence.	

BMDS 2113-Data Visualization Assignment

Graphing, Visualization, and research objective and summary	Graphing and visualization are omitted or not irrelevant. Hypothesis Discussions are omitted or confusing.	Limited graphing and visualization are presented. The information is unclear or confusing. Little discussions are presented.	Graphing and visualization are presented. The information is clear. Discussions of the research objective and summary are presented.	Graphing and visualization are presented systematically. The information is comprehensive and informative. All graphs and visualizations are linked to appropriately. The significance of the research objective and summary of the work is discussed.	
	0-5 poor	6-10 acceptable	11-15 admirable	16-20 Exceptional	
Presentation	Goal of research unclear, information included that does not support research claims in any way	There is a great deal of information that is not clearly integrated or connected to the research.	Sufficient information with many good points made, uneven balance and little consistency.	An abundance of material clearly related to the research is presented. Points are clearly made and evidence is used to support claims	1) 2) 3) 4) 5) 6)